Design of Experiments

STATS DOE ANALYSIS 3.1.0

Extension command for IBM SPSS Statistics

User Manual

Creating designs • Screening • Factorial analysis • Response surfaces • Mixtures
• Taguchi • Split-plot • Optimization • Next steps

Every example in this manual was run with the extension; the output shown is the output it produces.
The companion file `DOE_3.1.0_manual_examples.sps` contains the data and the syntax of every example.

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Chapter 1

About this manual

STATS DOE ANALYSIS brings Design of Experiments (DOE) to IBM SPSS Statistics. With one dialog (or one command) you can

- **create** an experimental design: full and fractional factorials, Plackett–Burman and definitive screening designs, central composite and Box–Behnken response-surface designs, general full factorials, Taguchi arrays, Latin hypercubes, D- and A-optimal designs, simplex and extreme-vertices mixture designs and split-plot designs;
- **plan** the number of runs with a power calculator;
- **analyze** the results: coded-factor regression, ANOVA with curvature and lack of fit, effects charts, alias structure, residual checks, Box–Cox transformations, term selection, response surfaces with canonical analysis, mixture models, Taguchi signal-to-noise ratios, split-plot (REML) analysis and the analysis of variability;
- **optimize** one or many responses with desirability functions or a profit/cost formula, check robustness, plan confirmation runs and see where predictions are uncertain;
- **decide what to do next**: a Next Steps table turns the results into recommendations with ready-to-run syntax;
- **explore** the results in an interactive HTML report with a Model and Optimizer Lab.

1.1 How this manual is organized

Chapter 2 installs the extension and runs a first example.

Chapter 3 is the place to start when you know your task by its industry name ("screening", "QbD design space", "robust design", . . .): it points to the right design, analysis and example.

Chapter 4 explains the ideas of DOE in plain words.

Chapter 5 documents every button and control of the dialog, with its syntax keyword and default.

Chapters 6 to 9 are 46 worked examples: creating designs, analyzing them, optimizing, and planning the next experiment. Each example gives the dialog steps, the syntax, the real output and an explanation of how to read it.

Chapter 10 strings the examples together into complete studies.

Chapters 11 and 12 explain the HTML report and every output table.

Chapters 13 to 15 are the reference part: every keyword, every message, the formulas.

The index at the end lists designs, analyses, industry terms, dialog controls and keywords.

1.2 Conventions

- **Analyze > Descriptive Statistics > Design of Experiments** is a menu path.
- **[Model...]** is a button; Model terms is a control (label) in a dialog.
- `TERMS` is a syntax keyword; *reactor* is a dataset name.
- Syntax is shown in grey boxes. It can be pasted into a syntax window and run as shown, once the example datasets have been created (Appendix A).

Tip

Every example in this manual comes from one file, `DOE_3.1.0_manual_examples.sps`: it creates the datasets and contains each example's syntax in the order of this manual. Run it from the top to reproduce every table shown here.

Chapter 2

Getting started

2.1 Requirements

- IBM SPSS Statistics with the **R integration** (R Essentials for SPSS Statistics), installed for the R version your SPSS release supports.
- The R packages **AlgDesign** and **jsonlite**. They are listed in the extension bundle, so SPSS offers to install them with the extension.
- Optional R packages, used when present:
  - **plotly**, **htmlwidgets**, **htmltools**: the interactive HTML report, the Model and Optimizer Lab and the uncertainty map in the report. Without them the SPSS Viewer output is complete but the HTML report is skipped (with a message).
  - **FrF2**: fractional factorial designs with named alias chains and design resolution. Without it, a two-level factorial is always full and the alias structure is shown as a correlation matrix.
  - **ggplot2**, **gridExtra**, **rsm**, **desirability**, **BsMD**: alternative chart styles and helpers of the legacy analysis.
  - **leaps**: best-subsets selection of the legacy analysis.

To install the optional packages, open the R console that SPSS uses (or any R session of the same version) and run

```
install.packages(c("plotly", "htmlwidgets", "htmltools", "FrF2"))
```

2.2 Installing the extension

1. In SPSS Statistics choose **Extensions > Install Local Extension Bundle**.
2. Select `STATS_DOE_ANALYSIS_v310.spe` and confirm. If SPSS asks to install the required R packages, accept.
3. Restart SPSS if the menu item does not appear.

The procedure is then on the menu **Analyze > Descriptive Statistics > Design of Experiments**, and the command `STATS DOE ANALYSIS` is available in syntax. `STATS DOE ANALYSIS /HELP.` prints a short syntax summary.

Tip

A second bundle, `STATS_DOE_ANALYSIS_v310_fallback_typed_names.spe`, offers the same procedure for SPSS versions whose dialog builder cannot show the Responses and Treat-as-categorical lists as variable pickers: there you type the variable names instead of dragging them. Install one bundle or the other, not both.

2.3 A first run

1. Open the file `DOE_3.1.0_manual_examples.sps` in a syntax window, select the lines from the top down to `DATASET NAME reactor.` and run them. The dataset *reactor* is created and active.
2. Choose **Analyze > Descriptive Statistics > Design of Experiments**.
3. Drag Temp, Time, Press and Catalyst into Factors (design columns) and Yield into Responses (measured results). Analyze the responses is ticked already.
4. Click **[OK]**.

The Viewer shows a *Summary of Results* in plain language followed by the model tables, and a link-like text block with the location of the HTML report. Example 7.1 explains every table of this run.

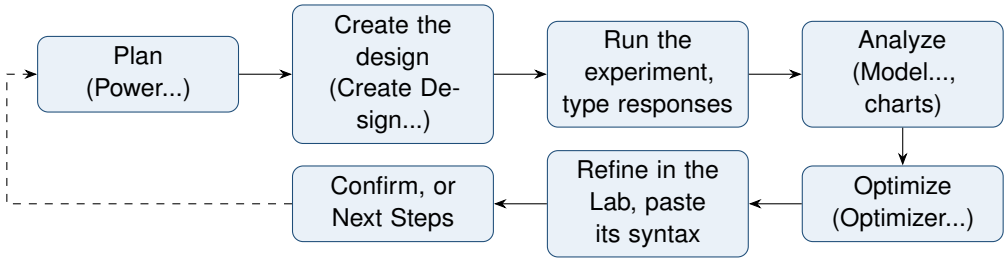
2.4 What one run can do

One `STATS DOE ANALYSIS` command does one of three things, decided by two switches in the main dialog:

Create a new design	Analyze the responses	What happens
ticked	unticked	a new design is built and saved to a new dataset
unticked	ticked	the active dataset is analyzed (and optimized, if asked)
unticked	unticked	only the tools that need no model: Power, Augment to CCD

When both are ticked, the new design is built and, because it has no responses yet, the analysis is skipped with a message.

2.5 The typical study



The dashed arrow is *sequential experimentation*: the results of one experiment (screening) decide the next one (characterization, then optimization). Chapter 10 shows complete sequences.

Chapter 3

Find your analysis

Industries and textbooks use different names for the same tasks. Look up your term here; the last column points to the worked example.

You call it	In this procedure	Example
Planning		
Sample size, number of runs, power analysis, "how many experiments?"	**[Power...]** (/POWER)	6.19, 6.20
Screening: which factors matter?		
Screening, vital few, factor screening, DMAIC Analyze/Improve screening	Plackett–Burman, fractional factorial, definitive screening	6.4, 6.3, 6.5
Many factors (more than 15), factor list in Excel	Read the factor table from the active dataset	6.16
Saturated design, unreplicated factorial, no error df, Daniel plot	Lenth's method, Pareto and normal plot of effects	7.5, 7.6
Confounding, aliasing, resolution, generators	Alias / confounding tables	6.3, 6.4, 7.5
Characterization: how do the factors act?		
Full factorial, 2^k, process characterization, interaction study	Factorial design; coded-factor analysis	6.1, 7.1
Replication, blocking, day-to-day / batch-to-batch variation, nuisance factor	Replicates, Blocks	6.2, 7.11
Curvature check, centre points, linearity of the region	Centre points; curvature term	6.1, 7.1
Comparing materials, suppliers, machines; multi-level categorical factors; ANOVA with Tukey	General full factorial, Treat as categorical, pairwise comparisons	6.8, 7.9, 7.10
Model reduction, stepwise, parsimony, hierarchical model	Model terms, Term selection, Lab	7.2, 7.3
Non-normal residuals, heteroscedasticity, log transform	Box–Cox	7.4
Optimization and design space		
Response surface methodology (RSM), optimization study, quadratic model, sweet spot	Central composite, Box–Behnken; canonical analysis	6.6, 6.7, 7.7, 7.8

You call it	In this procedure	Example
Quality by Design (QbD), design space, proven acceptable ranges, ICH Q8	RSM + overlaid contour plot + uncertainty map + robustness	7.7, 8.2, 7.17
Sequential experimentation, augmenting a design, adding star points	Augment to CCD; Next Steps	6.21, 9.2
Path of steepest ascent/descent, moving toward the optimum	Next Steps	9.1
Multiple response optimization, desirability, trade-off, compromise	Optimizer	8.2, 8.3
Economic optimization, cost minimization, profit maximization	Profit / cost goal	8.4, 8.5
Process window, operating window, specification limits	Optimizer limits; overlaid contours	8.2
Confirmation runs, verification runs	CONFIRMRUNS; Next Steps	8.2
Where can I trust the model? extrapolation risk	Uncertainty map	7.17
Robustness		
Robust design, robust parameter design, Taguchi methods, signal-to-noise	Taguchi array; S/N analysis	6.11, 7.14
Variance reduction, dispersion effects, reducing variability	Analyze variability	7.16
Sensitivity to setting errors, tolerance analysis	ROBUST, ROBUSTSD	8.2
Formulations		
Mixture design, formulation, recipe, blend, Scheffé model	Simplex-lattice, simplex-centroid; mixture analysis	6.12, 6.13, 7.13
Constrained mixture, ingredient limits, extreme vertices	Extreme-vertices design	6.14
Restricted randomization and special situations		
Hard-to-change factor, split-plot, oven/furnace/reactor setups, whole plots	HTC; split-plot (REML) analysis	6.18, 7.15
Categorical factor in an RSM study (supplier, catalyst type)	CCD with categorical factor	6.17
Custom design, irregular region, odd run budget, optimal design	Optimal design (D/A), Constraints	6.9, Sec. 6.22
Computer experiment, simulation, space filling	Latin hypercube	6.10
Existing (historical) data, factors not named	Factor detection; FACTORVARS	7.12

3.1 Which design should I use?

Goal	Factors	Design
Find the few important factors	5–30	Plackett–Burman (main effects only); definitive screening (also curvature); fractional factorial of resolution IV (needs FrF2)
Understand effects and interactions	2–5	Full factorial with centre points; replicate if run-to-run noise is large
Find the optimum (curvature)	2–5	Central composite (rotatable, 5 levels) or Box–Behnken (3 levels, no corners); or augment an earlier factorial
Several categorical levels	any	General full factorial
Recipe proportions	2–6 components	Simplex-lattice / simplex-centroid; extreme vertices when components have limits
A factor that is hard to change	2–5	Factorial or full factorial with `HTC` (split-plot)
Insensitivity to noise	2–7	Taguchi array with replicated responses; or any replicated design + Analyze variability
Odd budget, constraints, special model	any	Optimal (D or A) design
Simulation / computer model	any	Latin hypercube

Chapter 4

DOE in plain words

4.1 Why design experiments?

Changing one factor at a time is slow and misses interactions: the best temperature may depend on the pressure. A designed experiment changes several factors together in a planned pattern, so that each factor's effect, and the interactions between factors, can be separated with the fewest runs and with a known precision.

4.2 The vocabulary

Factor A setting you control: temperature, time, supplier. *Numeric* factors have a range (Low to High); *categorical* (text) factors have levels (A, B, C).

Level A value a factor takes in the experiment. Two-level designs use Low and High.

Response What you measure: yield, strength, cost. A study can have several responses.

Run One trial at one combination of factor settings. The *design* is the list of runs.

Replicate A complete repetition of a run (set up again, run again, measure again). Replicates show the run-to-run variation (*pure error*). Measuring the same sample twice is a *repeat*, not a replicate.

Randomization Running the runs in random order, so that drifts and unknown influences spread over all factors instead of biasing one. `RunOrder` is the order to run; `StdOrder` the textbook order.

Block A group of runs done under similar conditions (one day, one batch). Blocking removes the difference between groups from the comparison of factors.

Centre point A run with every numeric factor at the middle of its range. Centre points estimate pure error and test for curvature.

Coded units Each numeric factor rescaled so that Low $= -1$ and High $= +1$: $x = (\text{setting} - \text{midpoint})/\text{half-range}$. Coefficients in coded units are directly comparable, whatever the factor's units.

Main effect The average change in the response when a factor moves from Low to High. In coded units, effect $= 2\times$ coefficient.

Interaction When the effect of one factor depends on the setting of another. Shown as non-parallel lines in an interaction plot.

Curvature The response is not a straight line between Low and High (it has a maximum or minimum inside, or bends). Two-level designs detect it with centre points; response-surface designs estimate it with squared terms.

Aliasing / confounding In a fractional design two effects are estimated by the same contrast: you see their sum. "A + BCD" means the A estimate contains the BCD interaction.

Resolution How severe the aliasing is: III (main effects aliased with two-factor interactions), IV (main effects clear; two-factor interactions aliased with each other), V (both clear).

Model An equation for the response: $y = b_0 + b_1x_1 + b_2x_2 + b_{12}x_1x_2 + b_{11}x_1^2 + \ldots$ First-order models have main effects (and interactions); second-order (quadratic) models add squares.

Residual Observed minus fitted value. Residual checks test the model's assumptions.

Desirability A 0–1 score of how well a response meets its goal; the composite desirability D combines all responses.

4.3 How to read a p-value (Sig.)

Each term has a test of "this term has no effect". *Sig.* is the probability of an estimate at least this large if the term really had no effect. A small value (below α, usually 0.05) means the effect is unlikely to be noise: the term is *significant*. A large value does not prove there is no effect; it means the experiment could not detect one. Significance is about evidence, not importance: always look at the size of the effect as well.

4.4 How good is the model?

S the residual standard deviation, in response units: the typical size of the scatter around the model.

R-squared the share of the variation explained by the model. It always increases when terms are added.

Adjusted R-squared R-squared corrected for the number of terms; it falls when a useless term is added.

Predicted R-squared how well the model predicts each run when that run is left out (from the PRESS statistic). Much lower than R-squared means over-fitting.

Lack of fit compares the scatter around the model with the scatter between replicates. Significant lack of fit means the model's form misses something real.

4.5 The sequential approach

Experienced experimenters rarely spend the whole budget on one large experiment. A common rule is to spend at most a quarter of the budget on the first experiment:

1. **Screening**: many factors, two levels, few runs; find the vital few.
2. **Characterization**: the vital few in a full factorial with centre points; estimate interactions; test curvature.
3. **Optimization**: add axial points (or run a CCD / Box–Behnken), fit a quadratic model, find the best settings.

4. **Confirmation**: run the best settings a few times and check the prediction interval.

If the first experiment is far from the optimum (no curvature, strong linear effects), move along the path of steepest ascent before optimizing. The Next Steps table (Chapter 9) tells you which of these situations your data are in.

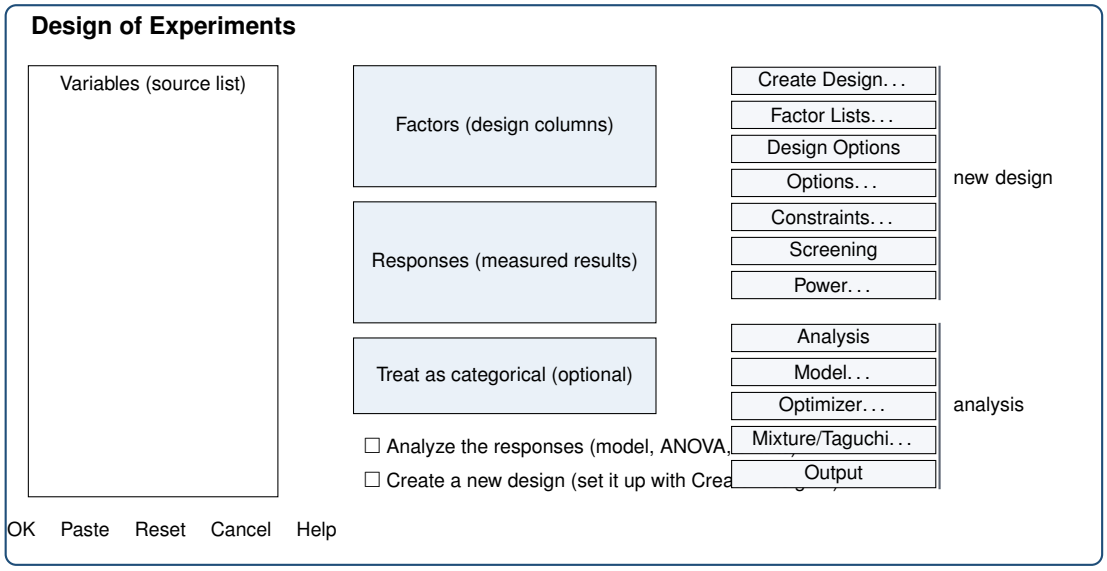
Chapter 5

The dialog, button by button

The dialog is opened with **Analyze > Descriptive Statistics > Design of Experiments**. It has a main window and twelve buttons. This chapter lists every control: its label as shown in the dialog, the syntax keyword it produces, its default and what it does. Controls are greyed out when they do not apply (for example, the optimizer goal fields until Optimize Response is ticked).

5.1 Layout of the main dialog

The sketch below is schematic: it shows the groups of controls, not the exact pixel layout.



- **Left**: the variables of the active dataset.
- **Middle**: drag factor columns, responses and (optionally) numeric factors to be treated as categories. For a new design, leave the lists empty.
- **Right, top group**: setting up a new design (and the power calculator and augmentation).
- **Right, bottom group**: analysis, model, optimizer and output.
- **[Help]** opens the command's help page. The sub-dialogs' own help buttons open the same page.

5.2 How to fill in the text fields

Several fields take a short text in a fixed form. The rules below apply to the dialog and to the syntax alike.

General rules.

- Separate items with spaces. `Temp=190` and `Temp = 190` are both accepted.
- Factor and response names are not case-sensitive (`temp` finds `Temp`); neither are text levels (`b` finds `B`).
- Numbers use a decimal point (`0.15`), in the factor's natural units (not coded $-1/+1$).
- **Letter codes**: every factor also has a letter, shown in the *Factor Coding* table: A, B, C, ... in the order of the Factors list (or the factor names), skipping I. With more than 25 factors the codes are F1, F2, ...
- Lists that have "one value per response" follow the order of the Responses list; lists with "one value per factor" follow the order of the factor names. A single value applies to every factor where the field says so.

Model terms (`TERMS`)

You type	Meaning
`Temp Time Press`	three main effects
`A B C`	the same, with letter codes
`Temp*Time` or `A*B` or `AB`	two-factor interaction
`A*B*C` or `ABC`	three-factor interaction
`Temp*Temp` or `AA`	square (curvature) of a numeric factor; needs 3 or more levels of that factor
`CTPT`	centre-point (curvature) term of a two-level design with centre points
`LINEAR` / `2WAY` / `3WAY`	all main effects / all 2-factor / all 3-factor interactions
`SQUARES` / `QUADRATIC` / `FULL`	all squares / main effects + squares + 2-way interactions / every interaction
`LINEAR Temp*Time CTPT`	keywords and terms can be mixed
`Yield: A B A*B CTPT; Cost: A B C`	a different model per response (response name, colon, terms; responses separated by `;`)

- **List the main effects yourself.** `A*B` alone fits only the interaction; write `A B A*B`. A typed model is used exactly as typed, and a warning names the lower-order terms that are missing (the automatic term selection and the Lab keep models hierarchical).
- In a two-level design with centre points, a square such as `A*A` is the same column as `CTPT`; use `CTPT`. Individual squares need a response-surface design.
- Terms the design cannot estimate separately are dropped and listed in the Warnings and the alias table; misspelled terms are ignored with a warning.
- Model terms overrides Standard model. Leave both at their defaults for the model that suits the design.

Predict at settings (PREDICT)

You type	Meaning
`Temp=190 Time=25 Press=1 Catalyst=B`	one setting
`A=190 D=B`	the same with letter codes; Time and Press at their centres
`Temp=190 Time=25; Temp=160 Time=12`	two settings, separated by ;
`Temp=250`	outside the tested range: predicted, but flagged as extrapolation

Factors you leave out are set to their centre (numeric) or first level (text). The output (Example 7.2) gives the fit, its standard error, the confidence interval for the average at those settings and the prediction interval for one new run. With a centre-point term in the model, predictions between the corners and the centre assume no curvature (a note says so).

Hold factors at (HOLD) and Text factor low level (LOWLEVELS)

Both take `Factor=value` pairs separated by spaces.

- `Catalyst=B Time=20` in Hold factors at keeps Catalyst at B and Time at 20 while the optimizer searches the other factors (Example 8.3).
- `Catalyst=B` in Text factor low (-1) level codes B as -1 and A as $+1$. This only flips the sign of the Catalyst coefficients; the fit, the tests and the predictions do not change. By default the level that appears first in the data is -1.

Factor lists ([Factor Lists...])

With factor names `Temp Time Catalyst` typed in Or factor names:

Low Values `150 10 1`	Temp from 150, Time from 10, Catalyst coded from 1
High Values `200 30 2`	to 200, 30 and 2
Number of Levels `0 0 2`	0 = the design's default for that factor
Decimals for Levels `0 1 0`	Time rounded to 1 decimal
Factor Flags `NO NO YES`	Catalyst is categorical
Mixture Flags	leave blank unless the factors are proportions of a blend

Optimizer lists ([Analysis] and [Optimizer...])

For responses `Yield Cost` (in that order), to aim Yield at 64 within 60–68 and keep Cost low (fully satisfied at 16, unacceptable above 22):

Optimization goals `TARGET MINIMIZE`	one goal per response
Target values `64 16`	Target: the value aimed at; Minimize/Maximize: the value at which $d = 1$
Lower limits `60 0`	Target/Maximize: $d = 0$ below it. Cost is minimized, so its lower limit is not used: type any number (0) to keep the list in order
Upper limits `68 22`	Target/Minimize: $d = 0$ above it
Weights `1 1`	1 = straight-line desirability; 2 = stricter, 0.5 = more lenient
Importance `1 3`	Cost counts three times as much in D

Leave a whole list blank to use the defaults (observed range, weight 1, importance 1). Chapter 8 shows the desirability formulas.

Profit or cost formula (`PROFIT`) and requirements (`PROFITLIMITS`)

- Formula: response names stand for their predicted values, numeric factor names for their settings in natural units. Allowed: numbers, `+ - * / ^ ( )`, `exp log log10 sqrt abs min max ifelse` and comparisons. Text factors are compared with double quotes: `ifelse(Catalyst=="B", 0.5, 0)`. Example: `120*Yield - 3*Cost - 0.02*Temp`. Type it without surrounding quotes in the dialog.
- Requirements: `Yield>=60 Cost<=25` (responses only, >= or <=, separated by spaces).

Other fields

Hard-to-change factors factor names separated by spaces, e.g. `Oven`. At least one factor must remain easy to change.

Whole plots the name of one variable; blank when it is called WholePlot.

Uncertainty limit a number in response units (a half-width: `0.4` marks settings whose prediction is less precise than ± 0.4).

Effect sizes effect divided by the run-to-run standard deviation: to detect a change of 3 units when the standard deviation is 2, type `1.5`.

Lambda a power, e.g. `0.33` (cube root) or `-0.5`.

5.3 Technical choices explained

Choice (where)	In plain words
Criterion D / A (**[Options...]**)	D makes the coefficients jointly as precise as possible; A minimizes their average variance. D is the usual choice.
Initial Design: Random / Nullification	How the optimal-design search starts. Nullification builds the start point by point: slower, usually better.
Design Algorithm: Exact / Approximate	Exact gives a design of whole runs (use this). Approximate gives weights on candidate points, a theoretical tool.
Eval-Table Centering	Optimal designs: centre the factors before computing the design-quality measures. It does not change the runs and has nothing to do with centre-point runs.
Candidate points	The grid of possible runs an optimal design chooses from.
Repetitions	How many times the optimal search is repeated; the best result is kept.
Model Type (Optimal only)	What the design must be able to estimate: Linear (main effects), Quadratic (curvature, for optimization), Cubic, CubicS (cubic for mixtures).
Centre points / Curvature term (CTPT)	Runs at the middle of every numeric factor; their term tests whether the response bends inside the region.
Axial (star) points	Runs on the factor axes; they let a model estimate squared (curvature) terms.
Standard model	A preset model: main effects, interactions, squares, full quadratic.
Term selection	Letting p-values decide which terms stay (backward, forward, stepwise).
Alpha to enter / remove	The p-value thresholds for term selection.
Box–Cox	Fit a power of the response (Y^{λ}, log when $\lambda = 0$) when its spread grows with its size.

Choice (where)	In plain words
Pseudo-components	Rescaled mixture proportions used when every component has a lower limit, which makes the coefficients stable.
Mixture model (Scheffé)	Linear = pure components; Quadratic adds two-component blending; Special/Full cubic add more blending terms.
Signal-to-noise (Taguchi)	One number per run that rewards a good mean *and* a small spread; larger S/N is better.
Canonical analysis	Locates where the fitted surface is flat and says whether that is a peak, a valley or a saddle.
Pairwise comparisons (Tukey)	Which levels of a categorical factor differ, shown with letters.
Analyze variability	Which factors change the run-to-run spread, not the average.
Weight / Importance (optimizer)	Weight shapes one response's desirability curve; importance sets how much that response counts in the total.
Robustness check	Simulates small errors in the settings to see whether the optimum stays acceptable.
Uncertainty map	Where in the region the model's predictions are least precise.
Hard-to-change factor / whole plot	A factor changed only between groups of runs; each group is a whole plot.
Legacy model	The earlier raw-unit analysis, kept for compatibility; use the coded model for new work.

5.4 Every control

5.4.1 Main dialog

The main dialog holds the variable lists and the two switches that decide what a run does. All twelve buttons on the right open sub-dialogs; the upper group sets up a new design, the lower group controls the analysis.

Control	Keyword	Default	What it does
Factors (design columns)	`FACTORVARS`	blank	Drag the columns that hold the factor settings. Left empty, the factors saved with a design generated by this procedure are used, or detected from the data.
Responses (measured results)	`RESPONSEVAR`	blank	Drag one or more numeric response variables here. With several responses, each is analyzed and the optimizer balances all of them.
Treat as categorical (optional)	`CATEGORICAL`	blank	Drag numeric factors here to compare their settings as separate categories (each setting gets its own mean, e.g. 3 temperatures as 3 levels) instead of fitting a line. Text factors are always categorical.
Analyze the responses (model, ANOVA, charts)	`ANALYZE`	ticked	Conduct ANOVA and display model statistics Tick it for every analysis; untick it when you only build a design or only use **[Power...]**.

Control	Keyword	Default	What it does
Create a new design (set it up with Create Design...)	GENERATEDESIGN	unticked	Tick to build a new experimental design from the Create Design settings (saved to the Output Dataset). Leave unticked to analyze the variables on the left. When ticked, Output Dataset in **[Create Design...]** must be filled in.

5.4.2 Create Design... (dialog title: Create Design)

Everything needed to describe a new design: its type, its factors and the dataset that receives it. Fill in either the factor table (up to 15 rows), or Number of factors, or tick Read the factor table from the active dataset, or type the names in Or factor names and give the settings in **[Factor Lists...]**. Use one way, not two.

Control	Keyword	Default	What it does
Number of factors	NFACTORS	blank	How many factors the design has. Unnamed rows become A, B, C ... (I is skipped) with Low -1 and High +1, so you only type the rows you want to change. More than 15: the extra factors are added the same way.
Read the factor table from the active dataset	FACTORSFROMDATA	unticked	Any number of factors: type or paste (e.g. from Excel) one row per factor into the Data Editor with columns Name, Low, High, and optionally Levels and Categorical, then tick this box.
Factor table: Name / Low / High / Levels / Categorical (rows 1–15)	NAMEn, LOWn, HIGHn, LEVELSn, FACTORn	blank	One row per factor. A blank name becomes A, B, C, ...; with Number of factors, blank Low/High become $-1/+1$. Categorical rows need only Levels.
Or factor names (settings in Factor Lists...)	VARNAMES	blank	Names of the factors of the new design, separated by spaces (valid variable names). Their order is the order of every list in Factor Lists.
Output Dataset	DATASET	blank	
Design Type	DESIGNTYPE	Optimal Design (D/A) — k≥1	Select the type of experimental design Choices: Optimal Design (D/A) — k≥1 (OPTIMAL); Factorial Design — k≥2, 2^k runs (FACTORIAL); Plackett-Burman Design — k≥2 (PLACKETTBURMAN); Response Surface Design — k≥2 (RSM); Box-Behnken Design — k≥3 (BOXBEHNKEN); Central Composite Design — k≥2 (CCD); Taguchi Design — k≥1 (TAGUCHI); Latin Hypercube Sampling — k≥1 (LHS); Definitive Screening (DSD) — k≥4 (DSD); Full Factorial (any number of levels) (FULLFACTORIAL); Simplex lattice (mixture) (SIMPLEXLATTICE); Simplex centroid (mixture) (SIMPLEXCENTROID); Extreme vertices (constrained mixture) (EXTREMEVERTICES) In an analysis the procedure recognizes the design from the data; choose a type only to override that (for example Plackett-Burman or Taguchi).

Control	Keyword	Default	What it does
Model Type (Optimal Design Only)	MODEL	Linear	Optimal designs only: the model the design must estimate. Linear = main effects; Quadratic = adds squares and 2-way interactions (finds optima); Cubic; CubicS = cubic for mixtures; Custom = your own R formula. Other design types ignore this. Choices: Linear (LINEAR); Quadratic (QUAD); Cubic (CUBIC); CubicS (CUBICS); Custom Formula: (custom)
	FORMULA	blank	

5.4.3 Factor Lists... (dialog title: Factor Lists)

The list form of the factor settings: one value per factor named in Or factor names, separated by spaces, in the same order. A single value is applied to every factor.

Control	Keyword	Default	What it does
Low Values (required)	LOWS	blank	Low setting of each factor, in the order of the factor names, separated by spaces (e.g. 150 10 1). One value applies to all factors. Mixture components: lower bound of the proportion.
High Values (required)	HIGHS	blank	High setting of each factor, in the order of the factor names, separated by spaces (e.g. 200 30 3). One value applies to all factors. Mixture components: upper bound of the proportion.
Factor Midpoint (optional)	CENTERS	blank	Optional center setting of each factor (blank = midpoint of Low and High). Must lie between Low and High.
Number of Levels (optional)	LEVELS	blank	Optional number of levels of each factor (2 or more), in the order of the factor names. Use 0 to keep the design's default for a factor while setting others. Used by Full factorial, Optimal and categorical factors.
Decimals for Levels (optional)	ROUNDTOS	blank	Optional number of decimals for each factor's settings (e.g. 0 1 3 rounds the first factor to whole numbers, the second to 1 decimal, the third to 3).
Factor Flags (optional, YES/NO)	FACTORS	blank	Optional YES/NO for each factor: YES = categorical factor (levels 1 to Levels). One value applies to all. Example: YES NO NO
Mixture Flags (optional, YES/NO)	MIXTURES	blank	Optional YES/NO for each factor: YES = mixture component (components add to Mixture Sum in every run). Example: YES YES YES NO

5.4.4 Design Options (dialog title: Design Options)

Run order, blocking, replication and the settings of the special designs (lattice degree, axial blends, split-plot, extreme vertices).

Control	Keyword	Default	What it does
Randomize Run Order	RANDOMIZE	unticked	Randomize the execution order of experimental runs to reduce systematic bias
Number of Blocks	BLOCKS	blank	Split the runs into groups run under similar conditions (days, batches); the analysis removes the difference between groups. Blocks are made of whole replicates, so Replicates must be a multiple of Blocks. Exception: with the R package FrF2, a factorial can be blocked within one replicate.
Number of Trials	TRIALS	blank	Number of runs. Optimal: default = number of model terms + 5. Latin hypercube: default = 5 per factor (at least 10). Two-level factorial: fewer runs than 2^k asks for a fractional factorial (needs the R package FrF2). Other designs set their own run count. For a two-level factorial, fewer runs than 2^k asks for a fraction (needs FrF2).
Number of Replicates	REPLICATES	blank	Number of complete copies of the design. Each copy is a new set of runs (set up and measured again), which estimates run-to-run variation and makes tests more sensitive.
Lattice degree (simplex-lattice)	LATTICEDEGREE	2	Simplex lattice: proportions go in steps of 1/degree. Degree 2 gives pure components and 50:50 blends; degree 3 gives steps of 1/3.
Add axial blends and centroid (mixture designs)	AUGMENTMIX	unticked	Mixture designs: adds axial check blends (inside the region) and the overall centroid, so lack of fit of the mixture model can be tested.
Hard-to-change factors (split-plot)	HTC	blank	Split-plot (Factorial and Full factorial designs): names of the factors that are hard or slow to change, e.g. Oven. They change only between groups of runs (whole plots), numbered in a WholePlot column. Use 2 or more replicates; at least one factor must be easy to change. Only Factorial and Full factorial designs. Use two or more replicates so the whole-plot error can be estimated.
Extreme vertices: centroid degree	EVDEGREE	blank	Extreme-vertices designs: 0 = the vertices and the overall centroid; 1 = also the midpoints of the edges (default); 2 = also the centroids of the 2-dimensional faces.

5.4.5 Options... (dialog title: Options)

Search settings for optimal designs, plus the confounding table and the mixture total, which apply to every design.

Control	Keyword	Default	What it does
Criterion	CRITERION	D (generalized variance)	Optimal designs: D = the most precise coefficients overall (smallest joint confidence region); A = the smallest average variance of the coefficients. Choices: D (generalized variance) (D); A (average coefficient variance) (A)
Eval-Table Centering	CENTER	unticked	Optimal designs only: center the numeric factors before the design is evaluated (affects the evaluation measures, not the runs). Not related to center-point runs.
Initial Design	INITIAL	Random	Optimal designs: how the search starts. Random = a random set of candidate points. Nullification = builds the start point by point; usually finds a better design, but is slower. Choices: Random (RANDOM); Nullification (NULLIFICATION)
Design Algorithm	DESIGNALG	Exact	Optimal designs: Exact = a design with whole runs (normal choice). Approximate = an approximate-theory design with weights; for research use. Choices: Approximate (APPROX); Exact (EXACT)
Include constant term	CONSTANT	ticked	
Repetitions of Design Calculation	REPEATS	5	Optimal designs: how many independent searches to run; the best design is kept (default 5). Increase it when results change from run to run.
Number of Candidate Points to Generate	NUMCAND	blank	Optimal designs: number of candidate points the runs are chosen from (default 10 x number of model terms). More candidates = finer choice, slower search.
Display confounding table	CONFOUNDING	ticked	New designs: shows which effects are confounded (aliased) with each other, or the correlations between main effects and 2-way interactions when the design is not a regular fraction.
Mixture Sum (applies if any mixture variables)	MIXTURESUM	1	

5.4.6 Constraints... (dialog title: Region Constraints)

Restricts the candidate points of an optimal design with an R function (see Section 6.22).

Control	Keyword	Default	What it does
Filter design points		unticked	
Browse for a File	CONSTRAINTFUNC	blank	

5.4.7 Screening (dialog title: Screening Designs)

Centre points for two-level designs and the sequential augmentation of a factorial into a central composite design.

Control	Keyword	Default	What it does
Add Center-Point Runs	SCREENINGCENTERPTS	unticked	Two-level designs: add runs with every numeric factor at its midpoint. They estimate pure error and test for curvature.
Number of center points	SCREENINGCENTERPTSCOUNT	blank	Number of center-point runs to add (default 3).
Display screening effect ranking summary	SCREENINGSUMMARY	unticked	Ranks main effects by absolute magnitude with cumulative percent contribution (Pareto-style summary)
Augment Design to CCD	AUGMENTTOCCD	unticked	Sequential experiment: reads the two-level design and responses in the active dataset and adds axial (star) points on the faces of the cube plus center points, giving a face-centered central composite design in the Output Dataset. Existing runs and responses are kept.
Number of additional center points	AUGMENTCENTERPTS	blank	Number of center-point runs added with the axial points (default 3; 0 allowed).

5.4.8 Power... (dialog title: Power and Sample Size)

A planning calculator that needs no data: how many runs are needed to detect an effect of a given size.

Control	Keyword	Default	What it does
Display power and sample size table	POWERTABLE	unticked	Estimates statistical power (or required runs) for detecting a given effect size in an unreplicated two-level factorial design, before generating the design. Does not require a dataset.
Effect sizes (in standard deviation units)	POWEREFFECTSIZES	blank	Effects to detect, in standard-deviation units: effect divided by the run-to-run standard deviation, separated by spaces (e.g. 1 1.5 2). An effect of 3 units with a standard deviation of 2 is 1.5. Default 0.5 1 1.5 2.
Number of runs to evaluate	POWERRUNS	blank	Run counts to evaluate, separated by spaces (e.g. 8 16 32). Default 4 8 16 32. The required run count for the target power is shown in any case.
Significance level (alpha)	POWERALPHA	0.05	Significance level of the planned test (default 0.05).
Target power	POWERTARGET	0.8	Power you want: the chance of detecting the effect if it exists (default 0.8 = 80%). Used for the Required Run Size table.

5.4.9 Analysis (dialog title: Factorial Analysis)

Charts, the response optimizer switch and goals, and the variable selection of the legacy model.

Control	Keyword	Default	What it does
Create Diagnostic Plots	CREATEPLOTS	unticked	Generate main effects, interaction, and residual plots
Main Effects Plot	MAINEFFECTS	unticked	Show average response at each factor level
Cube Plot	CUBEPLOT	unticked	Display fitted means for 2-3 factor designs
Residual Plots	RESIDUALPLOTS	unticked	Display diagnostic plots for model validation
Curvature Check Plot	CURVATUREPLOT	unticked	Plots mean response at factorial (cube) points vs. center points; a sloped line suggests curvature in the response surface
Interaction Plot	INTERACTIONS	unticked	Show how factors interact with each other
Contour Plot	CONTOURPLOT	unticked	Show response surface for continuous factors
Pareto Chart	PARETOPLOT	unticked	Rank effects by magnitude with significance threshold
Optimize Response	OPTIMIZERESPONSE	unticked	Find optimal factor settings for the response
Optimization Goal	OPTIMIZATIONGOAL	Maximize	Goal for a single response, and the default for all responses when Optimization goals is blank. Choices: Maximize (MAXIMIZE); Minimize (MINIMIZE); Target value (TARGET)
Optimization goals (multi-response)	OPTIMIZATIONGOALS	blank	Several responses: one goal per response, in the order of Responses, separated by spaces: MAXIMIZE, MINIMIZE or TARGET (e.g. TARGET MINIMIZE). Blank = the single Optimization Goal for every response.
Target value	OPTIMIZATIONTARGET	blank	One response with goal Target: the value to aim for. Blank = the middle of the observed range.
Target values (multi-response)	OPTIMIZATIONTARGETS	blank	One value per response, in the order of Responses. Target: the value aimed at. Maximize: the value at which the response is fully satisfactory (d = 1). Minimize: the value at or below which d = 1. Blank = observed maximum (Maximize), minimum (Minimize) or middle of the range (Target).
Selection Criterion Trace	SELECTIONTRACE	unticked	AIC trace plot showing the selection criterion at each stepwise step (Stepwise only)
Predictive Fit Profile	FITPROFILE	unticked	Dual R^2 / Adj. R^2 profile vs model size showing the overfitting gap (Best Subsets only)
Perform Variable Selection	VARSELECT	unticked	Performs stepwise or best-subsets variable selection on the fitted model to identify which factors contribute most to the response
Model Parsimony Plot	PARSIMONYPLOT	unticked	Mallows Cp vs number of parameters; points below the Cp=p line have low bias (Best Subsets only)
Factor Inclusion Map	FACTORMAP	unticked	Heatmap grid of which factors appear in each model size with stability percentages (Best Subsets only)
Selection Method	VARSELMETHOD	Stepwise	Selection Method Choices: Stepwise (STEPWISE); Best subsets (BESTSUBSETS)

Control	Keyword	Default	What it does
Stepwise Direction	STEPDIR	Both	Stepwise Direction Choices: Both (BOTH); Forward (FORWARD); Backward (BACKWARD)

5.4.10 Model... (dialog title: Analysis Model (Coded Factors))

The analysis model: which terms, how terms are selected, transformations, the extra tables and predictions.

Control	Keyword	Default	What it does
Analysis model	ANALYSISMODEL	Coded factors (recommended)	Coded factors (recommended): factors coded -1/+1, ANOVA, curvature and lack of fit, all 3.x tables and the Lab. Legacy: the earlier raw-unit model and tables, kept for compatibility. Choices: Coded factors (recommended) (CODED); Legacy (raw units, all interactions) (LEGACY)
Model terms	TERMS	blank	Exact model, overrides Standard model. Separate terms with spaces. A term is a factor name or its letter code from the Factor Coding table (A, B, C ... in the order of the Factors list; I is skipped), joined with * for interactions: Temp Time Temp*Time, or A B A*B (AB also works). Square of a numeric factor: Temp*Temp. CTPT = center-point (curvature) term. Shortcuts: LINEAR (all main effects), 2WAY, 3WAY, SQUARES, QUADRATIC, FULL. List main effects together with their interactions (the model is used as typed; a warning names missing ones). Own model per response: Yield: A B A*B CTPT; Cost: A B C
Text factor low (-1) level	LOWLEVELS	blank	Which level of a text factor is coded -1, as Factor=level (e.g. Catalyst=B; several separated by spaces). By default the level that appears first in the data is coded -1. Only changes the sign of the coded coefficients, not the fit.
Term selection	SELECTION	None (use all terms)	Let p-values choose the terms. Backward: start from the model, remove the least significant term while its p-value is above Alpha to remove. Forward: add the most significant term while its p-value is below Alpha to enter. Stepwise: forward steps with a backward check. Interactions always keep their main effects. Choices: None (use all terms) (NONE); Backward elimination (BACKWARD); Forward selection (FORWARD); Stepwise (STEPWISE)
Alpha to enter	ALPHAENTER	0.15	Forward and stepwise selection: a term is added when its p-value is below this (default 0.15).

Control	Keyword	Default	What it does
Alpha to remove	ALPHAREMOVE	0.10	Backward and stepwise selection: a term is removed when its p-value is above this (default 0.10).
Standard model	TERMSET	Default for the design	A whole model in one choice: Default for the design (two-level: main effects + 2-way interactions, plus the curvature term when there are center points; response surface: full quadratic), Main effects only, + 2-way interactions, up to 3-way, Main effects + squares, Full quadratic, All interactions. Model terms, when filled in, overrides it. To add or remove single terms, use the Lab in the HTML report. Choices: Default for the design (ts_default); Main effects only (ts_linear); Main effects + 2-way interactions (ts_2way); Up to 3-way interactions (ts_3way); Main effects + squares (ts_linsq); Full quadratic (main, squares, 2-way) (ts_quad); All interactions (full factorial model) (ts_full)
Significance level (alpha)	ALPHA	0.05	Significance level for the reference line of the Pareto chart, the highlighting of significant terms and the wording of the Summary of Results and Next Steps (default 0.05; 0.10 is common for screening).
Confidence level (%)	CONFLEVEL	95	Confidence level in percent for the confidence and prediction intervals of predictions and the optimizer (default 95).
Include center-point (curvature) term	CENTERTERM	ticked	Two-level designs with center points: adds a center-point (curvature) term (CTPT) that tests whether the center runs differ from what the corners predict. With it, the optimizer can only evaluate corners and the center.
Box-Cox transformation	BOXCOX	None	Models a transformed response Y^lambda when the spread grows with the level of the response or the residuals are skewed. Automatic picks the best lambda (rounded to 0.5) and shows its confidence interval; lambda 0 = natural log. Needs every response value > 0. Predictions and the optimizer are reported in original units. Choices: None (NONE); Automatic (best rounded lambda) (AUTO); Natural log (lambda = 0) (LOG); Square root (lambda = 0.5) (SQRT); Inverse (lambda = -1) (INVERSE); Other lambda: (CUSTOM)
Lambda	LAMBDA	blank	Power for Other lambda: the response is modeled as Y^lambda (e.g. 0.33 for cube root, -0.5 for inverse square root).

Control	Keyword	Default	What it does
Analyze variability (factors that change the spread)	VARIABILITY	unticked	Finds factors that change the spread of the response: models ln(standard deviation) of settings run two or more times and reports, per factor, how many times larger the standard deviation is at the high setting than at the low one. Needs at least two replicated settings.
Whole plots (split-plot; blank = WholePlot column)	WHOLEPLOT	blank	Split-plot analysis: the variable that numbers the whole plots (groups of runs sharing one setting of the hard-to-change factors). Leave blank when the column is named WholePlot, as in designs created with Hard-to-change factors. Needed only when the whole-plot column has another name than WholePlot.
Factor coding table	FACTORINFO	ticked	Shows which level of each factor is coded -1 and +1
Equation in natural units	EQUATION	ticked	Regression coefficients for the factors in their original units (one column per level of each text factor)
Alias structure (fractional designs)	ALIASTABLE	ticked	Fractional designs: lists which effects are estimated together (aliased), e.g. A + BCD means the A estimate also contains the BCD interaction.
Unusual observations	DIAGTABLE	ticked	Unusual observations (standardized residual above 2 or high leverage) and residual checks: Anderson-Darling normality, Durbin-Watson in run order, largest Cook's distance.
Normal plot of standardized effects	EFFECTSPLOT	unticked	Normal plot of the standardized effects: unimportant effects fall on a straight line; significant effects lie away from it. Useful when there are no error degrees of freedom.
Pairwise comparisons of categorical levels	GROUPING	unticked	Tukey comparisons of the fitted means of each categorical factor: levels that share a letter are not significantly different.
Stationary point / canonical analysis	CANONICAL	unticked	Quadratic models: the stationary point (where the surface is flat), whether it is a maximum, minimum or saddle (from the eigenvalues), and whether it lies inside the studied region.
3-D surface plots	SURFACEPLOT	unticked	Fitted response surface for each pair of numeric factors
Plain-language summary of results	INTERPRET	ticked	A short table above the results that explains in words what they mean: which terms matter, how good the model is, curvature, lack of fit and residual checks

Control	Keyword	Default	What it does
Predict at settings (optional)	PREDICT	blank	Settings to predict at, as Factor=value in natural units (e.g. Temp=190 Time=25 Catalyst=B). Factor names or letter codes; text levels as in the data (case does not matter). Separate several settings with ; (Temp=190 Time=25; Temp=160 Time=12). Factors left out are set to their center (numeric) or first level (text). Settings outside the tested range are flagged as extrapolation.

5.4.11 Optimizer... (dialog title: Response Optimization Settings)

Limits, weights and importance of each response for the desirability optimizer; the profit or cost goal; held factors; confirmation and robustness.

Control	Keyword	Default	What it does
Lower limits	LOWERS	blank	One value per response, in the order of Responses. Maximize: d = 0 at or below this value. Target: lowest acceptable value. Minimize: not used, but type any number (e.g. 0) to keep the list in order. Leave the whole list blank to use the observed minimum of each response.
Upper limits	UPPERS	blank	One value per response, in the order of Responses. Minimize: d = 0 at or above this value. Target: highest acceptable value. Maximize: not used, but type any number (e.g. 0) to keep the list in order. Leave the whole list blank to use the observed maximum of each response.
Weights (0.1 - 10)	WEIGHTS	blank	One value per response (0.1 to 10, default 1): shape of the desirability curve between the limit and the target. 1 = straight line; above 1 = d stays low until close to the target (stricter); below 1 = d rises quickly (more lenient).
Importance (0.1 - 10)	IMPORTANCE	blank	One value per response (0.1 to 10, default 1): how much each response counts in the composite desirability D. Importance 3 counts three times as much as importance 1.
Profit / cost formula	PROFIT	blank	A formula to maximize or minimize instead of the desirability goals (the desirability optimizer still runs if Optimize Response is ticked). Use response names (their predicted values), numeric factor names (natural units), numbers, + - * / ^ ( ) and exp, log, log10, sqrt, abs, min, max, ifelse. Text factors in comparisons with double quotes: ifelse(Catalyst=="B", 0.5, 0). Example: 120*Yield - 3*Cost - 0.02*Temp Write it without quotes in the dialog; the syntax puts single quotes around it.

Control	Keyword	Default	What it does
Formula goal	PROFITGOAL	Maximize (profit)	Choices: Maximize (profit) (maximize); Minimize (cost) (minimize)
Requirements (optional), e.g. Yield>=60 Cost<=25	PROFITLIMITS	blank	Requirements every solution of the profit formula must meet, written Response>=value or Response<=value (predicted responses only), separated by spaces, e.g. Yield>=60 Cost<=25.
Hold factors at	HOLD	blank	Factors fixed during the search, as Factor=value pairs separated by spaces, e.g. Catalyst=B Time=20. Text factors take a level name; numeric factors a value in natural units.
Number of solutions to show	SOLUTIONS	1	How many distinct best settings to list (1 to 50). Useful to see alternatives with nearly the same desirability.
Confirmation runs planned	CONFIRMRUNS	blank	Number of confirmation runs you plan at the optimum. Gives the interval in which their average should fall if the model is right.
Overlaid contour plot	OVERLAY	unticked	Shows the settings where every response is within its limits
Robustness check of the optimum	ROBUST	unticked	Simulates small errors in the factor settings plus run-to-run variation around the optimum
Setting error (% of half-range)	ROBUSTSD	5	Size of the setting errors simulated around the optimum: the standard deviation of each numeric factor as a percentage of its half-range (High - Low)/2. Default 5 (%).

5.4.12 Mixture/Taguchi... (dialog title: Mixture and Taguchi Analysis)

Settings that only apply when the data is a mixture or a Taguchi experiment.

Control	Keyword	Default	What it does
Mixture model	MIXMODEL	Quadratic	Scheffe mixture model: Linear = each pure component; Quadratic = adds 2-component blending terms; Special cubic = adds 3-component terms; Full cubic = adds further 2-component terms. Choose a model the design supports. Choices: Linear (LINEAR); Quadratic (QUADRATIC); Special cubic (SPECIALCUBIC); Full cubic (FULLCUBIC)
Component scale	MIXCOMPS	Automatic	Automatic uses pseudo-components when some component never goes down to 0 (rescales the constrained region to 0-1 for stable coefficients); Proportions uses the proportions as they are. Choices: Automatic (AUTO); Pseudo-components (PSEUDO); Proportions (PROPORTIONS)

Control	Keyword	Default	What it does
Response trace plot	TRACEPLOT	unticked	How the response changes as each component moves away from the reference blend
Signal-to-noise ratio (Taguchi)	SNTYPE	None / automatic	Taguchi analysis with signal-to-noise ratios from the replicate columns in Responses: Larger is better, Smaller is better, or Nominal is best (two versions: mean^2/s^2 or -log s^2). None = ordinary regression analysis. Choices: None / automatic (NONE); Larger is better (LARGER); Smaller is better (SMALLER); Nominal is best (mean^2/s^2) (NOMINAL); Nominal is best (-log s^2) (NOMINAL2)

5.4.13 Output (dialog title: Output)

Extra tables and the features of the HTML report.

Control	Keyword	Default	What it does
Effects & Significance Table	EFFECTSTABLE	unticked	Rank effect magnitudes and flag significance using Lenth's pseudo standard error method
Residual Normality & Autocorrelation Tests	RESIDUALTESTS	unticked	Shapiro-Wilk normality test and Durbin-Watson autocorrelation statistic for model residuals
Multicollinearity (VIF) Table	VIFTABLE	unticked	Variance Inflation Factors for each model term to assess multicollinearity
Optimization Detail Table	OPTDETAILTABLE	unticked	Confidence and prediction intervals for the predicted response at the optimal settings (requires Optimize Response)
Interactive Model and Optimizer Lab in the HTML report	LAB	ticked	Adds an interactive section to the HTML report: add or remove model terms and move factor settings with immediate results, then copy the syntax that reproduces the result. Needs the HTML report (EXPORTHTML=YES, the default).
Next steps: recommendations with ready-to-run syntax	ADVISOR	ticked	After the analysis, lists what the results imply for the next experiment (curvature, lack of fit, inactive factors, region to move to, path of steepest ascent, replication, transformation, confirmation runs), each with syntax you can run.
Uncertainty map (where predictions are least reliable)	UNCERTAINTYMAP	unticked	Charts the half-width of the prediction interval over pairs of factors and reports how much of the region is less certain than any run of the experiment.
Uncertainty limit (half-width; blank = largest at the runs)	UNCERTAINTYLIMIT	blank	In response units: settings whose prediction interval is wider than plus or minus this value are shaded as uncertain. Blank = the widest interval at any run of the experiment.

Control	Keyword	Default	What it does
Display confounding table	CONFOUNDING	ticked	New designs: shows which effects are confounded (aliased) with each other, or the correlations between main effects and 2-way interactions when the design is not a regular fraction.

Chapter 6

Creating designs

To create a design, tick Create a new design in the main dialog, untick Analyze the responses, and fill in **[Create Design...]**: the design type, the factors and the name of the Output Dataset. The procedure writes the design to that new dataset and shows:

- **Settings and Results Summary**: design type, number of runs, factors, output dataset (for optimal designs also the search settings);
- **Variable Specifications**: each factor's Low, High, Center, Levels, rounding and flags;
- **Design Worksheet**: the runs (the first 20 are shown; the dataset has all of them);
- where it applies, the alias / confounding structure and messages in a **Warnings** table;
- an HTML design report with pairwise scatter plots of the design points.

The dataset contains the factor columns and bookkeeping columns: `StdOrder` (standard order), `RunOrder` (the order to run, randomized when Randomize Run Order is ticked), `Reps` (replicate number) and, depending on the design, `Block`, `CenterPt` (0 = centre run), `PtType` (1 = cube or vertex, -1 = axial, 0 = centre, 2 = edge or binary blend) and `WholePlot`. Keep these columns: the analysis uses them. The design also stores its factor list and categorical factors as data file attributes (`DOE_Factors`, `DOE_Categorical`), so the later analysis needs no factor list. Save the dataset as `.sav` to keep them.

Three ways to enter factors.

1. The **factor table** in **[Create Design...]**: one row per factor (up to 15) with Name, Low, High, Levels and Categorical. Optionally Number of factors fills unnamed rows with A, B, C, ... and $-1/+1$ (Example 6.15).
2. The **lists**: type names in Or factor names and the settings in **[Factor Lists...]** (Examples 6.2, 6.5). Needed for centres, rounding and mixture flags, and for more than 15 factors.
3. A **factor sheet** in the active dataset: tick Read the factor table from the active dataset (Example 6.16).

6.1 Two-level full factorial with center points (factor table)

Example C01

Also known as / search terms: two-level factorial design, full factorial design: two-level, center points, randomization, factor table, characterization study

When to use it. You have a handful of factors (2 to about 5) and want every main effect and every interaction estimated cleanly, plus a check for curvature. Typical names: *full factorial*, 2^k *design*, *characterization study*, *DOE for process understanding*.

Data. No data is needed: the design is created from the factor table. Any dataset may be active.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.

2. In the main dialog, tick **Create a new design (set it up with Create Design...)**; untick **Analyze the responses (model, ANOVA, charts)** (no responses exist yet).

3. Click **[Create Design...]**: Factor table, row 1: Name `Temp`, Low 150, High 200. Factor table, row 2: Name `Time`, Low 10, High 30. Factor table, row 3: Name `Press`, Low 1, High 3. Factor table, row 4: Name `Stir`, Low 100, High 300. **Design Type**: select *Factorial Design – k≥2, 2^k runs*. **Output Dataset**: `plan_factorial`. Click **[Continue]**.

4. Click **[Design Options]**: tick **Randomize Run Order**. Click **[Continue]**.

5. Click **[Screening]**: tick **Add Center-Point Runs**. **Number of center points**: type `4`. Click **[Continue]**.

6. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=YES DESIGNTYPE=FACTORIAL
  NAME1=Temp LOW1=150 HIGH1=200 NAME2=Time LOW2=10 HIGH2=30
  NAME3=Press LOW3=1 HIGH3=3 NAME4=Stir LOW4=100 HIGH4=300
  RANDOMIZE=YES SCREENINGCENTERPTS=YES SCREENINGCENTERPTSCOUNT=4
  /SAVE DATASET=plan_factorial.
```

Output.

Settings and Results Summary

	Summary
Design Type	Full / Fractional Factorial
Number of Runs	20
Factors	Temp, Time, Press, Stir
Output Dataset	plan_factorial

Results computed by AlgDesign / base R

Variable Specifications

Variable	Low	High	Center	Levels	Round	Factor	Mixture
Temp	150	200	175	2	0	Yes	No
Time	10	30	20	2	0	Yes	No
Press	1	3	2	2	0	Yes	No
Stir	100	300	200	2	0	Yes	No

Constraint function: None

Design Worksheet — 20 runs

StdOrder	RunOrder	Reps	Temp	Time	Press	Stir	CenterPt
9	1	1	150	10	1	300	1
12	2	1	200	30	1	300	1
4	3	1	200	30	1	100	1
19	4	1	175	20	2	200	0
17	5	1	175	20	2	200	0
2	6	1	200	10	1	100	1
3	7	1	150	30	1	100	1
8	8	1	200	30	3	100	1
18	9	1	175	20	2	200	0
14	10	1	200	10	3	300	1
11	11	1	150	30	1	300	1
13	12	1	150	10	3	300	1
15	13	1	150	30	3	300	1
1	14	1	150	10	1	100	1
20	15	1	175	20	2	200	0
16	16	1	200	30	3	300	1
10	17	1	200	10	1	300	1
5	18	1	150	10	3	100	1
6	19	1	200	10	3	100	1
7	20	1	150	30	3	100	1

Full design saved to output dataset. Add your response column, then re-run with RESPONSEVAR= and ANALYZE=YES.

Alias / Confounding Structure

	Resolution
Design Resolution	Full

This design uses the complete (unfractionated) set of factor-level combinations: all main effects and interactions are independently estimable. No aliasing or confounding exists.

How to read the results. The worksheet has $2^4 = 16$ corner runs plus the 4 requested centre points (`CenterPt`=0), 20 runs in all. `StdOrder` is the textbook (Yates) order and `RunOrder` the randomized order in which to carry out the runs; run them in `RunOrder`, so that slow drifts (warm-up, tool wear, ambient temperature) do not masquerade as factor effects. The alias table confirms a full factorial: no effect is confounded with another. The centre points cost four runs but buy two things: an estimate of pure error (run-to-run variation without assumptions) and a test for curvature, which tells you whether a straight-line model is enough. The new dataset *plan_factorial* is ready for a response column: run the experiment, type the results into it and analyze it as in Example 7.1.

Next. Analyze the completed experiment (Example 7.1).

6.2 Replicated and blocked factorial (lists)

Example C02

Also known as / search terms: replicates, blocking, factorial design: blocked, factor lists

When to use it. The same full factorial, but each run done twice (replicates) and the work split over two days or two batches (blocks), so that the day-to-day difference is removed from the comparison of the factors.

Data. No data is needed.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.

2. In the main dialog, tick **Create a new design (set it up with Create Design...)**; untick **Analyze the responses (model, ANOVA, charts)** (no responses exist yet).
3. Click **[Create Design...]**: **Design Type**: select *Factorial Design – k≥2, 2^k runs*. **Or factor names (settings in Factor Lists...)**: type `Temp Time Press`. **Output Dataset**: `plan_blocked`. Click **[Continue]**.
4. Click **[Factor Lists...]**: **Low Values (required)**: type `150 10 1`. **High Values (required)**: type `200 30 3`. **Factor Flags (optional, YES/NO)**: type `YES YES YES`. Click **[Continue]**.
5. Click **[Design Options]**: **Number of Replicates**: type `2`. **Number of Blocks**: type `2`. tick **Randomize Run Order**. Click **[Continue]**.
6. Click **[OK]** (or **[Paste]** to see the syntax first).

The factor names are typed in Or factor names and the settings in **[Factor Lists...]**; the factor table in **[Create Design...]** would do the same. Use one or the other.

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=YES DESIGNTYPE=FACTORIAL
  VARNAMES=Temp Time Press LOWS=150 10 1 HIGHS=200 30 3 FACTORS=YES YES YES
  REPLICATES=2 BLOCKS=2 RANDOMIZE=YES
  /SAVE DATASET=plan_blocked.
```

Output.

Settings and Results Summary

	Summary
Design Type	Full / Fractional Factorial
Number of Runs	16
Factors	Temp, Time, Press
Output Dataset	plan_blocked

Results computed by AlgDesign / base R

Design Worksheet — 16 runs

StdOrder	RunOrder	Reps	Temp	Time	Press	Block
2	1	1	200	10	1	1
8	2	1	200	30	3	1
7	3	1	150	30	3	1
1	4	1	150	10	1	1
4	5	1	200	30	1	1
6	6	1	200	10	3	1
5	7	1	150	10	3	1
3	8	1	150	30	1	1
16	9	2	200	30	3	2
9	10	2	150	10	1	2
14	11	2	200	10	3	2
15	12	2	150	30	3	2
11	13	2	150	30	1	2
10	14	2	200	10	1	2
13	15	2	150	10	3	2
12	16	2	200	30	1	2

Full design saved to output dataset. Add your response column, then re-run with RESPONSEVAR= and ANALYZE=YES.

Alias / Confounding Structure

	Resolution
Design Resolution	Full

This design uses the complete (unfractionated) set of factor-level combinations: all main effects and interactions are independently estimable. No aliasing or confounding exists.

How to read the results. Two replicates of the 2^3 design give 16 runs. With `BLOCKS=2` and two replicates, each replicate forms one block (column `Block`); within each block the run order is randomized. In the analysis the block term absorbs any shift between the blocks, so the factor effects are not biased by it and the error estimate is not inflated by it. Replicates give an honest estimate of run-to-run variation (pure error) and make every test more powerful. A replicate means a complete new run (set up the factors again and measure), not a second measurement of the same sample.

6.3 Fractional factorial (6 factors in 16 runs)

Example C03

Also known as / search terms: fractional factorial design, screening design: fractional factorial, resolution, aliasing

When to use it. Six or more factors and a budget of 16 runs: a *fractional factorial* (a 2^{k-p} design) runs a carefully chosen fraction of all combinations, so that main effects stay estimable while some interactions are confounded with each other.

Data. No data is needed.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, tick **Create a new design (set it up with Create Design...)**; untick **Analyze the responses (model, ANOVA, charts)** (no responses exist yet).
3. Click **[Create Design...]**: **Design Type**: select *Factorial Design – k≥2, 2^k runs*. **Or factor names (settings in Factor Lists...)**: type `A B C D E F`. **Output Dataset**: `plan_fraction`. Click **[Continue]**.
4. Click **[Factor Lists...]**: **Low Values (required)**: type `-1`. **High Values (required)**: type `1`. **Factor Flags (optional, YES/NO)**: type `YES`. Click **[Continue]**.
5. Click **[Design Options]**: **Number of Trials**: type `16`. Click **[Continue]**.
6. Click **[Options...]**: leave ticked **Display confounding table**. Click **[Continue]**.
7. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=YES DESIGNTYPE=FACTORIAL
  VARNAMES=A B C D E F LOWS=-1 HIGHS=1 FACTORS=YES TRIALS=16
  /OUTPUT CONFOUNDING=YES
  /SAVE DATASET=plan_fraction.
```

Fractional factorials are built by the R package **FrF2**. When it is installed, this syntax gives a 16-run resolution IV fraction (2^{6-2}): main effects are clear of two-factor interactions, which are confounded in chains such as AB = CE. The alias table then lists the chains and the design resolution. When FrF2 is missing, the procedure says so and builds the full factorial instead, as in the output below (64 runs). Install FrF2 from the R console used by SPSS: `install.packages("FrF2")`.

Output.

Warnings

Message
A fractional factorial (TRIALS=16) needs the R package FrF2, which is not installed; the full factorial with 64 runs was generated instead. Install FrF2 (for example with the R command install.packages("FrF2")) for fractions, or use a Plackett-Burman or Definitive screening design.

Settings and Results Summary

	Summary
Design Type	Full / Fractional Factorial
Number of Runs	64
Factors	A, B, C, D, E, F
Output Dataset	plan_fraction

Results computed by AlgDesign / base R

How to read the results. Read the warning first: it states that FrF2 is not installed on the computer that produced this output and that the full 64-run factorial was built instead of the 16-run fraction. Either install FrF2 and run the syntax again, or use a design that needs no extra package: a Plackett–Burman design (Example 6.4) or a definitive screening design (Example 6.5). With FrF2 available, check the *resolution* in the alias table: resolution III confounds main effects with two-factor interactions, IV keeps main effects clear, V keeps two-factor interactions clear of each other.

6.4 Plackett-Burman screening design

Example C04

Also known as / search terms: Plackett–Burman design, screening design: Plackett–Burman, partial aliasing

When to use it. Many factors (here 7), few runs (here 12), and the goal is only to find the few factors that matter: the classic *screening* situation. Plackett–Burman designs have a run count that is a multiple of 4.

Data. No data is needed.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.

2. In the main dialog, tick **Create a new design (set it up with Create Design...)**; untick **Analyze the responses (model, ANOVA, charts)** (no responses exist yet).

3. Click **[Create Design...]**: Factor table, row 1: Name `Temp`, Low 150, High 200. Factor table, row 2: Name `Time`, Low 10, High 30. Factor table, row 3: Name `Press`, Low 1, High 3. Factor table, row 4: Name `Stir`, Low 100, High 300. Factor table, row 5: Name `pH`, Low 5, High 8. Factor table, row 6: Name `Feed`, Low 1, High 2. Factor table, row 7: Name `Purge`, Low 0, High 10. **Design Type**: select *Plackett-Burman Design – k≥2*. **Output Dataset**: `plan_pb`. Click **[Continue]**.

4. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=YES DESIGNTYPE=PLACKETTBURMAN
  NAME1=Temp LOW1=150 HIGH1=200 NAME2=Time LOW2=10 HIGH2=30 NAME3=Press LOW3
    ↪ =1 HIGH3=3
  NAME4=Stir LOW4=100 HIGH4=300 NAME5=pH LOW5=5 HIGH5=8 NAME6=Feed LOW6=1
    ↪ HIGH6=2
  NAME7=Purge LOW7=0 HIGH7=10
  /SAVE DATASET=plan_pb.
```

Output.

Design Worksheet — 12 runs

StdOrder	RunOrder	Reps	Temp	Time	Press	Stir	pH	Feed	Purge
1	1	1	200	30	1	300	8	2	0
2	2	1	200	10	3	300	8	1	0
3	3	1	150	30	3	300	5	1	0
4	4	1	200	30	3	100	5	1	10
5	5	1	200	30	1	100	5	2	0
6	6	1	200	10	1	100	8	1	10
7	7	1	150	10	1	300	5	2	10
8	8	1	150	10	3	100	8	2	0
9	9	1	150	30	1	300	8	1	10
10	10	1	200	10	3	300	5	2	10
11	11	1	150	30	3	100	8	2	10
12	12	1	150	10	1	100	5	1	0

Full design saved to output dataset. Add your response column, then re-run with RESPONSEVAR= and ANALYZE=YES.

Alias / Confounding Structure

	Resolution
Design Resolution	not applicable

Shown as a correlation matrix because the optional FrF2 R package is not installed (install it for named alias chains and a formal design resolution instead). Magnitude indicates degree of confounding between main effects and two-factor interactions.

Partial Aliasing: Main Effects vs Two-Factor Interactions

	Temp:Time	Temp:Press	Temp:Stir	Temp:pH	Temp:Feed	Temp:Purge	Time:Press	Time:Stir	Time:pH	Time:Feed	Time:Purge	Press:Stir	Press:pH	Press:Feed	Press:Purge	Stir:pH	Stir:Feed	Stir:Purge	pH:Feed	pH:Purge	Feed:Purge
Temp	0	0	0	0	0	0	-0.33333	-0.33333	-0.33333	0.33333	-0.33333	0.33333	-0.33333	-0.33333	0.33333	0.33333	0.33333	-0.33333	-0.33333	-0.33333	-0.33333
Time	0	-0.33333	-0.33333	-0.33333	0.33333	-0.33333	0	0	0	0	0	-0.33333	-0.33333	-0.33333	0.33333	0.33333	-0.33333	-0.33333	0.33333	0.33333	-0.33333
Press	-0.33333	0	0.33333	-0.33333	-0.33333	0.33333	0	-0.33333	-0.33333	-0.33333	0.33333	0	0	0	0	-0.33333	-0.33333	-0.33333	0.33333	-0.33333	0.33333
Stir	-0.33333	0.33333	0	0.33333	0.33333	-0.33333	-0.33333	0	0.33333	-0.33333	-0.33333	0	-0.33333	-0.33333	-0.33333	0	0	0	-0.33333	-0.33333	0.33333
pH	-0.33333	-0.33333	0.33333	0	-0.33333	-0.33333	-0.33333	0.33333	0	0.33333	0.33333	-0.33333	0	0.33333	-0.33333	0	-0.33333	-0.33333	0	0	-0.33333
Feed	0.33333	-0.33333	0.33333	-0.33333	0	-0.33333	-0.33333	-0.33333	0.33333	0	-0.33333	-0.33333	0.33333	0	0.33333	-0.33333	0	0.33333	0	-0.33333	0
Purge	-0.33333	0.33333	-0.33333	-0.33333	-0.33333	0	0.33333	-0.33333	0.33333	-0.33333	0	-0.33333	-0.33333	0.33333	0	-0.33333	0.33333	0	-0.33333	0	0

Rows: main effects. Columns: two-factor interactions. Values: correlation coefficients (degree of confounding).

How to read the results. Seven factors in 12 runs. Every main effect is estimated independently of every other main effect (the columns are orthogonal). Two-factor interactions are not confounded completely but *partially*: the partial aliasing table shows correlations of $\pm 1/3$ between each main effect and the interactions that do not contain it. A large interaction would therefore leak one third of its size into several main effects. This is why Plackett–Burman designs are analyzed with main effects only (Example 7.6) and used when interactions are expected to be small. The five spare columns of the 12-run array give error degrees of freedom for the analysis.

6.5 Definitive screening design

Example C05

Also known as / search terms: definitive screening design, screening design: definitive, DSD

When to use it. Screen 6 factors in 13 runs and, unlike two-level designs, also detect curvature in each factor. Definitive screening designs (Jones and Nachtsheim) have $2k + 1$ runs and three levels per factor.

Data. No data is needed.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, tick **Create a new design (set it up with Create Design...)**; untick **Analyze the responses (model, ANOVA, charts)** (no responses exist yet).

3. Click **[Create Design...]**: **Design Type**: select *Definitive Screening (DSD) – k≥4*. **Or factor names (settings in Factor Lists...)**: type `Temp Time Press Stir pH Feed`. **Output Dataset**: `plan_dsd`. Click **[Continue]**.
4. Click **[Factor Lists...]**: **Low Values (required)**: type `150 10 1 100 5 1`. **High Values (required)**: type `200 30 3 300 8 2`. Click **[Continue]**.
5. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=YES DESIGNTYPE=DSD
  VARNAMES=Temp Time Press Stir pH Feed LOWS=150 10 1 100 5 1 HIGHS=200 30 3
    ↪ 300 8 2
  /SAVE DATASET=plan_dsd.
```

Output.

Design Worksheet — 13 runs

StdOrder	RunOrder	Reps	Temp	Time	Press	Stir	pH	Feed
1	1	1	175	20	2	200	6.5	1.5
2	2	1	175	30	1	100	8	2
3	3	1	200	20	3	100	5	2
4	4	1	150	30	2	300	5	2
5	5	1	150	10	3	200	8	2
6	6	1	200	10	1	300	6.5	2
7	7	1	200	30	3	300	8	1.5
8	8	1	175	10	3	300	5	1
9	9	1	150	20	1	300	8	1
10	10	1	200	10	2	100	8	1
11	11	1	200	30	1	200	5	1
12	12	1	150	30	3	100	6.5	1
13	13	1	150	10	1	100	5	1.5

Full design saved to output dataset. Add your response column, then re-run with RESPONSEVAR= and ANALYZE=YES.

How to read the results. Each factor appears at three levels: low, middle (centre) and high. Run 1 is the overall centre point; the other runs come in fold-over pairs, which makes every main effect independent of every two-factor interaction and of every quadratic effect. The analysis fits main effects and squared terms by default; a significant squared term tells you a factor has an optimum inside its range. Definitive screening designs are an efficient first experiment when you suspect curvature but do not yet know which factors are active.

6.6 Central composite design (response surface)

Example C06

Also known as / search terms: central composite design, response surface methodology: central composite, CCD, rotatable design, axial points

When to use it. Optimization: you know the 2–5 important factors and need a model with curvature (a quadratic *response surface*) to locate the best settings. The central composite design is the standard choice.

Data. No data is needed.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.

2. In the main dialog, tick **Create a new design (set it up with Create Design...)**; untick **Analyze the responses (model, ANOVA, charts)** (no responses exist yet).
3. Click **[Create Design...]**: Factor table, row 1: Name `Speed`, Low 80, High 120. Factor table, row 2: Name `Feed`, Low 0.15, High 0.25. Factor table, row 3: Name `Depth`, Low 1.5, High 2.5. **Design Type**: select *Central Composite Design – k≥2*. **Output Dataset**: `plan_ccd`. Click **[Continue]**.
4. Click **[Design Options]**: tick **Randomize Run Order**. Click **[Continue]**.
5. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=YES DESIGNTYPE=CCD
  NAME1=Speed LOW1=80 HIGH1=120 NAME2=Feed LOW2=0.15 HIGH2=0.25 NAME3=Depth
    ↪ LOW3=1.5 HIGH3=2.5
  RANDOMIZE=YES
  /SAVE DATASET=plan_ccd.
```

Output.

Design Worksheet — 17 runs

StdOrder	RunOrder	Reps	Speed	Feed	Depth
9	1	1	133.64	0.2	2
17	2	1	100	0.2	2
13	3	1	100	0.2	2.8409
11	4	1	100	0.28409	2
3	5	1	80	0.25	1.5
5	6	1	80	0.15	2.5
7	7	1	80	0.25	2.5
15	8	1	100	0.2	2
6	9	1	120	0.15	2.5
4	10	1	120	0.25	1.5
2	11	1	120	0.15	1.5
8	12	1	120	0.25	2.5
1	13	1	80	0.15	1.5
14	14	1	100	0.2	1.1591
10	15	1	66.364	0.2	2
16	16	1	100	0.2	2
12	17	1	100	0.11591	2

Full design saved to output dataset. Add your response column, then re-run with RESPONSEVAR= and ANALYZE=YES.

How to read the results. 17 runs: 8 cube (factorial) points, 6 axial (star) points and 3 centre points. The axial points lie at $\alpha = 1.682$ coded units from the centre (here Speed 66.36 and 133.64), which makes the design *rotatable*: the prediction precision depends only on the distance from the centre. Axial settings are therefore outside the Low–High range typed in; if that is not possible on your equipment, use a face-centred design (augmenting a factorial, Example 6.21) or a Box–Behnken design (Example 6.7). The axial values are exact; round them to settings your equipment can achieve and type the actual settings in the dataset before the analysis.

6.7 Box-Behnken design

Example C07

Also known as / search terms: Box–Behnken design, response surface methodology: Box–Behnken

When to use it. A response-surface design with only three levels per factor and no runs at the extreme corners. Choose it when corner combinations are unsafe, impossible or expensive.

Data. No data is needed.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, tick **Create a new design (set it up with Create Design...)**; untick **Analyze the responses (model, ANOVA, charts)** (no responses exist yet).
3. Click **[Create Design...]**: **Design Type**: select *Box-Behnken Design – $k \geq 3$*. **Or factor names (settings in Factor Lists...)**: type `Dose TempC Ratio`. **Output Dataset**: `plan_bb`. Click **[Continue]**.
4. Click **[Factor Lists...]**: **Low Values (required)**: type `5 40 0.5`. **High Values (required)**: type `15 60 1.5`. Click **[Continue]**.
5. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=YES DESIGNTYPE=BOXBEHNKEN
  VARNAMES=Dose TempC Ratio LOWS=5 40 0.5 HIGHS=15 60 1.5
  /SAVE DATASET=plan_bb.
```

Output.

Design Worksheet — 15 runs

StdOrder	RunOrder	Reps	Dose	TempC	Ratio
1	1	1	5	40	1
2	2	1	5	60	1
3	3	1	15	40	1
4	4	1	15	60	1
5	5	1	5	50	0.5
6	6	1	5	50	1.5
7	7	1	15	50	0.5
8	8	1	15	50	1.5
9	9	1	10	40	0.5
10	10	1	10	40	1.5
11	11	1	10	60	0.5
12	12	1	10	60	1.5
13	13	1	10	50	1
14	14	1	10	50	1
15	15	1	10	50	1

Full design saved to output dataset. Add your response column, then re-run with RESPONSEVAR= and ANALYZE=YES.

How to read the results. 15 runs for 3 factors: each run sets two factors at their extremes and the third at its centre (the midpoints of the cube's edges), plus 3 centre points. Every factor stays within the typed Low–High range and no run has all factors at an extreme. The design supports the full quadratic model (Example 7.8). Compared with a CCD it has fewer levels per factor and no corners, but predicts less well at the corners of the region.

6.8 General full factorial with a 3-level categorical factor

Example C08

Also known as / search terms: general full factorial design, full factorial design: general, categorical factor, multi-level factorial

When to use it. Factors with more than two levels, or text factors such as material, supplier or machine: the general full factorial runs every combination of every level.

Data. No data is needed.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, tick **Create a new design (set it up with Create Design...)**; untick **Analyze the responses (model, ANOVA, charts)** (no responses exist yet).
3. Click **[Create Design...]**: Factor table, row 1: Name `Material`, Levels 3, tick Categorical. Factor table, row 2: Name `TestTemp`, Low 20, High 80, Levels 2. **Design Type**: select *Full Factorial (any number of levels)*. **Output Dataset**: `plan_full`. Click **[Continue]**.
4. Click **[Design Options]**: **Number of Replicates**: type `2`. Click **[Continue]**.
5. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=YES DESIGNTYPE=FULLFACTORIAL
  NAME1=Material LEVELS1=3 FACTOR1=YES NAME2=TestTemp LOW2=20 HIGH2=80
    ↪ LEVELS2=2
  REPLICATES=2
  /SAVE DATASET=plan_full.
```

Output.

Settings and Results Summary

	Summary
Design Type	General Full Factorial
Number of Runs	12
Factors	Material, TestTemp
Output Dataset	plan_full

Results computed by AlgDesign / base R

Variable Specifications

Variable	Low	High	Center	Levels	Round	Factor	Mixture
Material	1	3	2	3	0	Yes	No
TestTemp	20	80	50	2	0	No	No

Constraint function: None

Design Worksheet — 12 runs

StdOrder	RunOrder	Reps	Material	TestTemp
1	1	1	1	20
2	2	1	2	20
3	3	1	3	20
4	4	1	1	80
5	5	1	2	80
6	6	1	3	80
7	7	2	1	20
8	8	2	2	20
9	9	2	3	20
10	10	2	1	80
11	11	2	2	80
12	12	2	3	80

Full design saved to output dataset. Add your response column, then re-run with RESPONSEVAR= and ANALYZE=YES.

How to read the results. Material has 3 levels (coded 1, 2, 3; replace the codes with the real names in the dataset if you wish), TestTemp 2 levels: $3 \times 2 = 6$ combinations, twice (Reps 1 and 2), 12

runs. The analysis treats Material as categorical and compares its levels (Example 7.9). Replication is important here: without it, a general factorial with an interaction term has no error degrees of freedom.

6.9 D-optimal design for a quadratic model

Example C09

Also known as / search terms: optimal design, D-optimal design, A-optimal design, custom design, computer-generated design

When to use it. Irregular situations that standard designs do not cover: a run budget that is not a standard size, a mix of factor types, or a model you specify. The *D-optimal* design chooses, from a grid of candidate points, the runs that estimate the model's coefficients most precisely.

Data. No data is needed.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, tick **Create a new design (set it up with Create Design...)**; untick **Analyze the responses (model, ANOVA, charts)** (no responses exist yet).
3. Click **[Create Design...]**: **Design Type**: select *Optimal Design (D/A) – k≥1*. **Or factor names (settings in Factor Lists...)**: type `Temp Press Speed`. **Model Type (Optimal Design Only)**: select *Quadratic*. **Output Dataset**: `plan_optimal`. Click **[Continue]**.
4. Click **[Factor Lists...]**: **Low Values (required)**: type `100 1 10`. **High Values (required)**: type `200 5 50`. **Number of Levels (optional)**: type `3 3 3`. Click **[Continue]**.
5. Click **[Design Options]**: **Number of Trials**: type `15`. Click **[Continue]**.
6. Click **[Options...]**: **Criterion**: select *D (generalized variance)*. **Initial Design**: select *Nullification*. **Repetitions of Design Calculation**: type `10`. Click **[Continue]**.
7. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=YES DESIGNTYPE=OPTIMAL
  VARNAMES=Temp Press Speed LOWS=100 1 10 HIGHS=200 5 50 LEVELS=3 3 3
  MODEL=QUAD TRIALS=15
  /OPTIONS CRITERION=D INITIAL=NULLIFICATION REPEATS=10
  /SAVE DATASET=plan_optimal.
```

Optimal designs are computed by the R package **AlgDesign**, which is installed with the extension. The output of this example was not reproduced in the manual's build environment, so no numbers are shown; the description below matches the output you get in SPSS.

Output.

How to read the results. The procedure builds the candidate grid (3 levels of each factor, $3^3 = 27$ points), runs the D-optimal exchange search 10 times starting from nullification, and keeps the best 15-run design for the full quadratic model in Temp, Press and Speed (10 coefficients, so 5 runs are left for error). The output contains the *Settings and Results Summary* (design type, candidate points, runs, model, criterion), the *Variable Specifications*, the *Design Worksheet*, design-evaluation measures (D, A, G efficiency and the average prediction variance) and, when requested, the confounding

table. Higher D-efficiency is better; compare designs of the same size. Increase `REPEATS` when the result changes noticeably between runs.

6.10 Latin hypercube (computer experiments)

Example C10

Also known as / search terms: Latin hypercube design, space-filling design, computer experiments, simulation experiments

When to use it. Computer simulations and other deterministic or nearly noise-free experiments, where the aim is to explore the whole region evenly rather than to estimate effects at two levels.

Data. No data is needed.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.

2. In the main dialog, tick **Create a new design (set it up with Create Design...)**; untick **Analyze the responses (model, ANOVA, charts)** (no responses exist yet).

3. Click **[Create Design...]**: **Design Type**: select *Latin Hypercube Sampling – k≥1*. **Or factor names (settings in Factor Lists...)**: type `Load Speed Gap`. **Output Dataset**: `plan_lhs`. Click **[Continue]**.

4. Click **[Factor Lists...]**: **Low Values (required)**: type `10 100 0.1`. **High Values (required)**: type `50 500 0.5`. **Decimals for Levels (optional)**: type `1 0 3`. Click **[Continue]**.

5. Click **[Design Options]**: **Number of Trials**: type `20`. Click **[Continue]**.

6. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=YES DESIGNTYPE=LHS
  VARNAMES=Load Speed Gap LOWS=10 100 0.1 HIGHS=50 500 0.5 TRIALS=20 ROUNDTOS
    ↪ =1 0 3
  /SAVE DATASET=plan_lhs.
```

Output.

Design Worksheet — 20 runs

StdOrder	RunOrder	Reps	Load	Speed	Gap
1	1	1	17.7	483	0.36
2	2	1	23.5	333	0.23
3	3	1	11.2	213	0.35
4	4	1	14	250	0.337
5	5	1	35.2	462	0.405
6	6	1	46.3	263	0.111
7	7	1	31.3	232	0.45
8	8	1	43	424	0.196
9	9	1	36.8	441	0.215
10	10	1	15	291	0.388
11	11	1	45.6	366	0.269
12	12	1	18.3	152	0.478
13	13	1	26.7	133	0.299
14	14	1	40.4	405	0.147
15	15	1	21.8	316	0.421
16	16	1	38.6	346	0.128
17	17	1	33.2	118	0.489
18	18	1	28.4	195	0.249
19	19	1	48.7	397	0.16
20	20	1	24.4	175	0.31

Full design saved to output dataset. Add your response column, then re-run with RESPONSEVAR= and ANALYZE=YES.

How to read the results. 20 runs, each factor divided into 20 equal intervals with exactly one run in each interval, so every factor is sampled evenly over its range. `ROUNDTOS=1 0 3` rounds Load to one decimal, Speed to whole numbers and Gap to three decimals. The design changes from run to run (it is random); fit the results with a flexible model (quadratic or higher) or a surrogate model.

6.11 Taguchi orthogonal array

Example C11

Also known as / search terms: Taguchi design, orthogonal array, robust parameter design, L8 array

When to use it. Taguchi orthogonal arrays for robust parameter design: find settings that make the product or process insensitive to noise. The analysis uses signal-to-noise ratios (Example 7.14).

Data. No data is needed.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, tick **Create a new design (set it up with Create Design...)**; untick **Analyze the responses (model, ANOVA, charts)** (no responses exist yet).
3. Click **[Create Design...]**: **Design Type**: select *Taguchi Design – k≥1*. **Or factor names (settings in Factor Lists...)**: type `A B C D`. **Output Dataset**: `plan_taguchi`. Click **[Continue]**.
4. Click **[Factor Lists...]**: **Low Values (required)**: type `1`. **High Values (required)**: type `2`. **Factor Flags (optional, YES/NO)**: type `YES`. Click **[Continue]**.
5. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=YES DESIGNTYPE=TAGUCHI
  VARNAMES=A B C D LOWS=1 HIGHS=2 FACTORS=YES
  /SAVE DATASET=plan_taguchi.
```

Output.

Design Worksheet — 8 runs

StdOrder	RunOrder	Reps	A	B	C	D
1	1	1	2	2	2	2
2	2	1	1	2	1	2
3	3	1	2	1	1	2
4	4	1	1	1	2	2
5	5	1	2	2	2	1
6	6	1	1	2	1	1
7	7	1	2	1	1	1
8	8	1	1	1	2	1

Full design saved to output dataset. Add your response column, then re-run with RESPONSEVAR= and ANALYZE=YES.

Warnings

Message
Taguchi design rounded up to the nearest valid orthogonal array: 8 runs (smallest array that stays orthogonal for 4 factors).

How to read the results. Four two-level factors need at least 5 runs; the smallest orthogonal array for them is the L8 (8 runs), and the warning says so. Each column is balanced (four runs at each level) and every pair of columns is orthogonal. In a robust design each row is run several times under different noise conditions; give each replicate its own response column (Y1, Y2, . . .).

6.12 Simplex-lattice mixture design

Example C12

Also known as / search terms: mixture design: simplex-lattice, simplex-lattice design, formulation, axial check blends

When to use it. Formulations: the factors are proportions of ingredients that add to 100%, so they cannot be varied independently. Simplex designs spread the blends over the triangle (three components) or tetrahedron (four).

Data. No data is needed.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, tick **Create a new design (set it up with Create Design...)**; untick **Analyze the responses (model, ANOVA, charts)** (no responses exist yet).
3. Click **[Create Design...]**: **Design Type**: select *Simplex lattice (mixture)*. **Or factor names (settings in Factor Lists...)**: type `Polymer Plasticizer Filler`. **Output Dataset**: `plan_lattice`. Click **[Continue]**.
4. Click **[Design Options]**: **Lattice degree (simplex-lattice)**: type `2`. tick **Add axial blends and centroid (mixture designs)**. Click **[Continue]**.
5. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=YES DESIGNTYPE=SIMPLEXLATTICE
  VARNAMES=Polymer Plasticizer Filler LATTICEDEGREE=2 AUGMENTMIX=YES
  /SAVE DATASET=plan_lattice.
```

Output.

Settings and Results Summary

	Summary
Design Type	Simplex-Lattice Mixture Design
Number of Runs	10
Factors	Polymer, Plasticizer, Filler
Output Dataset	plan_lattice

Results computed by AlgDesign / base R

Variable Specifications

Variable	Low	High	Center	Levels	Round	Factor	Mixture
Polymer				2	0	No	Yes
Plasticizer				2	0	No	Yes
Filler				2	0	No	Yes

Constraint function: None

Design Worksheet — 10 runs

StdOrder	RunOrder	Reps	Polymer	Plasticizer	Filler	PtType
1	1	1	1	0	0	1
2	2	1	0	1	0	1
3	3	1	0	0	1	1
4	4	1	0.5	0.5	0	2
5	5	1	0.5	0	0.5	2
6	6	1	0	0.5	0.5	2
7	7	1	0.66667	0.16667	0.16667	-1
8	8	1	0.16667	0.66667	0.16667	-1
9	9	1	0.16667	0.16667	0.66667	-1
10	10	1	0.33333	0.33333	0.33333	0

Full design saved to output dataset. Add your response column, then re-run with RESPONSEVAR= and ANALYZE=YES.

How to read the results. Lattice degree 2 gives the pure components (`PtType`=1) and the 50:50 binary blends (`PtType`=2); `AUGMENTMIX=YES` adds the axial check blends (2/3, 1/6, 1/6; `PtType`=−1) and the overall centroid (`PtType`=0): 10 blends. The Variable Specifications show no Low and High because a simplex design always covers the full range 0 to 1. The check blends lie inside the triangle and test the fitted model where the lattice has no runs.

6.13 Simplex-centroid mixture design

Example C13

Also known as / search terms: mixture design: simplex-centroid, simplex-centroid design

When to use it. The simplex-centroid design: every pure component, every equal two-way blend, every equal three-way blend and so on, up to the overall centroid. It supports the special cubic mixture model.

Data. No data is needed.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, tick **Create a new design (set it up with Create Design...)**; untick **Analyze the responses (model, ANOVA, charts)** (no responses exist yet).

3. Click **[Create Design...]**: **Design Type**: select *Simplex centroid (mixture)*. **Or factor names (settings in Factor Lists...)**: type `Polymer Plasticizer Filler`. **Output Dataset**: `plan_centroid`. Click **[Continue]**.
4. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=YES DESIGNTYPE=SIMPLEXCENTROID
  VARNAMES=Polymer Plasticizer Filler
  /SAVE DATASET=plan_centroid.
```

Output.

Design Worksheet — 7 runs

StdOrder	RunOrder	Reps	Polymer	Plasticizer	Filler	PtType
1	1	1	1	0	0	1
2	2	1	0	1	0	1
3	3	1	0	0	1	1
4	4	1	0.5	0.5	0	2
5	5	1	0.5	0	0.5	2
6	6	1	0	0.5	0.5	2
7	7	1	0.33333	0.33333	0.33333	0

Full design saved to output dataset. Add your response column, then re-run with RESPONSEVAR= and ANALYZE=YES.

How to read the results. For three components: 3 pure components, 3 binary 50:50 blends and the 1/3:1/3:1/3 centroid, 7 blends. This is the design behind the *blend* data analyzed in Example 7.13 (where the pure components were run twice).

6.14 Extreme-vertices design (constrained mixture)

Example C14

Also known as / search terms: mixture design: extreme vertices, extreme-vertices design, constrained mixture, formulation

When to use it. Real formulations rarely span 0–100% of every ingredient: each has a lower and an upper limit. The feasible blends then form an irregular region whose corners (vertices) are found by the McLean–Anderson algorithm.

Data. No data is needed. The limits are the classic flare-composition example: magnesium 40–60%, sodium nitrate 10–50%, strontium nitrate 10–50%, binder 3–8%.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, tick **Create a new design (set it up with Create Design...)**; untick **Analyze the responses (model, ANOVA, charts)** (no responses exist yet).
3. Click **[Create Design...]**: **Design Type**: select *Extreme vertices (constrained mixture)*. **Or factor names (settings in Factor Lists...)**: type `Mg NaNO3 SrNO3 Binder`. **Output Dataset**: `plan_ev`. Click **[Continue]**.
4. Click **[Factor Lists...]**: **Low Values (required)**: type `0.40 0.10 0.10 0.03`. **High Values (required)**: type `0.60 0.50 0.50 0.08`. Click **[Continue]**.
5. Click **[Design Options]**: **Extreme vertices: centroid degree**: type `1`. Click **[Continue]**.
6. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=YES DESIGNTYPE=EXTREMEVERTICES
  VARNAMES=Mg NaNO3 SrNO3 Binder LOWS=0.40 0.10 0.10 0.03 HIGHS=0.60 0.50
    ↪ 0.50 0.08
  EVDEGREE=1
  /SAVE DATASET=plan_ev.
```

Output.

Warnings

Message
Bounds implied by the others were used for NaNO3 [0.1, 0.47], SrNO3 [0.1, 0.47] (the stated bounds cannot all be reached when the components sum to 1).
Extreme-vertices design: 8 vertices, 12 face centroid(s), 1 overall centroid (21 blends).

Design Worksheet — 21 runs (showing first 20)

StdOrder	RunOrder	Reps	Mg	NaNO3	SrNO3	Binder	PtType
1	1	1	0.4	0.47	0.1	0.03	1
2	2	1	0.4	0.1	0.47	0.03	1
3	3	1	0.6	0.27	0.1	0.03	1
4	4	1	0.4	0.42	0.1	0.08	1
5	5	1	0.6	0.22	0.1	0.08	1
6	6	1	0.6	0.1	0.27	0.03	1
7	7	1	0.4	0.1	0.42	0.08	1
8	8	1	0.6	0.1	0.22	0.08	1
9	9	1	0.4	0.1	0.445	0.055	2
10	10	1	0.6	0.1	0.245	0.055	2
11	11	1	0.4	0.445	0.1	0.055	2
12	12	1	0.6	0.245	0.1	0.055	2
13	13	1	0.4	0.285	0.285	0.03	2
14	14	1	0.6	0.185	0.185	0.03	2
15	15	1	0.4	0.26	0.26	0.08	2
16	16	1	0.6	0.16	0.16	0.08	2
17	17	1	0.5	0.1	0.37	0.03	2
18	18	1	0.5	0.1	0.32	0.08	2
19	19	1	0.5	0.37	0.1	0.03	2
20	20	1	0.5	0.32	0.1	0.08	2

Full design saved to output dataset. Add your response column, then re-run with RESPONSEVAR= and ANALYZE=YES.

How to read the results. The first warning shows that the stated upper limit of 0.50 for NaNO3 and SrNO3 can never be reached (with Mg at least 0.40 and the others at their minimum, at most $1 - 0.40 - 0.10 - 0.03 = 0.47$ remains), so the implied limit 0.47 was used. The region has 8 vertices (`PtType`=1) and 12 edges; with `EVDEGREE=1` the edge midpoints (`PtType`=2) and the overall centroid (`PtType`=0) are added: 21 blends. Analyze the results with the mixture models (Example 7.13); the procedure uses L-pseudo-components so the coefficients are well conditioned.

6.15 Number of factors with default names and ranges

Example C15

Also known as / search terms: number of factors, default factor names

When to use it. A quick way to set up many factors: give the number of factors and type only the rows that differ from the defaults.

Data. No data is needed.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, tick **Create a new design (set it up with Create Design...)**; untick **Analyze the responses (model, ANOVA, charts)** (no responses exist yet).
3. Click **[Create Design...]**: Factor table, row 2: Name `Temp`, Low 150, High 200. **Design Type**: select *Factorial Design – k≥2, 2^k runs*. **Number of factors**: type `5`. **Output Dataset**: `plan_nfactors`. Click **[Continue]**.
4. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=YES DESIGNTYPE=FACTORIAL NFACTORS=5
  NAME2=Temp LOW2=150 HIGH2=200
  /SAVE DATASET=plan_nfactors.
```

Output.

Variable Specifications

Variable	Low	High	Center	Levels	Round	Factor	Mixture
A	-1	1	0	2	0	Yes	No
Temp	150	200	175	2	0	Yes	No
C	-1	1	0	2	0	Yes	No
D	-1	1	0	2	0	Yes	No
E	-1	1	0	2	0	Yes	No

Constraint function: None

Design Worksheet — 32 runs (showing first 20)

StdOrder	RunOrder	Reps	A	Temp	C	D	E
1	1	1	-1	150	-1	-1	-1
2	2	1	1	150	-1	-1	-1
3	3	1	-1	200	-1	-1	-1
4	4	1	1	200	-1	-1	-1
5	5	1	-1	150	1	-1	-1
6	6	1	1	150	1	-1	-1
7	7	1	-1	200	1	-1	-1
8	8	1	1	200	1	-1	-1
… 12 more rows							

Full design saved to output dataset. Add your response column, then re-run with RESPONSEVAR= and ANALYZE=YES.

How to read the results. Five factors were requested; row 2 was named Temp with its real range, the others became A, C, D, E (named after their position) with the coded range −1 to +1. Use this for a quick plan, then replace the coded settings with real ones.

6.16 Factor table read from the active dataset (any number of factors)

Example C16

Also known as / search terms: factor table from data, many factors, screening design: from a factor sheet

When to use it. More than 15 factors, or a factor list kept in a spreadsheet: type or paste the list into the Data Editor (one row per factor) and let the procedure read it.

Data. The active dataset *factorsheet* has one row per factor with columns Name, Low and High (optional: Levels, Categorical).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, tick **Create a new design (set it up with Create Design...)**; untick **Analyze the responses (model, ANOVA, charts)** (no responses exist yet).
3. Click **[Create Design...]**: **Design Type**: select *Plackett-Burman Design – $k \geq 2$*. tick **Read the factor table from the active dataset**. **Output Dataset**: `plan_sheet`. Click **[Continue]**.
4. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=YES DESIGNTYPE=PLACKETTBURMAN
    ↪ FACTORSFROMDATA=YES
  /SAVE DATASET=plan_sheet.
```

Output.

Variable Specifications

Variable	Low	High	Center	Levels	Round	Factor	Mixture
Temp	150	200	175	2	0	No	No
Time	10	30	20	2	0	No	No
Press	1	3	2	2	0	No	No
Speed	80	120	100	2	0	No	No
Feed	0.1	0.3	0.2	2	1	No	No
Depth	1	3	2	2	0	No	No
Coolant	0	1	0.5	2	0	No	No
Tool	0	1	0.5	2	0	No	No
Angle	5	15	10	2	0	No	No
Humid	30	60	45	2	0	No	No
Load	1	5	3	2	0	No	No
Ratio	0.2	0.8	0.5	2	0	No	No
Mix	1	4	2.5	2	0	No	No
Cure	10	20	15	2	0	No	No

Constraint function: None

Design Worksheet — 20 runs

StdOrder	RunOrder	Reps	Temp	Time	Press	Speed	Feed	Depth	Coolant	Tool	Angle	Humid	Load	Ratio	Mix	Cure
1	1	1	200	30	1	80	0.3	3	1	1	5	60	1	0.8	1	10
2	2	1	200	10	1	120	0.3	3	1	0	15	30	5	0.2	1	10
3	3	1	150	10	3	120	0.3	3	0	1	5	60	1	0.2	1	10
4	4	1	150	30	3	120	0.3	1	1	0	15	30	1	0.2	1	20
5	5	1	200	30	3	120	0.1	3	0	1	5	30	1	0.2	4	20
6	6	1	200	30	3	80	0.3	1	1	0	5	30	1	0.8	4	10
7	7	1	200	30	1	120	0.1	3	0	0	5	30	5	0.8	1	20
8	8	1	200	10	3	80	0.3	1	0	0	5	60	5	0.2	4	20

… 12 more rows

Full design saved to output dataset. Add your response column, then re-run with RESPONSEVAR= and ANALYZE=YES.

How to read the results. All 14 factors were read from the sheet. With 14 factors the Plackett–Burman design needs 20 runs (the next multiple of 4 above 14). The variable specifications repeat the sheet, so you can check that every row was read as intended.

6.17 Central composite design with a categorical factor

Example C17

Also known as / search terms: response surface methodology: categorical factor, central composite design: with categorical factor, categorical factor

When to use it. A response-surface study in which one factor is categorical, for example two numeric process settings and three suppliers of raw material.

Data. No data is needed.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, tick **Create a new design (set it up with Create Design...)**; untick **Analyze the responses (model, ANOVA, charts)** (no responses exist yet).
3. Click **[Create Design...]**: Factor table, row 1: Name `Temp`, Low 150, High 200. Factor table, row 2: Name `Time`, Low 10, High 30. Factor table, row 3: Name `Supplier`, Levels 3, tick Categorical. **Design Type**: select *Central Composite Design – k≥2*. **Output Dataset**: `plan_ccdcat`. Click **[Continue]**.
4. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=YES DESIGNTYPE=CCD
  NAME1=Temp LOW1=150 HIGH1=200 NAME2=Time LOW2=10 HIGH2=30
  NAME3=Supplier LEVELS3=3 FACTOR3=YES
  /SAVE DATASET=plan_ccdcat.
```

Output.

Warnings

Message
Supplier: the central composite design in Temp, Time was run once for each of the 3 combinations of the categorical levels (33 runs). Analyze it with Supplier as categorical factors (recorded with the design).

Design Worksheet — 33 runs (showing first 20)

StdOrder	RunOrder	Reps	Temp	Time	Supplier
1	1	1	150	10	1
2	2	1	200	10	1
3	3	1	150	30	1
4	4	1	200	30	1
5	5	1	210.36	20	1
6	6	1	139.64	20	1
7	7	1	175	34.142	1
8	8	1	175	5.8579	1
9	9	1	175	20	1
10	10	1	175	20	1
11	11	1	175	20	1
12	12	1	150	10	2
… 8 more rows					

Full design saved to output dataset. Add your response column, then re-run with RESPONSEVAR= and ANALYZE=YES.

How to read the results. The central composite design in Temp and Time (11 runs: 4 cube, 4 axial at $\alpha = 1.414$, 3 centre) is run once for each Supplier: 33 runs. The design records that Supplier is categorical, so the analysis treats it so automatically. The model then contains the quadratic surface in Temp and Time, the Supplier effect and the Supplier-by-factor interactions the design supports, allowing the surface to shift between suppliers.

6.18 Split-plot design (hard-to-change factor)

Example C18

Also known as / search terms: split-plot design, hard-to-change factor, whole plot, restricted randomization

When to use it. One factor is expensive or slow to change (an oven temperature, a reactor setup, a mould): you cannot reset it for every run. A *split-plot* design changes it only between groups of runs (whole plots) and randomizes the easy-to-change factors inside each group.

Data. No data is needed.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, tick **Create a new design (set it up with Create Design...)**; untick **Analyze the responses (model, ANOVA, charts)** (no responses exist yet).
3. Click **[Create Design...]**: **Design Type**: select *Full Factorial (any number of levels)*. **Or factor names (settings in Factor Lists...)**: type `Oven Press Mix`. **Output Dataset**: `plan_split`. Click **[Continue]**.
4. Click **[Factor Lists...]**: **Low Values (required)**: type `150 1 1`. **High Values (required)**: type `200 3 2`. **Number of Levels (optional)**: type `2 2 2`. Click **[Continue]**.
5. Click **[Design Options]**: **Number of Replicates**: type `3`. tick **Randomize Run Order**. **Hard-to-change factors (split-plot)**: type `Oven`. Click **[Continue]**.
6. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=YES DESIGNTYPE=FULLFACTORIAL
  VARNAMES=Oven Press Mix LOWS=150 1 1 HIGHS=200 3 2 LEVELS=2 2 2
  REPLICATES=3 RANDOMIZE=YES HTC=Oven
  /SAVE DATASET=plan_split.
```

Output.

Warnings

Message
Split-plot design: 6 whole plots (one setup of Oven each) with 4 runs per whole plot. Run the whole plots in the order of WholePlot; change Oven only between whole plots.

Design Worksheet — 24 runs (showing first 20)

StdOrder	RunOrder	Reps	Oven	Press	Mix	WholePlot
12	1	2	200	3	1	1
14	2	2	200	1	2	1
16	3	2	200	3	2	1
10	4	2	200	1	1	1
24	5	3	200	3	2	2
18	6	3	200	1	1	2
20	7	3	200	3	1	2
22	8	3	200	1	2	2
11	9	2	150	3	1	3
9	10	2	150	1	1	3
15	11	2	150	3	2	3
13	12	2	150	1	2	3
… 8 more rows						

Full design saved to output dataset. Add your response column, then re-run with RESPONSEVAR= and ANALYZE=YES.

How to read the results. Oven is the hard-to-change factor. Each replicate needs Oven at 150 and at 200 once: 3 replicates × 2 settings = 6 whole plots of 4 runs each. `WholePlot` numbers the oven setups in the order to run them; inside each, Press and Mix are run in random order. Change Oven

only when `WholePlot` changes. Keep the `WholePlot` column: the analysis uses it to estimate the two error terms (Example 7.15).

6.19 Power and sample size

Example C19

Also known as / search terms: power analysis, sample size, number of runs

When to use it. Before running an experiment: how many runs are needed to detect an effect of a given size? Effect sizes are in standard-deviation units: $d = 1$ means an effect as large as the run-to-run standard deviation.

Data. No data is needed.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, untick **Analyze the responses (model, ANOVA, charts)**.
3. Click **[Power...]**: tick **Display power and sample size table**. **Effect sizes (in standard deviation units)**: type `1 1.5 2`. **Number of runs to evaluate**: type `8 16 32`. **Significance level (alpha)**: type `0.05`. **Target power**: type `0.8`. Click **[Continue]**.
4. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO
  /POWER POWERTABLE=YES POWEREFFECTSIZES=1 1.5 2 POWERRUNS=8 16 32 POWERALPHA
    ↪ =0.05 POWERTARGET=0.8.
```

Output.

Power Table (Effect Size in Standard-Deviation Units)

Runs	d=1	d=1.5	d=2
8	0.2232	0.43	0.6569
16	0.4612	0.7965	0.9602
32	0.7814	0.984	0.9998

Power to detect an effect of size d = effect/sigma at alpha=0.05, assuming an unreplicated 2-level design with df=Runs-2.

Required Run Size for 80% Power

Effect Size (d)	Required Runs
1	34
1.5	18
2	12

Minimum number of runs to achieve 80% power at alpha=0.05 for each effect size. Blank/NA means not achievable within 512 runs.

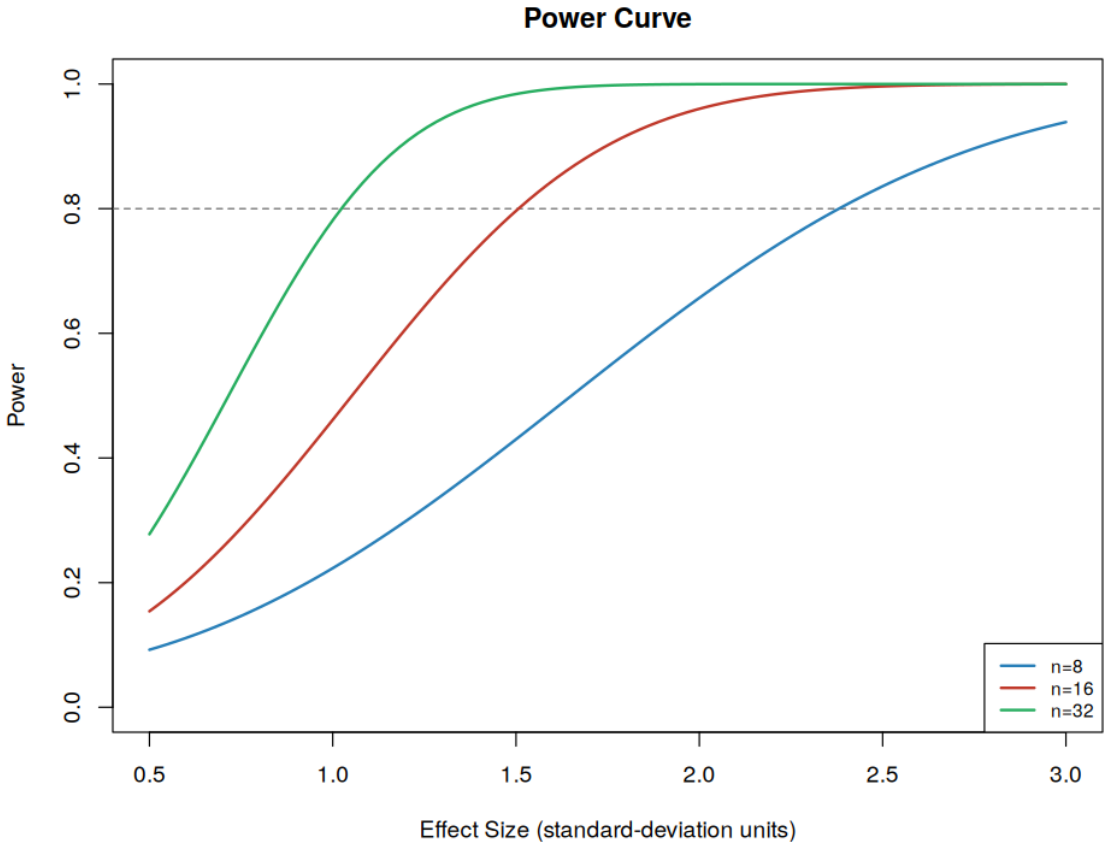


Figure 6.1: Power curves for 8, 16 and 32 runs.

How to read the results. With 16 runs the chance of detecting an effect of 1.5 standard deviations is 0.80, and of 2 standard deviations 0.96; with only 8 runs it is 0.43 and 0.66. To reach 80% power for an effect of 1 standard deviation you need 34 runs. Use this before choosing replicates: if your process standard deviation is 2 units and you must detect a change of 3 units ($d = 1.5$), 18 runs give 80% power.

6.20 Runs needed for a target power

Example C19B

Also known as / search terms: power analysis: target power

When to use it. The required number of runs for a stricter target (90% power) and a range of effect sizes; runs to evaluate are left at their defaults.

Data. No data is needed.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, untick **Analyze the responses (model, ANOVA, charts)**.
3. Click **[Power...]**: tick **Display power and sample size table**. **Effect sizes (in standard deviation units)**: type `0.5 1 1.5 2`. **Target power**: type `0.9`. Click **[Continue]**.
4. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO
  /POWER POWERTABLE=YES POWEREFFECTSIZES=0.5 1 1.5 2 POWERTARGET=0.9.
```

Output.

Required Run Size for 90% Power

Effect Size (d)	Required Runs
0.5	172
1	46
1.5	22
2	14

Minimum number of runs to achieve 90% power at alpha=0.05 for each effect size. Blank/NA means not achievable within 512 runs.

Power Table (Effect Size in Standard-Deviation Units)

Runs	d=0.5	d=1	d=1.5	d=2
4	0.0615	0.0952	0.1487	0.2183
8	0.0923	0.2232	0.43	0.6569
16	0.154	0.4612	0.7965	0.9602
32	0.2777	0.7814	0.984	0.9998

Power to detect an effect of size d = effect/sigma at alpha=0.05, assuming an unreplicated 2-level design with df=Runs-2.

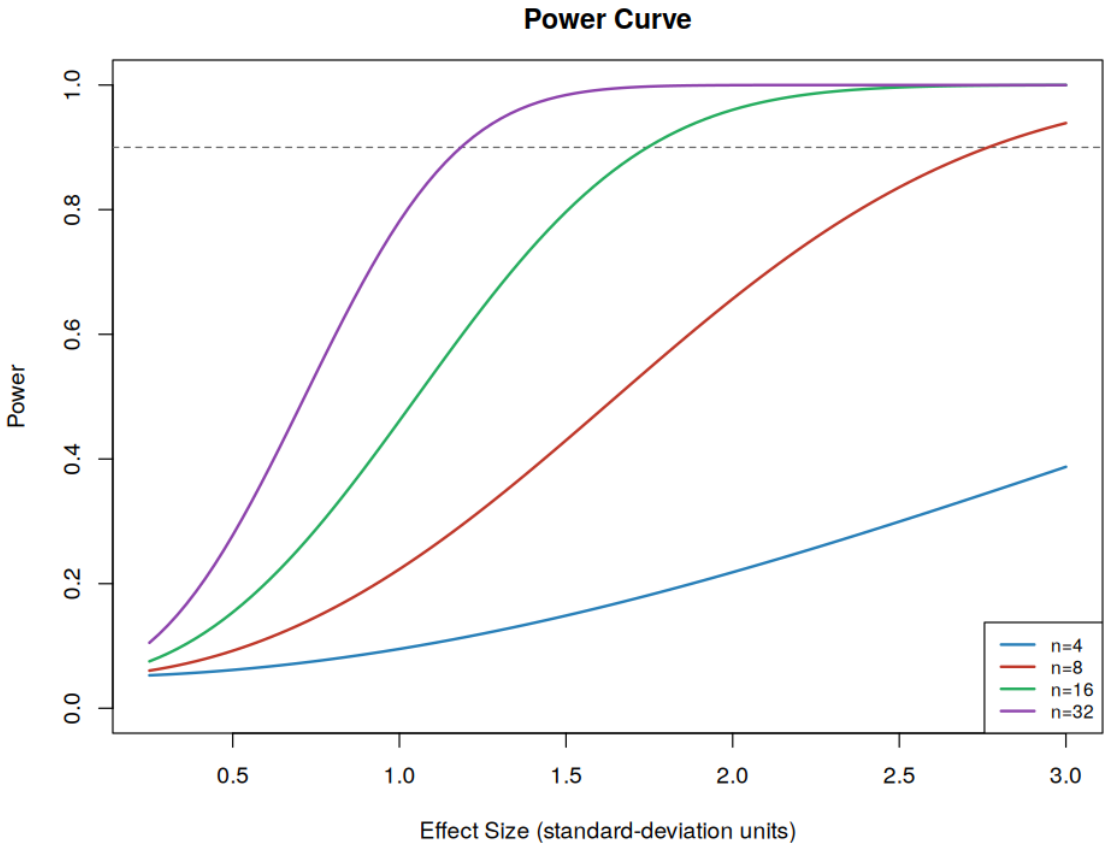


Figure 6.2: Power curves for 4 to 32 runs; dashed line: target power 0.9.

How to read the results. Detecting half a standard deviation with 90% power needs 172 runs; one standard deviation 46 runs; two standard deviations only 14 runs. Small effects are expensive to detect: halving the effect size roughly quadruples the runs needed.

6.21 Augment a factorial to a central composite design

Example C20

Also known as / search terms: sequential experimentation, augment to central composite, face-centred central composite, fold-over

When to use it. A factorial with centre points showed curvature: add axial points to turn it into a central composite design, keeping all the runs already done.

Data. Dataset *firstorder*: a 2^3 factorial with 3 centre points and response Y.

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.

2. In the main dialog, drag A, B, C into **Factors (design columns)**; drag Y into **Responses (measured results)**; untick **Analyze the responses (model, ANOVA, charts)**.

3. Click **[Screening]**: tick **Augment Design to CCD**. **Number of additional center points**: type `3`. Click **[Continue]**.

4. Click **[Create Design...]**: **Output Dataset**: `plan_augmented`. Click **[Continue]**.

5. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO AUGMENTTOCCD=YES AUGMENTCENTERPTS=3
  FACTORVARS=A B C RESPONSEVAR=Y
  /SAVE DATASET=plan_augmented.
```

Output.

Augment to Central Composite Design — Summary

	Value
Original Runs Kept	11
Axial (Star) Runs Added	6
Center Runs Added	3
Total Runs	20
Alpha (Axial Distance)	1.0000
Output Dataset	plan_augmented

The original runs and any response values already entered were kept as-is. Fill in the response column for the new (axial/center) rows in the output dataset, then analyze the output dataset (ANALYZE=YES): the full quadratic model is fitted automatically.

Output dataset plan_augmented (20 rows)

A	B	C	StdOrder	Y	PtType	Block
10	1	100	1	48.69	Cube	1
20	1	100	2	54.91	Cube	1
10	3	100	3	44.61	Cube	1
20	3	100	4	50.56	Cube	1
10	1	200	5	48.62	Cube	1
20	1	200	6	55.27	Cube	1
10	3	200	7	44.37	Cube	1
20	3	200	8	50.65	Cube	1
15	2	150	9	54.19	Center	1
15	2	150	10	53.93	Center	1
15	2	150	11	54.62	Center	1
20	2	150	12		Axial	2
10	2	150	13		Axial	2
15	3	150	14		Axial	2
15	1	150	15		Axial	2
15	2	200	16		Axial	2
15	2	100	17		Axial	2
15	2	150	18		Center	2
15	2	150	19		Center	2
15	2	150	20		Center	2

How to read the results. The 11 existing runs keep their response values. Six axial points (one at each face of the cube, distance 1, so the new settings stay within the original Low–High range) and 3 new centre points are added: 20 runs, with `PtType` marking cube, axial and centre runs (the original centre runs are recognized and labelled Center) and a `Block` column separating the original runs (block 1) from the new ones (block 2), so that any shift between the two stages is not mistaken for a factor effect. Run the 9 new rows, type their Y values, and analyze the dataset: the full quadratic model is fitted automatically.

6.22 Constraints for optimal designs

Sometimes not every combination of settings is allowed: a high temperature together with a long time may burn the product. For *optimal* designs, **[Constraints...]** restricts the candidate points with an R function named `dfilter`, saved in a text file. It receives the settings of one candidate point (in the order of the factor names) and returns `TRUE` to keep the point:

```
dfilter = function(x) {
    # x[1] = Temp, x[2] = Time: exclude hot-and-long combinations
    return(!(x[1] > 180 && x[2] > 25))
}
```

In the dialog tick Filter design points and browse to the file. In syntax add the file name, for example

```
CONSTRAINTFUNC="C:/designs/filter.R"
```

The optimal search then chooses its runs among the allowed points only. Constraints are not applied to the standard designs, whose structure is fixed.

Chapter 7

Analyzing experiments

To analyze an experiment, make its dataset active, drag the factor columns into Factors and the responses into Responses, keep Analyze the responses ticked, and click **[OK]**. Leave Factors empty for a design created by this procedure: its factor list is stored in the dataset.

What the procedure decides for you. The analysis uses the *coded-factor model*: numeric factors are coded -1 (Low) to $+1$ (High), a two-level text factor $-1/+1$, a text factor with more levels by effect coding. The kind of design is recognized from the data and sets the default model:

Data look like	Default model
two-level factorial (with or without centre points)	main effects + two-factor interactions (+ centre-point term)
Plackett–Burman (or `DESIGNTYPE=PLACKETTBURMAN`)	main effects
definitive screening	main effects + squares
central composite, Box–Behnken, 3+ numeric levels	full quadratic
general factorial (categorical with 3+ levels)	main effects + two-factor interactions
mixture components (sum constant)	Scheffé quadratic (`MIXMODEL`)
Taguchi (`SNTYPE` or `DESIGNTYPE=TAGUCHI`)	signal-to-noise response tables
`WholePlot` column present	split-plot REML analysis

Terms that cannot be estimated (aliased with terms already in the model) are dropped and listed in the alias structure. Change the model with **[Model...]**: Standard model for a whole model, Model terms for exact terms, Term selection to let p-values decide.

The tables, in order. For each response: Summary of Results (plain language), Factor Coding, Coded Coefficients, Model Summary, Analysis of Variance, Regression Equation in Natural Units, Alias Structure (fractional designs), Unusual Observations, Residual Checks; then requested extras (predictions, Tukey comparisons, canonical analysis, variability, uncertainty), the optimizer, and the Next Steps table. Chapter 12 describes each table.

7.1 Two-level factorial with center points: default analysis and charts

Example A01

Also known as / search terms: factorial analysis, two-level factorial design: analysis, curvature test, Pareto chart, normal plot of effects, main effects plot, interaction plot, cube plot, residual plots, ANOVA, coded coefficients

When to use it. The standard analysis of a two-level factorial: which factors and interactions matter, how large their effects are, is there curvature, and do the model's assumptions hold.

Data. Dataset *reactor* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.

2. In the main dialog, drag Temp, Time, Press, Catalyst into **Factors (design columns)**; drag Yield into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).

3. Click **[Analysis]**: tick **Create Diagnostic Plots**. tick **Pareto Chart**. tick **Main Effects Plot**. tick **Interaction Plot**. tick **Cube Plot**. tick **Residual Plots**. Click **[Continue]**.

4. Click **[Model...]**: tick **Normal plot of standardized effects**. Click **[Continue]**.

5. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=Temp Time Press Catalyst
  RESPONSEVAR=Yield ANALYZE=YES
  /OUTPUT PARETOPLOT=YES MAINEFFECTS=YES INTERACTIONS=YES CUBEPLOT=YES
    ↪ RESIDUALPLOTS=YES
  /FITMODEL EFFECTSPLOT=YES.
```

Output.

Summary of Results: Yield

	Finding
Overall	The model is statistically significant (Sig. 0.000): the factors explain real changes in Yield.
Model fit	The model explains 99.2% of the variation in Yield (adjusted 98.1%). Predicted R-squared 96.8%: it should predict new runs well.
Significant terms	5 of 10 terms are significant at alpha = 0.05: Temp, Time, Temp*Time, Press, Catalyst (strongest first). Not significant: Temp*Press, Temp*Catalyst, Time*Press, Time*Catalyst, Press*Catalyst.
Largest effects	Averaged over the other factors: Temp from 150 to 200 changes Yield by 9.546 (the size depends on Time: see the interaction plot); Time from 10 to 30 changes Yield by 6.338 (the size depends on Temp: see the interaction plot); Press from 1 to 3 changes Yield by -4.644.
Curvature	The center points differ from the corners (Sig. 0.000): the response curves inside the region. A response surface design (e.g. augment to a central composite) is needed to model it.
Lack of fit	No evidence of lack of fit (Sig. 0.982).
Residuals	Residuals are not normal (Anderson-Darling Sig. 0.023); 1 run(s) with a large standardized residual (> 2).

Plain-language reading of the tables below. Decisions use the alpha level shown; check the charts before acting.

Factor Coding: Yield

Code	Factor	Type	Coded -1	Coded +1
A	Temp	Numeric	150	200
B	Time	Numeric	10	30
C	Press	Numeric	1	3
D	Catalyst	Text/categorical	A	B

Design treated as two-level factorial. Runs used: 20 (center points: 4).

Coded Coefficients: Yield

	Effect	Coefficient	SE Coef.	t	Sig.	VIF
Constant		60.643	0.22745	266.63	0.000	
Temp	9.5465	4.7732	0.22745	20.99	0.000	1.00
Time	6.3378	3.1689	0.22745	13.93	0.000	1.00
Press	-4.6442	-2.3221	0.22745	-10.21	0.000	1.00
Catalyst	2.9838	1.4919	0.20343	7.33	0.000	1.00
Temp*Time	5.1662	2.5831	0.22745	11.36	0.000	1.00
Temp*Press	0.18975	0.094875	0.22745	0.42	0.688	1.00
Temp*Catalyst	0.57025	0.28512	0.22745	1.25	0.245	1.00
Time*Press	-0.7075	-0.35375	0.22745	-1.56	0.158	1.00
Time*Catalyst	0.48	0.24	0.22745	1.06	0.322	1.00
Press*Catalyst	-0.0375	-0.01875	0.22745	-0.08	0.936	1.00
Center Point		3.8253	0.50859	7.52	0.000	1.00

Coded units. Effect = 2 x coefficient. Error df = 8; terms with Sig. < 0.05 are significant at alpha = 0.05.

Model Summary: Yield

	S	R-squared	Adj. R-squared	Pred. R-squared	PRESS
Yield	0.90979	99.19%	98.08%	96.85%	25.86

S = residual standard deviation. Predicted R-squared uses leave-one-out (PRESS) residuals; shown as 0 when negative.

Analysis of Variance: Yield

	df	Adj. SS	Adj. MS	F	Sig.
Model	11	813.96	73.996	89.40	0.000
Linear	4	656	164	198.14	0.000
Temp	1	364.54	364.54	440.42	0.000
Time	1	160.67	160.67	194.11	0.000
Press	1	86.276	86.276	104.23	0.000
Catalyst	1	44.515	44.515	53.78	0.000
2-Way Interactions	6	111.13	18.522	22.38	0.000
Temp*Time	1	106.76	106.76	128.98	0.000
Temp*Press	1	0.14402	0.14402	0.17	0.688
Temp*Catalyst	1	1.3007	1.3007	1.57	0.245
Time*Press	1	2.0022	2.0022	2.42	0.158
Time*Catalyst	1	0.9216	0.9216	1.11	0.322
Press*Catalyst	1	0.005625	0.005625	0.01	0.936
Curvature	1	46.824	46.824	56.57	0.000
Error	8	6.6217	0.82771		
Lack-of-Fit	6	1.723	0.28716	0.12	0.982
Pure Error	2	4.8987	2.4494		
Total	19	820.58			

Adjusted (partial) sums of squares: each term is tested given all other terms in the model.

Residual Checks: Yield

	Statistic	Reference	Conclusion
Normality (Anderson-Darling)	0.855	Sig. = 0.023	Residuals are not normal (Sig. < 0.05): p-values may be inaccurate; consider Box-Cox.
Independence (Durbin-Watson)	1.591	about 2 = independent	No sign of correlation between neighbouring runs.
Influence (Cook's distance)	0.323	above 1 = influential	No single run dominates the fit.

Checks on the residuals (observed minus fitted). Durbin-Watson uses the RunOrder column.

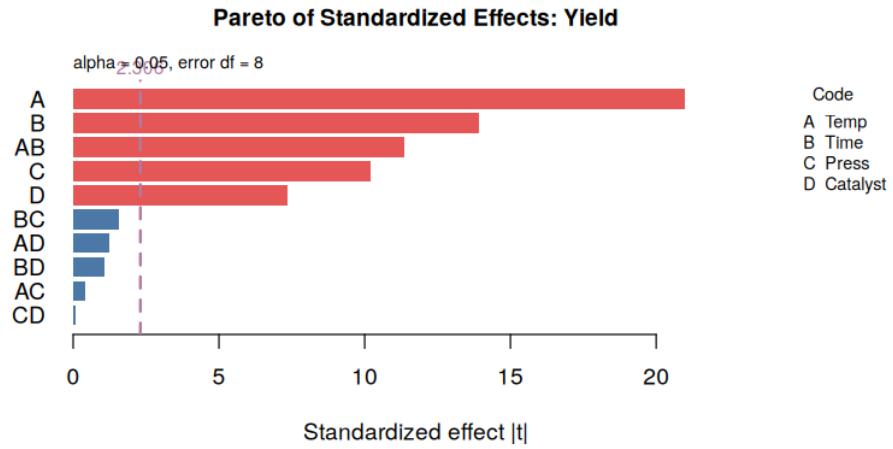


Figure 7.1: Pareto chart of standardized effects (letters as in the Factor Coding table).

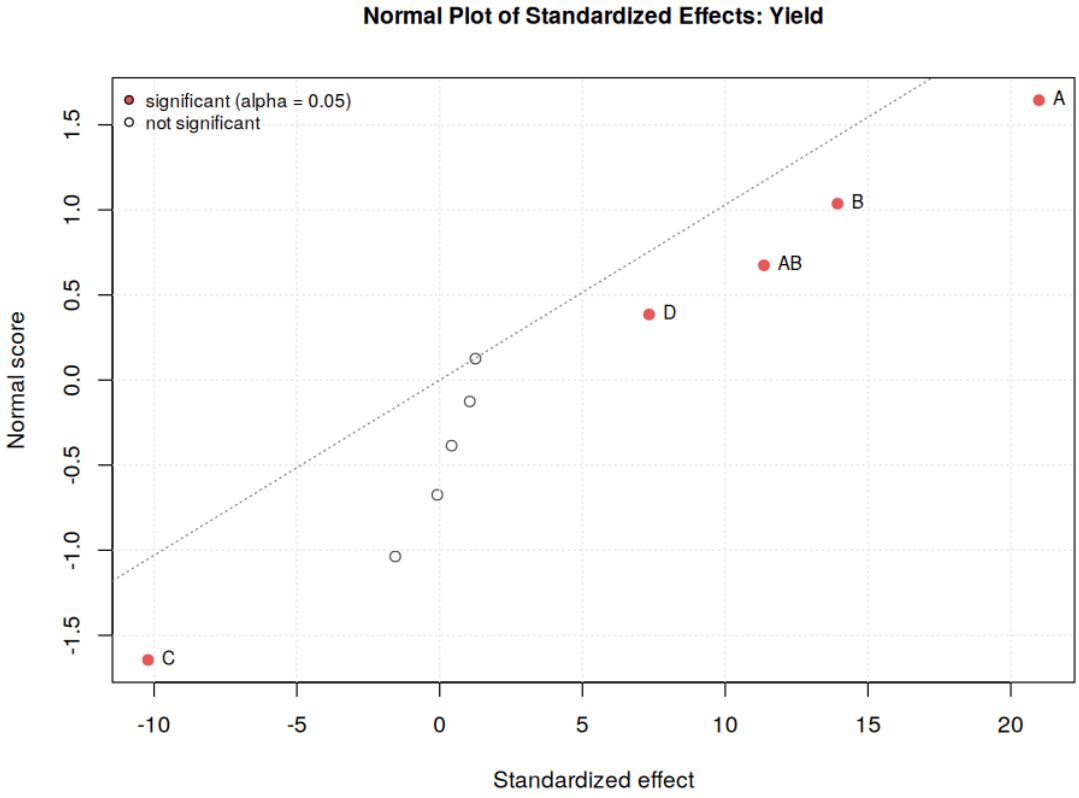


Figure 7.2: Normal plot of standardized effects.

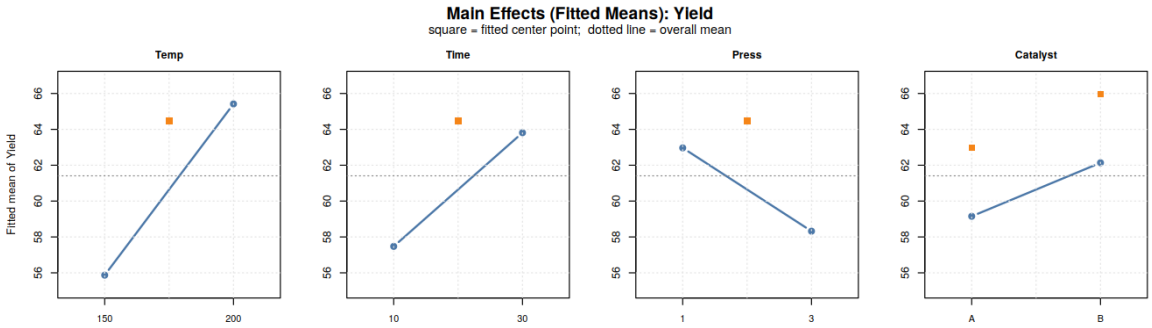


Figure 7.3: Main effects plot (fitted means; square = fitted centre point).

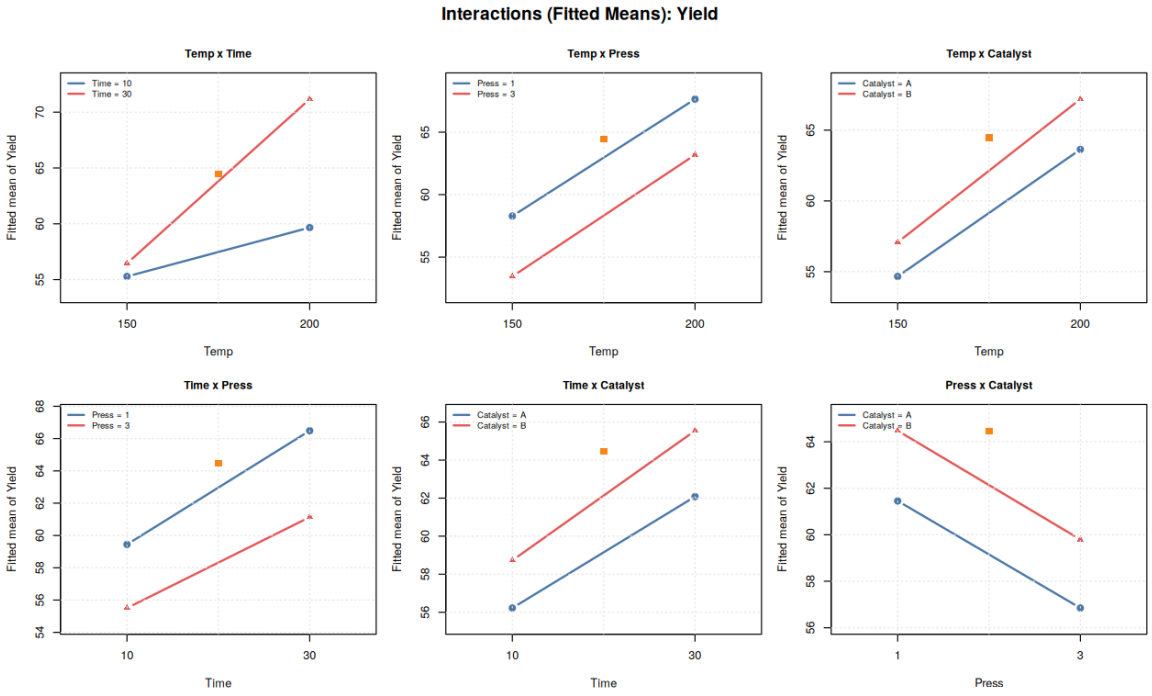


Figure 7.4: Interaction plots.

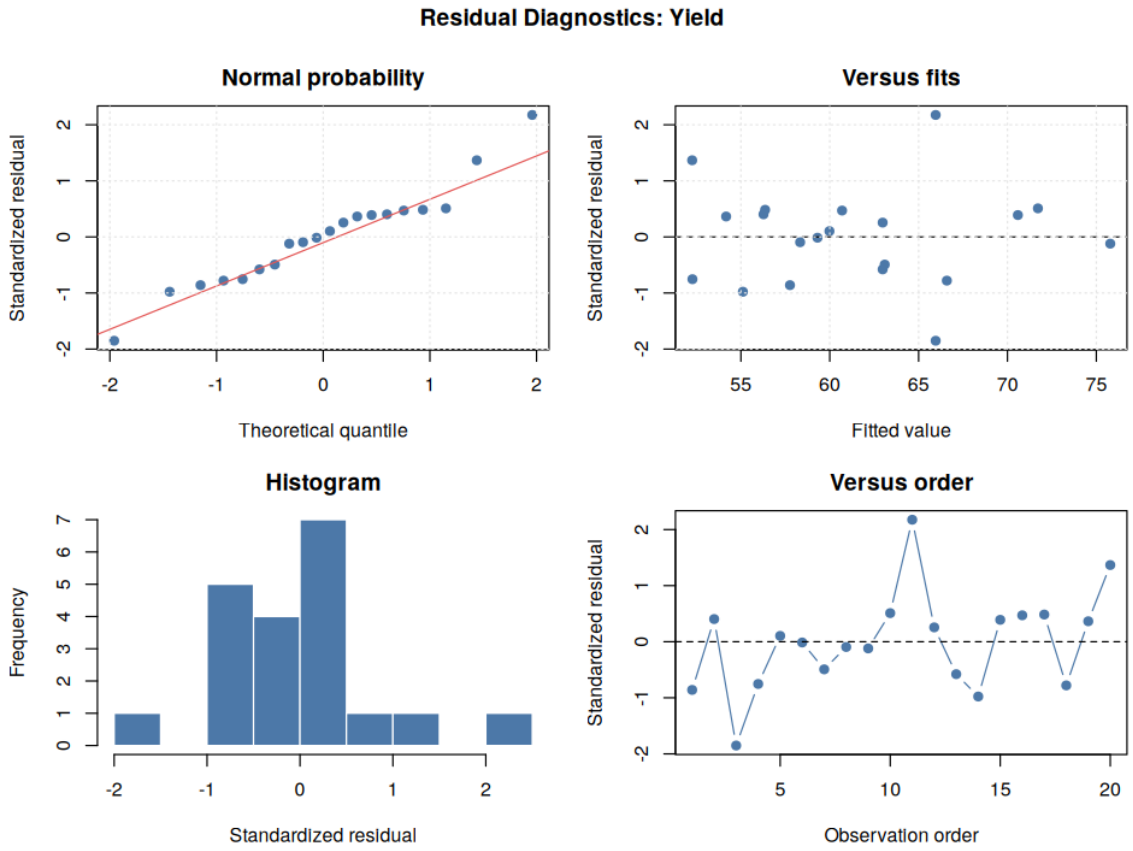


Figure 7.5: Residual plots.

How to read the results. Start with the *Summary of Results*: it states the conclusions in words.

- **Is anything going on?** The model is significant (Sig. 0.000) and explains 99.2% of the variation (R-squared). Predicted R-squared, 96.8%, measures how well the model predicts runs it did not see; when it is far below R-squared the model is over-fitted.

- **Which effects matter?** In the Coded Coefficients, Temp, Time, Press, Catalyst and Temp*Time have Sig. < 0.05. The *Effect* column is the change in Yield from the low to the high setting: raising Temp from 150 to 200 raises Yield by 9.55 on average, Press from 1 to 3 lowers it by 4.64. The Pareto chart shows the same ranking with the critical value (dashed line); the normal plot marks the significant effects away from the line of the unimportant ones.

- **Interaction.** Temp*Time is large (effect 5.17): the Temp effect depends on Time. The interaction plot shows it: at Time 30 the Temp line is much steeper than at Time 10. When an interaction is significant, interpret the two factors together, not their main effects alone.

- **Curvature.** The Center Point term (coefficient 3.83, Sig. 0.000) says the centre runs average 3.8 units higher than the corners predict: the response bends inside the region. A two-level model cannot locate an optimum inside the region; the Next Steps table suggests adding runs at intermediate levels (with a text factor present, a response-surface design built for the numeric factors).

- **Lack of fit and residuals.** No evidence of lack of fit (Sig. 0.982). The Anderson–Darling test flags non-normal residuals (Sig. 0.023) and one run (case 20) has a standardized residual of 2.18. Check that run for a recording error first; with 20 runs, one moderately large residual is not unusual.

- **VIF** = 1.00 everywhere: the design is orthogonal, so each effect is estimated independently of the others.

Next. Simplify the model (Example 7.2) or let the procedure choose terms (Example 7.3); model the curvature with a response-surface design (Examples 6.21 and 7.7).

7.2 Own model per response and predictions

Example A02

Also known as / search terms: model terms, reduced model, multiple responses, prediction interval, confidence interval, prediction at settings

When to use it. Fit your own model to each response (keep only the terms that matter) and predict the response at settings of interest, with confidence and prediction intervals.

Data. Dataset *reactor* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag Temp, Time, Press, Catalyst into **Factors (design columns)**; drag Yield, Cost into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Model...]**: **Model terms**: type `Yield: Temp Time Press Catalyst Temp*Time CTPT; Cost: Temp Time Press Temp*Press`. **Predict at settings (optional)**: type `Temp=190 Time=25 Press=1 Catalyst=B; Temp=160 Time=12 Press=3 Catalyst=A`. Click **[Continue]**.
4. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=Temp Time Press Catalyst
  RESPONSEVAR=Yield Cost ANALYZE=YES
  /FITMODEL TERMS="Yield: Temp Time Press Catalyst Temp*Time CTPT; Cost: Temp
   ↪  Time Press Temp*Press"
    PREDICT="Temp=190 Time=25 Press=1 Catalyst=B; Temp=160 Time=12 Press=3
   ↪ Catalyst=A".
```

Output.

Coded Coefficients: Yield

	Effect	Coefficient	SE Coef.	t	Sig.	VIF
Constant		60.643	0.22992	263.75	0.000	
Temp	9.5465	4.7732	0.22992	20.76	0.000	1.00
Time	6.3378	3.1689	0.22992	13.78	0.000	1.00
Press	-4.6442	-2.3221	0.22992	-10.10	0.000	1.00
Catalyst	2.9838	1.4919	0.20565	7.25	0.000	1.00
Temp*Time	5.1662	2.5831	0.22992	11.23	0.000	1.00
Center Point		3.8253	0.51413	7.44	0.000	1.00

Coded units. Effect = 2 x coefficient. Error df = 13; terms with Sig. < 0.05 are significant at alpha = 0.05.

Model Summary: Yield

	S	R-squared	Adj. R-squared	Pred. R-squared	PRESS
Yield	0.9197	98.66%	98.04%	96.96%	24.975

S = residual standard deviation. Predicted R-squared uses leave-one-out (PRESS) residuals; shown as 0 when negative.

Prediction: Yield

	Fit	SE Fit	95% CI (mean)	95% PI (single run)	Note
Temp=190, Time=25, Press=1, Catalyst=B	69.681	0.4301	(68.751, 70.61)	(67.487, 71.874)	curvature term assumed 0 away from the center
Temp=160, Time=12, Press=3, Catalyst=A	52.67	0.4616	(51.673, 53.667)	(50.447, 54.893)	curvature term assumed 0 away from the center

CI = interval for the average response at these settings; PI = interval for one new run. Factors not listed are at their center (numeric) or first level (text).

Coded Coefficients: Cost

	Effect	Coefficient	SE Coef.	t	Sig.	VIF
Constant		19.789	0.13026	151.92	0.000	
Temp	4.3116	2.1558	0.14563	14.80	0.000	1.00
Time	-1.5816	-0.79081	0.14563	-5.43	0.000	1.00
Press	1.5489	0.77444	0.14563	5.32	0.000	1.00
Temp*Press	1.4319	0.71594	0.14563	4.92	0.000	1.00

Coded units. Effect = 2 x coefficient. Error df = 15; terms with Sig. < 0.05 are significant at alpha = 0.05.

Prediction: Cost

	Fit	SE Fit	95% CI (mean)	95% PI (single run)	Note
Temp=190, Time=25, Press=1, Catalyst=B	19.484	0.2424	(18.967, 20)	(18.139, 20.828)	
Temp=160, Time=12, Press=3, Catalyst=A	19.474	0.2589	(18.922, 20.025)	(18.115, 20.832)	

CI = interval for the average response at these settings; PI = interval for one new run. Factors not listed are at their center (numeric) or first level (text).

Next Steps

Curvature. The center points differ from the corners for Yield (Sig. 0.000): the response bends inside the region, so a two-level model cannot locate the best settings.

→ *Add runs at intermediate levels of the numeric factors (for example a response-surface design) so the curvature can be modelled.*

Residuals. Cost: residuals are not normal (Anderson-Darling Sig. 0.018).

→ *Check the flagged runs for recording errors first; then try a Box-Cox transformation.*

```
STATS DOE ANALYSIS GENERATEDESIGN=NO VARNAMES=Temp Time Press Catalyst
   ↪ RESPONSEVAR=Cost
  LOWS=150 10 1 0 HIGHS=200 30 3 0
  ANALYZE=YES
  /FITMODEL TERMS="Cost: Temp Time Press Temp*Press" SELECTION=NONE
   ↪ CENTERTERM=YES ALPHA=0.05 CONFLEVEL=95 BOXCOX=AUTO.
```

Generated from the results above. Each syntax can be pasted into a syntax window; dataset names such as design_ccd are suggestions. Judgement still applies: check that each step makes sense for your process.

How to read the results. `TERMS` gives each response its own model, separated by semicolons. For Yield the reduced model keeps the five significant terms and the curvature term (CTPT); for Cost, the three main effects and Temp*Press.

- Removing the five unimportant interactions barely changed R-squared (99.2% to 98.7%), raised the error degrees of freedom from 8 to 13 and slightly improved predicted R-squared (97.0%): the simpler model predicts at least as well.
- **Predictions.** At Temp=190, Time=25, Press=1, Catalyst=B the fitted Yield is 69.7. The 95% confidence interval (68.8, 70.6) is where the *average* yield at these settings lies; the prediction interval (67.5, 71.9) is where a *single new run* will fall, and is wider because it includes run-to-run variation. The note "curvature term assumed 0 away from the center" is a reminder that, with curvature present, predictions between the corners and the centre are approximate.

- The Next Steps table repeats the curvature finding for Yield and, for Cost, flags non-normal residuals with a Box–Cox syntax to try.

7.3 Automatic term selection (stepwise)

Example A03

Also known as / search terms: stepwise regression, term selection, model selection: stepwise, hierarchy

When to use it. Let the procedure choose the terms by their p-values. Stepwise selection adds the most significant term and removes terms that become non-significant, keeping the model hierarchical (an interaction always keeps its main effects).

Data. Dataset *reactor* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag Temp, Time, Press, Catalyst into **Factors (design columns)**; drag Cost into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Model...]**: **Term selection**: select *Stepwise*. **Alpha to enter**: type `0.15`. **Alpha to remove**: type `0.10`. Click **[Continue]**.
4. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=Temp Time Press Catalyst
  RESPONSEVAR=Cost ANALYZE=YES
  /FITMODEL SELECTION=STEPWISE ALPHAENTER=0.15 ALPHAREMOVE=0.10.
```

Output.

Term Selection: Cost

Action	Term	P-Value
Entered	Temp	0.000
Entered	Temp*Press	0.011
Entered	Time	0.000
Entered	Temp*Catalyst	0.017

Stepwise selection. Alpha to enter = 0.15, alpha to remove = 0.1; hierarchy kept (an interaction always keeps its lower-order terms).

Coded Coefficients: Cost

	Effect	Coefficient	SE Coef.	t	Sig.	VIF
Constant		19.731	0.11393	173.19	0.000	
Temp	4.3116	2.1558	0.11393	18.92	0.000	1.00
Time	-1.5816	-0.79081	0.11393	-6.94	0.000	1.00
Press	1.5489	0.77444	0.11393	6.80	0.000	1.00
Catalyst	-0.3802	-0.1901	0.1019	-1.87	0.087	1.00
Temp*Press	1.4319	0.71594	0.11393	6.28	0.000	1.00
Temp*Catalyst	0.63287	0.31644	0.11393	2.78	0.017	1.00
Center Point		0.29219	0.25475	1.15	0.274	1.00

Coded units. Effect = 2 x coefficient. Error df = 12; terms with Sig. < 0.05 are significant at alpha = 0.05.

Model Summary: Cost

	S	R-squared	Adj. R-squared	Pred. R-squared	PRESS
Cost	0.45572	97.68%	96.32%	93.06%	7.4408

S = residual standard deviation. Predicted R-squared uses leave-one-out (PRESS) residuals; shown as 0 when negative.

How to read the results. Four steps: Temp, Temp*Press, Time and Temp*Catalyst entered. Press and Catalyst were added by the hierarchy rule (they belong to the selected interactions), which is why Catalyst appears with Sig. 0.087. The curvature term stays because the design has centre points (it is not significant here: Sig. 0.274, so Cost is adequately described by a first-order model). Compare with Example 7.2: stepwise also found Temp*Catalyst (Sig. 0.017), and adjusted R-squared rose from 94.0% to 96.3%. Use selection as a guide, not as the final word: check that the selected model makes engineering sense.

7.4 Box-Cox transformation

Example A04

Also known as / search terms: Box–Cox transformation, transformation: log, non-constant variance, lack of fit

When to use it. The response spreads more at high values than at low ones (typical for viscosity, concentrations, times to failure), or the residuals are skewed. A Box–Cox transformation models Y^{λ} instead of Y; $\lambda = 0$ means the logarithm.

Data. Dataset *reactor* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.

2. In the main dialog, drag Temp, Time, Press, Catalyst into **Factors (design columns)**; drag Visc into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).

3. Click **[Model...]**: **Box-Cox transformation**: select *Automatic (best rounded lambda)*. **Standard model**: select *Main effects only*. Click **[Continue]**.

4. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=Temp Time Press Catalyst
  RESPONSEVAR=Visc ANALYZE=YES
  /FITMODEL BOXCOX=AUTO TERMSET=ts_linear.
```

Output.

Summary of Results: Visc

	Finding
Overall	The model is statistically significant (Sig. 0.000): the factors explain real changes in Visc (on the Box-Cox scale, lambda = 0).
Model fit	The model explains 98.0% of the variation in Visc (adjusted 97.4%). Predicted R-squared 96.1%: it should predict new runs well.
Significant terms	2 of 4 terms are significant at alpha = 0.05: Temp, Time (strongest first). Not significant: Press, Catalyst.
Largest effects	Averaged over the other factors: Temp from 150 to 200 changes Visc by 0.8012; Time from 10 to 30 changes Visc by 0.3464.
Lack of fit	Significant lack of fit (Sig. 0.038): the model misses something real; consider adding terms or a transformation.
Residuals	Residuals are consistent with a normal distribution (Anderson-Darling Sig. 0.198); 1 run(s) with a large standardized residual (> 2).

Plain-language reading of the tables below. Decisions use the alpha level shown; check the charts before acting.

Box-Cox Transformation: Visc

	Best lambda	Rounded lambda	95% CI	Lambda used
Visc	0.04	0.0	(-0.28, 0.36)	0.00

The model is fitted to Y^lambda (natural log when lambda = 0); tables and effect charts are on that scale, the optimizer reports original units. Lambda = 1 means no transformation is needed.

Coded Coefficients: Visc

	Effect	Coefficient	SE Coef.	t	Sig.	VIF
Constant		3.0063	0.014506	207.25	0.000	
Temp	0.80123	0.40061	0.016218	24.70	0.000	1.00
Time	0.34638	0.17319	0.016218	10.68	0.000	1.00
Press	0.001697	0.00084851	0.016218	0.05	0.959	1.00
Catalyst	0.0040875	0.0020438	0.014506	0.14	0.890	1.00

Coded units. Effect = 2 x coefficient. Error df = 15; terms with Sig. < 0.05 are significant at alpha = 0.05.

Analysis of Variance: Visc

	df	Adj. SS	Adj. MS	F	Sig.
Model	4	3.0479	0.76197	181.06	0.000
Linear	4	3.0479	0.76197	181.06	0.000
Temp	1	2.5679	2.5679	610.19	0.000
Time	1	0.47992	0.47992	114.04	0.000
Press	1	0.000011519	0.000011519	0.00	0.959
Catalyst	1	0.000083539	0.000083539	0.02	0.890
Error	15	0.063124	0.0042083		
Lack-of-Fit	13	0.062753	0.0048272	26.04	0.038
Pure Error	2	0.00037074	0.00018537		
Total	19	3.111			

Adjusted (partial) sums of squares: each term is tested given all other terms in the model.

Next Steps

Lack of fit. Visc: significant lack of fit (Sig. 0.038); the model leaves out terms this design can estimate (Temp*Time Temp*Press Temp*Catalyst Time*Press Time*Catalyst Press*Catalyst CTPT).

→ *Refit with the design's standard model (the terms left out are added back), then check the lack of fit again.*

```
STATS DOE ANALYSIS GENERATEDESIGN=NO VARNAMES=Temp Time Press Catalyst
   ↪ RESPONSEVAR=Visc
  LOWS=150 10 1 0 HIGHS=200 30 3 0
  ANALYZE=YES
  /FITMODEL TERMS="Visc: Temp Time Press Catalyst Temp*Time Temp*Press Temp*
   ↪ Catalyst Time*Press Time*Catalyst Press*Catalyst CTPT" SELECTION=NONE
   ↪ CENTERTERM=YES ALPHA=0.05 CONFLEVEL=95 BOXCOX=LOG.
```

Inactive factors. Press, Catalyst: no term containing these factors is significant for any response (alpha = 0.05).

→ Refit without them to check the simpler model; in the next experiment fix them at the most convenient or cheapest level and spend the runs on the active factors. Not significant is not proof of no effect: with few runs a small effect can be missed.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO VARNAMES=Temp Time Press Catalyst
   ↪ RESPONSEVAR=Visc
  LOWS=150 10 1 0 HIGHS=200 30 3 0
  ANALYZE=YES
  /FITMODEL TERMS="Visc: Temp Time" SELECTION=NONE CENTERTERM=YES ALPHA=0.05
   ↪ CONFLEVEL=95 BOXCOX=LOG.
```

Generated from the results above. Each syntax can be pasted into a syntax window; dataset names such as design_ccd are suggestions. Judgement still applies: check that each step makes sense for your process.

How to read the results.

- **Box–Cox table.** The best λ is 0.04 with a 95% interval from -0.28 to 0.36; the rounded value 0 (natural log) lies inside it and was used. $\lambda = 1$ (no transformation) is outside the interval, so a transformation is needed.
- On the log scale Temp (effect 0.80) and Time (0.35) are the active factors: raising Temp from 150 to 200 multiplies viscosity by $e^{0.80} \approx 2.2$. Press and Catalyst have no effect.
- **Lack of fit** is significant (Sig. 0.038). Here the model was restricted to main effects (`TERMSET=ts_linear`); the Next Steps table points out that the design can estimate the interactions and the curvature term that were left out and gives the syntax that adds them back. A significant lack of fit with only 2 pure-error degrees of freedom should be taken seriously but not over-interpreted.
- The optimizer and predictions always report results back in original units.

7.5 Fractional factorial screening: aliasing and Lenth's method

Example A05

Also known as / search terms: fractional factorial design: analysis, Lenth's method, saturated model, alias structure, half fraction

When to use it. A screening experiment with no spare runs: the model uses every degree of freedom, so there is no error estimate and no p-values. Lenth's method judges effects against a robust estimate of noise taken from the small effects.

Data. Dataset *screen5* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag A, B, C, D, E into **Factors (design columns)**; drag Response into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Analysis]**: tick **Create Diagnostic Plots**. tick **Pareto Chart**. Click **[Continue]**.
4. Click **[Model...]**: tick **Normal plot of standardized effects**. Click **[Continue]**.
5. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=A B C D E RESPONSEVAR=
    ↪ Response ANALYZE=YES
  /OUTPUT PARETOPLOT=YES
  /FITMODEL EFFECTSPLOT=YES.
```

Output.

Summary of Results: Response

	Finding
Model fit	The model explains 100.0% of the variation in Response (adjusted -).
Significant terms	No error degrees of freedom, so terms were judged with Lenth's method: 3 of 15 stand out: A, B*C, E.
Largest effects	Averaged over the other factors: A from -1 to 1 changes Response by 8.187 (the size depends on B, C, D, E: see the interaction plot); E from -1 to 1 changes Response by -4.651 (the size depends on A, B, C, D: see the interaction plot); D from -1 to 1 changes Response by -0.6892 (the size depends on A, B, C, E: see the interaction plot).

Plain-language reading of the tables below. Decisions use the alpha level shown; check the charts before acting.

Coded Coefficients: Response

	Effect	Coefficient	SE Coef.	t	Sig.	VIF
Constant		49.687				
A	8.1868	4.0934				1.00
B	-0.05825	-0.029125				1.00
C	0.043	0.0215				1.00
D	-0.68925	-0.34462				1.00
E	-4.6512	-2.3256				1.00
A*B	0.659	0.3295				1.00
A*C	-0.09875	-0.049375				1.00
A*D	-0.4995	-0.24975				1.00
A*E	-0.55	-0.275				1.00
B*C	6.2492	3.1246				1.00
B*D	0.375	0.1875				1.00
B*E	-0.6945	-0.34725				1.00
C*D	0.14675	0.073375				1.00
C*E	-0.20725	-0.10362				1.00
D*E	-0.1695	-0.08475				1.00

Coded units. No error degrees of freedom: standard errors and tests are unavailable; judge effects with the Pareto or normal plot of effects (Lenth's method).

Alias Structure: Response

	Aliased With
A	+ BCDE
B	+ ACDE
C	+ ABDE
D	+ ABCE
E	+ ABCD
AB	+ CDE
AC	+ BDE
AD	+ BCE
AE	+ BCD
BC	+ ADE
BD	+ ACE
BE	+ ACD
CD	+ ABE
CE	+ ABD
DE	+ ABC

A = A; B = B; C = C; D = D; E = E

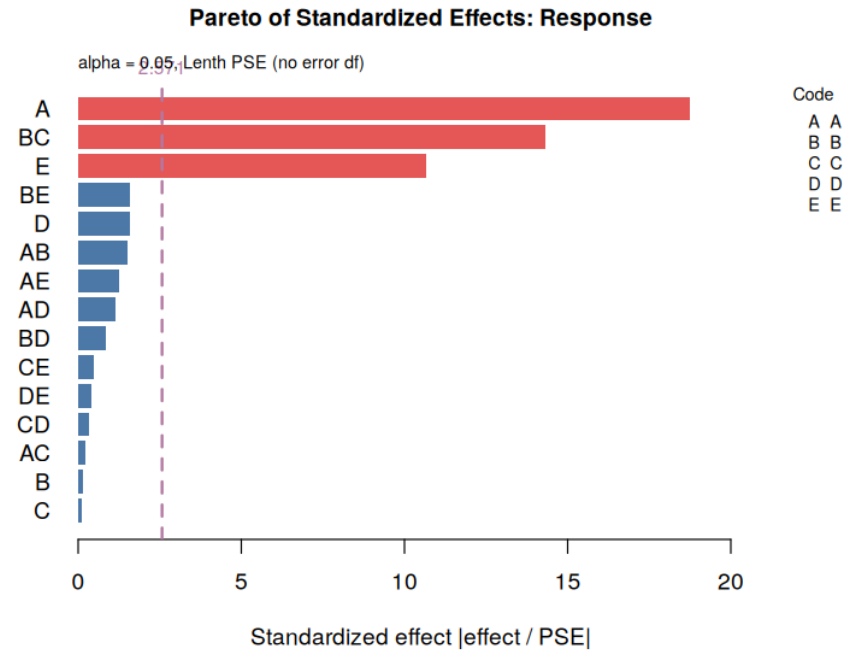


Figure 7.6: Pareto chart with Lenth's margin of error.

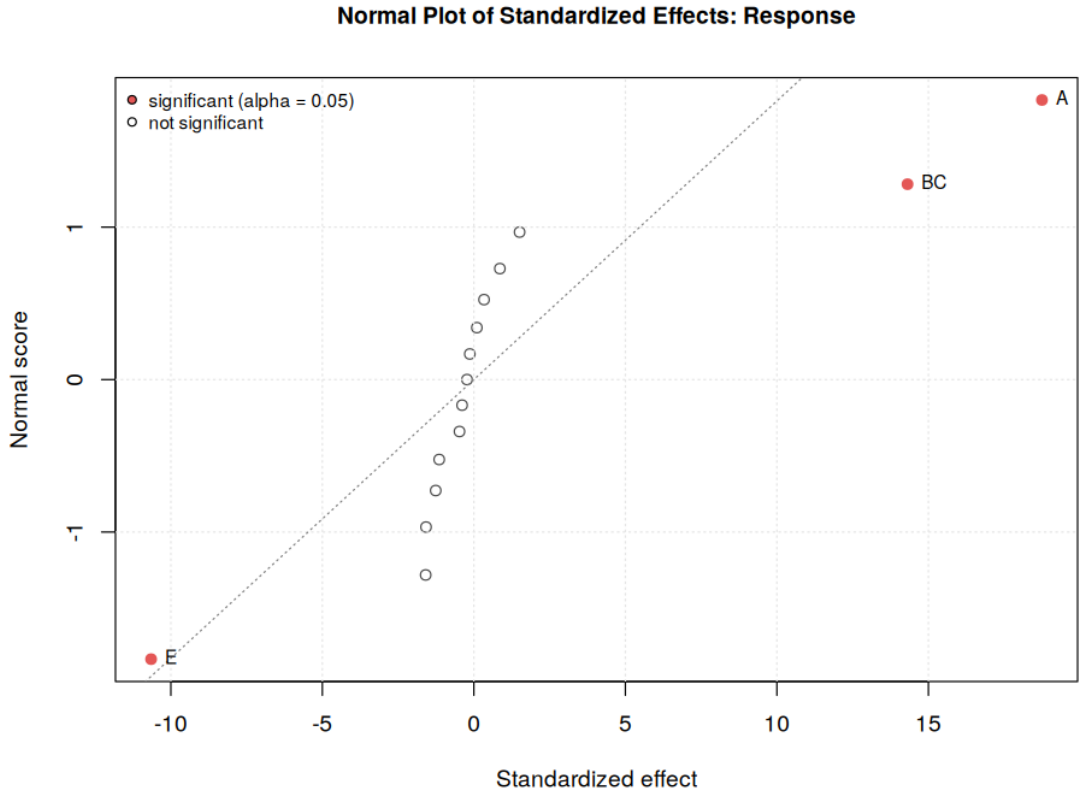


Figure 7.7: Normal plot of effects.

How to read the results.

- The *Alias Structure* identifies the design as a half fraction 2^{5-1} with defining relation I =

ABCDE (resolution V): each main effect is aliased only with a four-factor interaction (A + BCDE) and each two-factor interaction with a three-factor interaction (AB + CDE). Higher-order interactions are usually negligible, so all main effects and two-factor interactions can be estimated.

- With 16 runs and 16 coefficients, no error degrees of freedom remain: SE, t and Sig. are blank. Lenth's method (Pareto chart) finds three effects that stand out: A (effect 8.19), B*C (6.25) and E (-4.65). On the normal plot they lie far from the straight line through the small effects.

- B and C individually do nothing, but their interaction is large: in screening, do not drop a factor whose interactions are active.

Next. Refit with only A, E, B, C and B*C to obtain error degrees of freedom and p-values; Next Steps suggests the same.

7.6 Plackett-Burman screening analysis

Example A06

Also known as / search terms: Plackett–Burman design: analysis, screening analysis, main effects model

When to use it. Find the active factors among 11 in a 12-run Plackett–Burman screen.

Data. Dataset *pb11* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag F1, F2, F3, F4, F5, F6, F7, F8, F9, F10, F11 into **Factors (design columns)**; drag Response into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Create Design...]**: **Design Type**: select *Plackett-Burman Design – k≥2*. Click **[Continue]**.
4. Click **[Analysis]**: tick **Create Diagnostic Plots**. tick **Pareto Chart**. Click **[Continue]**.
5. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO DESIGNTYPE=PLACKETTBURMAN
  FACTORVARS=F1 F2 F3 F4 F5 F6 F7 F8 F9 F10 F11 RESPONSEVAR=Response ANALYZE=
    ↪ YES
  /OUTPUT PARETOPLOT=YES.
```

Output.

Summary of Results: Response

	Finding
Model fit	The model explains 100.0% of the variation in Response (adjusted -).
Significant terms	No error degrees of freedom, so terms were judged with Lenth's method: 3 of 11 stand out: F1, F4, F7.

Plain-language reading of the tables below. Decisions use the alpha level shown; check the charts before acting.

Factor Coding: Response

Code	Factor	Type	Coded -1	Coded +1
A	F1	Numeric	-1	1
B	F2	Numeric	-1	1
C	F3	Numeric	-1	1
D	F4	Numeric	-1	1
E	F5	Numeric	-1	1
F	F6	Numeric	-1	1
G	F7	Numeric	-1	1
H	F8	Numeric	-1	1
J	F9	Numeric	-1	1
K	F10	Numeric	-1	1
L	F11	Numeric	-1	1

Design treated as screening (main effects). Runs used: 12 (center points: 0).

Coded Coefficients: Response

	Coefficient	SE Coef.	t	Sig.	VIF
Constant	9.9641				
F1	2.0611				1.00
F2	0.034583				1.00
F3	-0.0050833				1.00
F4	-1.2639				1.00
F5	0.22075				1.00
F6	0.14292				1.00
F7	0.96225				1.00
F8	0.15725				1.00
F9	-0.18742				1.00
F10	-0.054417				1.00
F11	-0.15292				1.00

Coded units. No error degrees of freedom: standard errors and tests are unavailable; judge effects with the Pareto or normal plot of effects (Lenth's method).

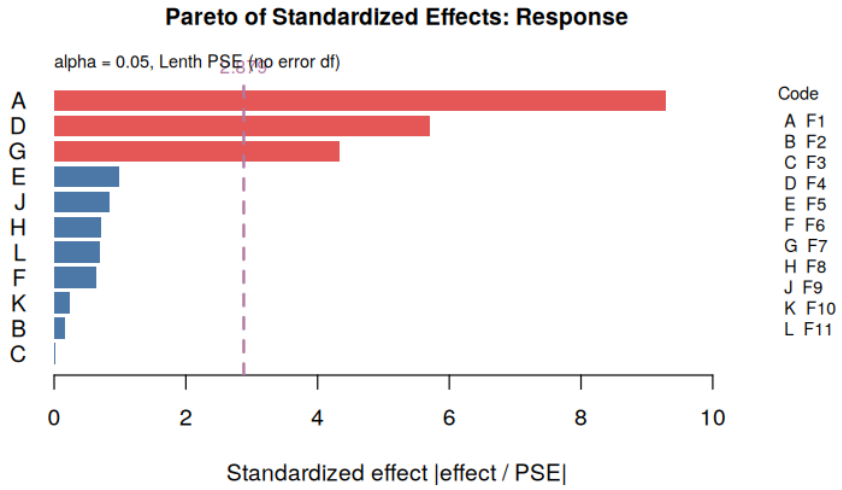


Figure 7.8: Pareto chart of the 11 effects with Lenth's margin of error.

How to read the results. The main-effects model for 11 factors uses all 12 runs, so Lenth's method is used. Three factors stand out: F1 (coefficient 2.06, i.e. effect 4.12), F4 (-1.26) and F7 (0.96); the other eight are small. In the next experiment fix the inactive factors at convenient levels and study F1, F4 and F7 in a full factorial with centre points, which can also estimate their interactions (partially aliased in this design, see Example 6.4).

7.7 Response surface (CCD): quadratic model, canonical analysis, contours

Example A07

Also known as / search terms: response surface methodology: analysis, quadratic model, canonical analysis, stationary point, contour plot, surface plot

When to use it. Fit the full quadratic model to a central composite design, find where the response is smallest, and draw the surface.

Data. Dataset *machining* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag Speed, Feed, Depth into **Factors (design columns)**; drag Rough into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Analysis]**: tick **Create Diagnostic Plots**. tick **Contour Plot**. Click **[Continue]**.
4. Click **[Model...]**: tick **Stationary point / canonical analysis**. tick **3-D surface plots**. Click **[Continue]**.
5. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=Speed Feed Depth RESPONSEVAR=
    ↪ Rough ANALYZE=YES
  /OUTPUT CONTOURPLOT=YES
  /FITMODEL CANONICAL=YES SURFACEPLOT=YES.
```

Output.

Summary of Results: Rough

	Finding
Overall	The model is statistically significant (Sig. 0.000): the factors explain real changes in Rough.
Model fit	The model explains 99.0% of the variation in Rough (adjusted 98.1%). Predicted R-squared 94.6%: it should predict new runs well.
Significant terms	6 of 9 terms are significant at alpha = 0.05: Feed, Speed, Speed*Speed, Depth, Speed*Feed, Feed*Feed (strongest first). Not significant: Depth*Depth, Speed*Depth, Feed*Depth.
Lack of fit	No evidence of lack of fit (Sig. 0.390).
Residuals	Residuals are consistent with a normal distribution (Anderson-Darling Sig. 0.141); 1 run(s) with a large standardized residual (> 2).

Plain-language reading of the tables below. Decisions use the alpha level shown; check the charts before acting.

Coded Coefficients: Rough

	Coefficient	SE Coef.	t	Sig.	VIF
Constant	2.9962	0.057363	52.23	0.000	
Speed	-0.60805	0.038057	-15.98	0.000	1.00
Feed	0.8277	0.038057	21.75	0.000	1.00
Depth	0.30597	0.038057	8.04	0.000	1.00
Speed*Speed	0.40242	0.037042	10.86	0.000	1.02
Feed*Feed	0.27091	0.037042	7.31	0.000	1.02
Depth*Depth	0.0053844	0.037042	0.15	0.887	1.02
Speed*Feed	-0.3733	0.049726	-7.51	0.000	1.00
Speed*Depth	-0.002525	0.049726	-0.05	0.961	1.00
Feed*Depth	0.03865	0.049726	0.78	0.455	1.00

Coded units. Effect = 2 x coefficient. Error df = 10; terms with Sig. < 0.05 are significant at alpha = 0.05.

Model Summary: Rough

	S	R-squared	Adj. R-squared	Pred. R-squared	PRESS
Rough	0.14065	99.02%	98.14%	94.64%	1.0799

S = residual standard deviation. Predicted R-squared uses leave-one-out (PRESS) residuals; shown as 0 when negative.

Analysis of Variance: Rough

	df	Adj. SS	Adj. MS	F	Sig.
Model	9	19.969	2.2187	112.16	0.000
Linear	3	15.686	5.2286	264.32	0.000
Speed	1	5.0499	5.0499	255.28	0.000
Feed	1	9.3572	9.3572	473.03	0.000
Depth	1	1.2786	1.2786	64.64	0.000
Square	3	3.1561	1.052	53.18	0.000
Speed*Speed	1	2.3346	2.3346	118.02	0.000
Feed*Feed	1	1.0581	1.0581	53.49	0.000
Depth*Depth	1	0.00041796	0.00041796	0.02	0.887
2-Way Interactions	3	1.1268	0.37561	18.99	0.000
Speed*Feed	1	1.1148	1.1148	56.36	0.000
Speed*Depth	1	0.000051005	0.000051005	0.00	0.961
Feed*Depth	1	0.011951	0.011951	0.60	0.455
Error	10	0.19781	0.019781		
Lack-of-Fit	5	0.11184	0.022368	1.30	0.390
Pure Error	5	0.085976	0.017195		
Total	19	20.166			

Adjusted (partial) sums of squares: each term is tested given all other terms in the model.

Canonical Analysis: Rough

	Speed	Feed	Depth	Predicted	Eigenvalues	Nature	Inside design region
Rough	132.86	0.30778	-15.882	-2.0826	0.5348 0.1404 0.003432	minimum	No

Stationary point of the fitted second-order surface (factor settings in natural units). All eigenvalues negative = maximum, all positive = minimum, mixed signs = saddle. If the point lies outside the design region, move the experiment in that direction rather than extrapolating.

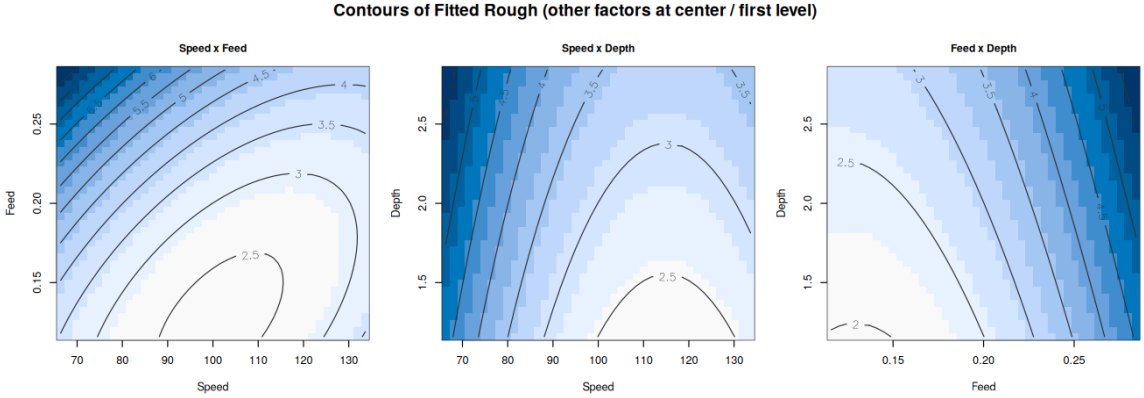


Figure 7.9: Contour plots of fitted roughness (third factor at its centre).

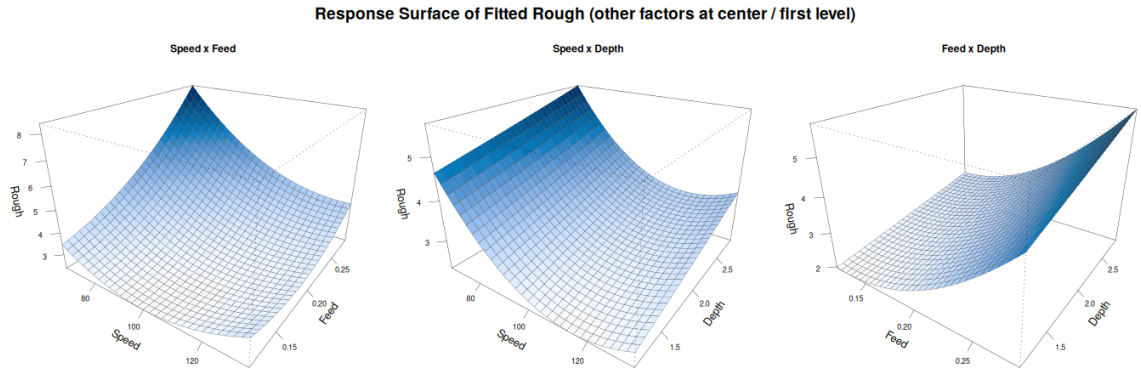


Figure 7.10: Response surface plots.

How to read the results.

- The quadratic model explains 99.0% of the variation, with no lack of fit (Sig. 0.390) and predicted R-squared 94.6%.
- Feed has the largest effect (coefficient 0.83: higher feed, rougher surface), then Speed (−0.61), Depth (0.31). The squares Speed*Speed (0.40) and Feed*Feed (0.27) are positive: roughness is U-shaped in Speed, with a minimum inside the range. Speed*Feed (−0.37) means that higher speed helps more at high feed.
- **Canonical analysis.** All three eigenvalues are positive (0.535, 0.140, 0.003), so the surface is bowl-shaped: the stationary point is a *minimum*. It lies at Depth = −15.9, far outside the region ("Inside design region: No"): the third eigenvalue is almost zero, so the surface is nearly flat in one direction and the exact minimum is poorly defined. In such cases do not extrapolate; use the optimizer (Example 8.1), which searches only inside the region.
- The contour plots show the low-roughness valley at Speed around 100–110 and low Feed.

7.8 Box-Behnken analysis

Example A08

Also known as / search terms: Box–Behnken design: analysis, saddle point, canonical analysis

When to use it. Analyze the Box–Behnken experiment on purification and characterize the fitted surface.

Data. Dataset *purify* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag Dose, TempC, Ratio into **Factors (design columns)**; drag Purity into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Create Design...]**: **Design Type**: select *Box-Behnken Design – k≥3*. Click **[Continue]**.
4. Click **[Model...]**: tick **Stationary point / canonical analysis**. Click **[Continue]**.
5. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO DESIGNTYPE=BOXBEHNKEN FACTORVARS=Dose
    ↪ TempC Ratio
  RESPONSEVAR=Purity ANALYZE=YES
  /FITMODEL CANONICAL=YES.
```

Output.

Summary of Results: Purity

	Finding
Overall	The model is statistically significant (Sig. 0.000): the factors explain real changes in Purity.
Model fit	The model explains 99.9% of the variation in Purity (adjusted 99.6%). Predicted R-squared 98.7%: it should predict new runs well.
Significant terms	7 of 9 terms are significant at alpha = 0.05: TempC, Dose, Dose*Dose, TempC*TempC, Dose*Ratio, Ratio, and 1 more (strongest first). Not significant: Ratio*Ratio, TempC*Ratio.
Lack of fit	No evidence of lack of fit (Sig. 0.597).
Residuals	Residuals are consistent with a normal distribution (Anderson-Darling Sig. 0.866).

Plain-language reading of the tables below. Decisions use the alpha level shown; check the charts before acting.

Coded Coefficients: Purity

	Coefficient	SE Coef.	t	Sig.	VIF
Constant	79.921	0.22395	356.87	0.000	
Dose	3.8419	0.13714	28.01	0.000	1.00
TempC	-6.0337	0.13714	-44.00	0.000	1.00
Ratio	1.6569	0.13714	12.08	0.000	1.00
Dose*Dose	-5.287	0.20186	-26.19	0.000	1.01
TempC*TempC	-2.5782	0.20186	-12.77	0.000	1.01
Ratio*Ratio	-0.26846	0.20186	-1.33	0.241	1.01
Dose*TempC	-0.538	0.19394	-2.77	0.039	1.00
Dose*Ratio	2.4638	0.19394	12.70	0.000	1.00
TempC*Ratio	0.1145	0.19394	0.59	0.581	1.00

Coded units. Effect = 2 x coefficient. Error df = 5; terms with Sig. < 0.05 are significant at alpha = 0.05.

Analysis of Variance: Purity

	df	Adj. SS	Adj. MS	F	Sig.
Model	9	577.82	64.203	426.72	0.000
Linear	3	431.29	143.76	955.52	0.000
Dose	1	118.08	118.08	784.82	0.000
TempC	1	291.25	291.25	1935.79	0.000
Ratio	1	21.962	21.962	145.97	0.000
Square	3	121.04	40.348	268.17	0.000
Dose*Dose	1	103.21	103.21	685.97	0.000
TempC*TempC	1	24.543	24.543	163.13	0.000
Ratio*Ratio	1	0.2661	0.2661	1.77	0.241
2-Way Interactions	3	25.49	8.4968	56.47	0.000
Dose*TempC	1	1.1578	1.1578	7.70	0.039
Dose*Ratio	1	24.28	24.28	161.38	0.000
TempC*Ratio	1	0.052441	0.052441	0.35	0.581
Error	5	0.75228	0.15046		
Lack-of-Fit	3	0.41011	0.1367	0.80	0.597
Pure Error	2	0.34217	0.17108		
Total	14	578.58			

Adjusted (partial) sums of squares: each term is tested given all other terms in the model.

Canonical Analysis: Purity

	Dose	TempC	Ratio	Predicted	Eigenvalues	Nature	Inside design region
Purity	-68.321	39.316	-33.51	-4.1245	0.01763 -2.553 -5.598	saddle point	No

Stationary point of the fitted second-order surface (factor settings in natural units). All eigenvalues negative = maximum, all positive = minimum, mixed signs = saddle. If the point lies outside the design region, move the experiment in that direction rather than extrapolating.

How to read the results. The model explains 99.9% (predicted 98.7%); lack of fit is not significant. TempC has the largest effect (−6.03: purity falls as temperature rises), then Dose (3.84) with strong curvature (Dose*Dose −5.29: an optimum dose inside the range), and the Dose*Ratio interaction (2.46). The eigenvalues have mixed signs (0.018, −2.55, −5.60): the stationary point is a *saddle* and lies outside the region, and the first eigenvalue is close to zero, meaning a ridge: along one direction purity hardly changes. A ridge is good news in practice: a range of settings gives nearly the same purity, so choose among them on cost or convenience. Find the best settings within the region with the optimizer.

7.9 General factorial with a categorical factor and Tukey grouping

Example A09

Also known as / search terms: general full factorial design: analysis, categorical factor: analysis, Tukey comparisons, multiple comparisons, letter grouping

When to use it. Compare the levels of a categorical factor (three materials) and check whether the effect of test temperature differs between materials.

Data. Dataset *materials* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag Material, TestTemp into **Factors (design columns)**; drag Strength into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Analysis]**: tick **Create Diagnostic Plots**. tick **Main Effects Plot**. tick **Interaction Plot**. Click **[Continue]**.
4. Click **[Model...]**: tick **Pairwise comparisons of categorical levels**. Click **[Continue]**.
5. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=Material TestTemp RESPONSEVAR
    ↪ =Strength ANALYZE=YES
  /OUTPUT MAINEFFECTS=YES INTERACTIONS=YES
  /FITMODEL GROUPING=YES.
```

Output.

Summary of Results: Strength

	Finding
Overall	The model is statistically significant (Sig. 0.000): the factors explain real changes in Strength.
Model fit	The model explains 98.9% of the variation in Strength (adjusted 98.0%). Predicted R-squared 95.7%: it should predict new runs well.
Significant terms	3 of 3 terms are significant at alpha = 0.05: Material, TestTemp, Material*TestTemp (strongest first).
Residuals	Residuals are consistent with a normal distribution (Anderson-Darling Sig. 0.597); possible run-order pattern (Durbin-Watson 3.06).

Plain-language reading of the tables below. Decisions use the alpha level shown; check the charts before acting.

Factor Coding: Strength

Code	Factor	Type	Coded -1	Coded +1
A	Material	Text/categorical	Steel	Alu, Ti
B	TestTemp	Numeric	20	80

Design treated as general factorial. Runs used: 12 (center points: 0).

Analysis of Variance: Strength

	df	Adj. SS	Adj. MS	F	Sig.
Model	5	907.55	181.51	111.38	0.000
Linear	3	881.66	293.89	180.33	0.000
Material	2	833.71	416.86	255.79	0.000
TestTemp	1	47.952	47.952	29.42	0.002
2-Way Interactions	2	25.886	12.943	7.94	0.021
Material*TestTemp	2	25.886	12.943	7.94	0.021
Error	6	9.7782	1.6297		
Total	11	917.33			

Adjusted (partial) sums of squares: each term is tested given all other terms in the model.

Pairwise Comparison (Tukey) of Material: Strength

	N	Fitted mean	Grouping
Steel	4	110.19	A
Ti	4	108.1	A
Alu	4	91.56	B

Levels that share a letter are not significantly different (family alpha = 0.05). Means are fitted means averaged over the other factors.

Regression Equation in Natural Units: Strength

	Material=Steel	Material=Alu	Material=Ti
Constant	104.9165	91.680833	103.26517
TestTemp	0.10555	-0.0024166667	0.096766667

Strength = sum of coefficient x term, with factors in their original units; one column per level of the text factor(s).

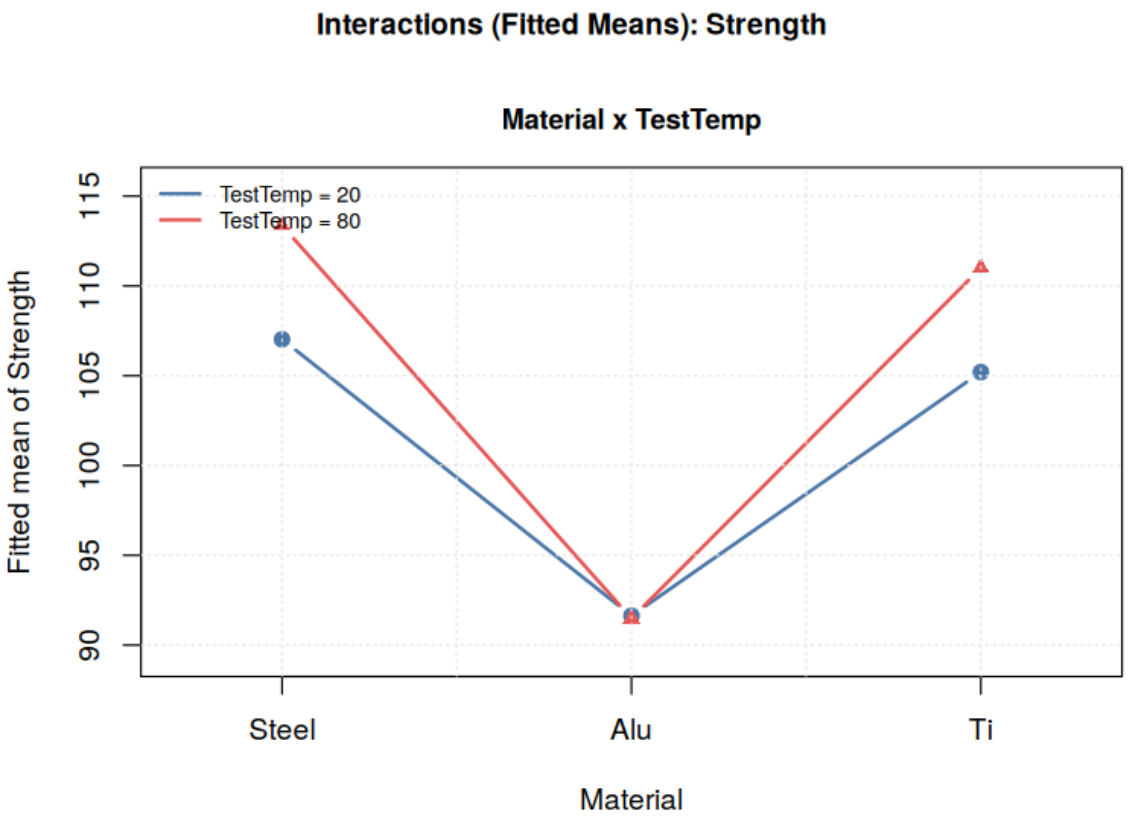


Figure 7.11: Interaction plot: Material × TestTemp.

How to read the results.

- Material is highly significant (F = 255.8), TestTemp too (Sig. 0.002), and so is their interaction (Sig. 0.021): temperature affects the materials differently.
- **Tukey grouping.** Steel (110.2) and Ti (108.1) share the letter A: their difference is not significant. Alu (91.6, group B) is clearly weaker.
- The regression equation, one column per material, shows the interaction directly: strength rises with temperature for Steel (+0.106 per degree) and Ti (+0.097) but not for Alu (−0.002).
- Coefficients of a 3-level factor are reported as "Material (1)" and "(2)": deviations of the first two levels from the average (effect coding). Use the equation and the fitted means for interpretation.
- Durbin–Watson 3.06 notes that consecutive rows alternate; in this dataset the rows are in standard order, not run order, so the check has no meaning here. Always keep a `RunOrder` column.

7.10 Two 3-level text factors, numeric factor as categorical

Example A10

Also known as / search terms: categorical factor: numeric factor as categorical, general full factorial design: three factors, Tukey comparisons

When to use it. Two text factors with three levels each and a numeric factor whose two settings are treated as categories.

Data. Dataset *suppliers* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag Supplier, Machine, Temp into **Factors (design columns)**; drag Thickness into **Responses (measured results)**; drag Temp into **Treat as categorical (optional)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Model...]**: tick **Pairwise comparisons of categorical levels**. Click **[Continue]**.
4. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=Supplier Machine Temp
    ↪ RESPONSEVAR=Thickness ANALYZE=YES
  /FITMODEL CATEGORICAL=Temp GROUPING=YES.
```

Output.

Analysis of Variance: Thickness

	df	Adj. SS	Adj. MS	F	Sig.
Model	13	127.88	9.8368	37.64	0.000
Linear	5	108.99	21.799	83.40	0.000
Supplier	2	37.525	18.763	71.79	0.000
Machine	2	37.946	18.973	72.59	0.000
Temp	1	33.522	33.522	128.26	0.000
2-Way Interactions	8	18.885	2.3607	9.03	0.000
Supplier*Machine	4	18.169	4.5424	17.38	0.000
Supplier*Temp	2	0.28057	0.14028	0.54	0.592
Machine*Temp	2	0.43527	0.21764	0.83	0.448
Error	22	5.7501	0.26137		
Lack-of-Fit	4	1.8355	0.45888	2.11	0.122
Pure Error	18	3.9146	0.21748		
Total	35	133.63			

Adjusted (partial) sums of squares: each term is tested given all other terms in the model.

Pairwise Comparison (Tukey) of Supplier: Thickness

	N	Fitted mean	Grouping
SupA	12	34.91	A
SupB	12	32.98	B
SupC	12	32.568	B

Levels that share a letter are not significantly different (family alpha = 0.05). Means are fitted means averaged over the other factors.

Pairwise Comparison (Tukey) of Machine: Thickness

	N	Fitted mean	Grouping
M2	12	34.795	A
M1	12	33.377	B
M3	12	32.287	C

Levels that share a letter are not significantly different (family alpha = 0.05). Means are fitted means averaged over the other factors.

Pairwise Comparison (Tukey) of Temp: Thickness

N	Fitted mean	Grouping
18	34.451	A
18	32.521	B

Levels that share a letter are not significantly different (family alpha = 0.05). Means are fitted means averaged over the other factors.

How to read the results. Supplier, Machine and Temp are all significant, and so is the Supplier*Machine interaction (Sig. 0.000): which machine is best depends on the supplier. The Tukey tables: SupA gives the thickest film (34.9; group A) while SupB and SupC do not differ (group B); all three machines differ (letters A, B, C); Temp 20 gives 1.9 units more than Temp 10. Because Supplier*Machine is significant, look at the combinations (interaction plot, or the regression equation table, which has a column per combination) before choosing a supplier on its average alone. Treating Temp as categorical makes no difference with two levels, but with three or more numeric levels it lets each level have its own mean instead of a straight line.

7.11 Blocked factorial with center points

Example A11

Also known as / search terms: blocking: analysis, block effect, centre points: curvature

When to use it. A factorial run on two days (blocks) with centre points: remove the day effect and test the factors and curvature.

Data. Dataset *shifts* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.

2. In the main dialog, drag A, B, C into **Factors (design columns)**; drag Strength into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).

3. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=A B C RESPONSEVAR=Strength
    ↪ ANALYZE=YES.
```

Output.

Summary of Results: Strength

	Finding
Overall	The model is statistically significant (Sig. 0.000): the factors explain real changes in Strength.
Model fit	The model explains 98.9% of the variation in Strength (adjusted 98.1%). Predicted R-squared 96.6%: it should predict new runs well.
Significant terms	3 of 6 terms are significant at alpha = 0.05: A, B, A*B (strongest first). Not significant: C, A*C, B*C.
Largest effects	Averaged over the other factors: A from -1 to 1 changes Strength by 5.96 (the size depends on B: see the interaction plot); B from -1 to 1 changes Strength by -1.815 (the size depends on A: see the interaction plot).
Curvature	The center points differ from the corners (Sig. 0.000): the response curves inside the region. A response surface design (e.g. augment to a central composite) is needed to model it.
Lack of fit	No evidence of lack of fit (Sig. 0.902).
Residuals	Residuals are consistent with a normal distribution (Anderson-Darling Sig. 0.364).

Plain-language reading of the tables below. Decisions use the alpha level shown; check the charts before acting.

Coded Coefficients: Strength

	Effect	Coefficient	SE Coef.	t	Sig.	VIF
Constant		20.392	0.10736	189.95	0.000	
Block 1		-0.43415	0.096023	-4.52	0.001	1.00
A	5.9598	2.9799	0.10736	27.76	0.000	1.00
B	-1.8152	-0.90762	0.10736	-8.45	0.000	1.00
C	0.10525	0.052625	0.10736	0.49	0.634	1.00
A*B	1.7145	0.85725	0.10736	7.99	0.000	1.00
A*C	-0.011	-0.0055	0.10736	-0.05	0.960	1.00
B*C	-0.136	-0.068	0.10736	-0.63	0.539	1.00
Center Point		1.7808	0.24006	7.42	0.000	1.00

Coded units. Effect = 2 x coefficient. Error df = 11; terms with Sig. < 0.05 are significant at alpha = 0.05.

Analysis of Variance: Strength

	df	Adj. SS	Adj. MS	F	Sig.
Model	8	181.05	22.631	122.72	0.000
Blocks	1	3.7697	3.7697	20.44	0.001
Linear	3	155.3	51.766	280.72	0.000
A	1	142.07	142.07	770.44	0.000
B	1	13.181	13.181	71.47	0.000
C	1	0.04431	0.04431	0.24	0.634
2-Way Interactions	3	11.833	3.9442	21.39	0.000
A*B	1	11.758	11.758	63.76	0.000
A*C	1	0.000484	0.000484	0.00	0.960
B*C	1	0.073984	0.073984	0.40	0.539
Curvature	1	10.147	10.147	55.03	0.000
Error	11	2.0285	0.18441		
Lack-of-Fit	9	1.2093	0.13437	0.33	0.902
Pure Error	2	0.81915	0.40957		
Total	19	183.08			

Adjusted (partial) sums of squares: each term is tested given all other terms in the model.

Next Steps

Curvature. The center points differ from the corners for Strength (Sig. 0.000): the response bends inside the region, so a two-level model cannot locate the best settings.

→ *Add axial (star) points and more center points to turn this design into a central composite design, run the new rows, then fit the full quadratic model.*

```
STATS DOE ANALYSIS GENERATEDESIGN=NO AUGMENTTOCCD=YES AUGMENTCENTERPTS=4
    ↪ FACTORVARS=A B C RESPONSEVAR=Strength
  /SAVE DATASET=design_ccd.
```

Inactive factors. C: no term containing this factor is significant for any response (alpha = 0.05).

→ *Refit without them to check the simpler model; in the next experiment fix them at the most convenient or cheapest level and spend the runs on the active factors. Not significant is not proof of no effect: with few runs a small effect can be missed.*

```
STATS DOE ANALYSIS GENERATEDESIGN=NO VARNAMES=A B C RESPONSEVAR=Strength
  LOWS=-1 -1 -1 HIGHS=1 1 1
  ANALYZE=YES
  /FITMODEL TERMS="Strength: A B A*B CTPT" SELECTION=NONE CENTERTERM=YES
    ↪ ALPHA=0.05 CONFLEVEL=95.
```

Generated from the results above. Each syntax can be pasted into a syntax window; dataset names such as design_ccd are suggestions. Judgement still applies: check that each step makes sense for your process.

How to read the results. The Blocks row (Sig. 0.001) shows that Day1 and Day2 differed: Block 1 is 0.43 units below the average. Had the blocks been ignored, this difference would have been

part of the error, making the factor tests less sensitive. A (effect 5.96), B (−1.82) and A*B (1.71) are active; C has no effect. Curvature is significant (centre-point coefficient 1.78). The Next Steps table recommends augmenting to a central composite design (all factors are numeric, so the augmentation syntax is given) and refitting without C.

7.12 Factors not named: detected from the data

Example A12

Also known as / search terms: factor detection, factors used, design information

When to use it. Analyze a dataset without naming the factors: the procedure uses the factor list saved with the design or, as here, detects the factor columns from the data.

Data. Dataset *machining* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.

2. In the main dialog, drag Life into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).

3. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO RESPONSEVAR=Life ANALYZE=YES.
```

Output.

Factors Used

	Type	Settings in the data
Speed	Numeric	66.36, 80, 100, 120, 133.64
Feed	Numeric	0.1159, 0.15, 0.2, 0.25, 0.2841
Depth	Numeric	1.159, 1.5, 2, 2.5, 2.841

VARNAMES was not given: these columns were detected as factors (2 to 7 distinct settings; order, block, center-point and response columns excluded). Check the list; if it is wrong, name the factors with VARNAMES.

Summary of Results: Life

	Finding
Overall	The model is statistically significant (Sig. 0.000): the factors explain real changes in Life.
Model fit	The model explains 98.4% of the variation in Life (adjusted 96.9%). Predicted R-squared 90.9%: it should predict new runs well.
Significant terms	4 of 9 terms are significant at alpha = 0.05: Speed, Feed, Speed*Speed, Depth*Depth (strongest first). Not significant: Depth, Feed*Feed, Speed*Feed, Speed*Depth, Feed*Depth.
Lack of fit	No evidence of lack of fit (Sig. 0.245).
Residuals	Residuals are consistent with a normal distribution (Anderson-Darling Sig. 0.803); 1 run(s) with a large standardized residual (> 2).

Plain-language reading of the tables below. Decisions use the alpha level shown; check the charts before acting.

Coded Coefficients: Life

	Coefficient	SE Coef.	t	Sig.	VIF
Constant	50.149	0.56949	88.06	0.000	
Speed	-8.2896	0.37782	-21.94	0.000	1.00
Feed	-3.0569	0.37782	-8.09	0.000	1.00
Depth	-0.44186	0.37782	-1.17	0.269	1.00
Speed*Speed	-2.0212	0.36775	-5.50	0.000	1.02
Feed*Feed	0.0052157	0.36775	0.01	0.989	1.02
Depth*Depth	1.1352	0.36775	3.09	0.012	1.02
Speed*Feed	-1.0522	0.49367	-2.13	0.059	1.00
Speed*Depth	0.007	0.49367	0.01	0.989	1.00
Feed*Depth	0.1325	0.49367	0.27	0.794	1.00

Coded units. Effect = 2 x coefficient. Error df = 10; terms with Sig. < 0.05 are significant at alpha = 0.05.

How to read the results. The *Factors Used* table lists Speed, Feed and Depth, each with five settings (the CCD levels), and explains the rule: columns with 2–7 distinct settings, excluding order, block, centre-point and response columns. Always check this table; when the guess is wrong, name the factors. Every design generated by the procedure stores its factor list in the dataset (data file attribute `DOE_Factors`), so analyses of your own designs need no factor list at all. Tool life is driven by Speed (coefficient −8.29) and Feed (−3.06) with curvature in Speed and Depth.

7.13 Mixture analysis (Scheffe model) with trace plot

Example A13

Also known as / search terms: mixture design: analysis, Scheffé model, response trace plot, formulation optimization, blending effects

When to use it. Model a property of a blend from the proportions of its components and find the best blend.

Data. Dataset *blend* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag Polymer, Plasticizer, Filler into **Factors (design columns)**; drag Elongation into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Analysis]**: tick **Optimize Response**; **Optimization Goal**: Maximize. Click **[Continue]**.
4. Click **[Mixture/Taguchi...]**: **Mixture model**: select *Quadratic*. tick **Response trace plot**. Click **[Continue]**.
5. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=Polymer Plasticizer Filler
   ↪ RESPONSEVAR=Elongation ANALYZE=YES
  OPTIMIZERESPONSE=YES OPTIMIZATIONGOAL=MAXIMIZE
  /FITMODEL MIXMODEL=QUADRATIC TRACEPLOT=YES.
```

Output.

Summary of Results: Elongation

	Finding
Model fit	The blend model explains 99.1% of the variation in Elongation; predicted R-squared 90.7%.
Components	The pure components give different responses (Linear Sig. 0.000).
Blending	Significant non-linear blending: Polymer*Plasticizer (the mix behaves differently from a simple average of its parts).
Lack of fit	No evidence of lack of fit (Sig. 0.362).
Residuals	Residuals are not normal (Anderson-Darling Sig. 0.020).

Plain-language reading of the tables below. Decisions use the alpha level shown; check the charts before acting.

Mixture Regression Coefficients (quadratic model): Elongation

	Coefficient	SE Coef.	t	Sig.
Polymer	9.8073	0.2564		
Plasticizer	14.212	0.2564		
Filler	7.7453	0.2564		
Polymer*Plasticizer	8.038	1.5205	5.29	0.006
Polymer*Filler	1.967	1.5205	1.29	0.265
Plasticizer*Filler	-0.21803	1.5205	-0.14	0.893

Scheffe model without a constant. A linear coefficient is the predicted response of the pure component, so its t-test (against 0) is not shown; the Linear row of the ANOVA tests whether the components blend differently.

Analysis of Variance for Mixture: Elongation

	df	Adj. SS	Adj. MS	F	Sig.
Regression	5	57.205	11.441	86.68	0.000
Linear	2	43.645	21.822	165.34	0.000
Quadratic	3	3.9622	1.3207	10.01	0.025
Polymer*Plasticizer	1	3.6883	3.6883	27.95	0.006
Polymer*Filler	1	0.22087	0.22087	1.67	0.265
Plasticizer*Filler	1	0.0027138	0.0027138	0.02	0.893
Error	4	0.52793	0.13198		
Lack-of-Fit	1	0.14647	0.14647	1.15	0.362
Pure Error	3	0.38146	0.12715		
Total	9	57.733			

Optimal Blend (composite D = 1.0000)

	Amount	Proportion
Polymer	0.20385	0.20385
Plasticizer	0.79573	0.79573
Filler	0.00041144	0.00041144

Search restricted to the range of each component in the data (process variables at their middle setting).

Predicted Responses at the Optimal Blend

	Fit	SE Fit	95% CI	95% PI	Desirability d
Elongation	14.615	0.2553	(13.906, 15.324)	(13.382, 15.848)	1.0000

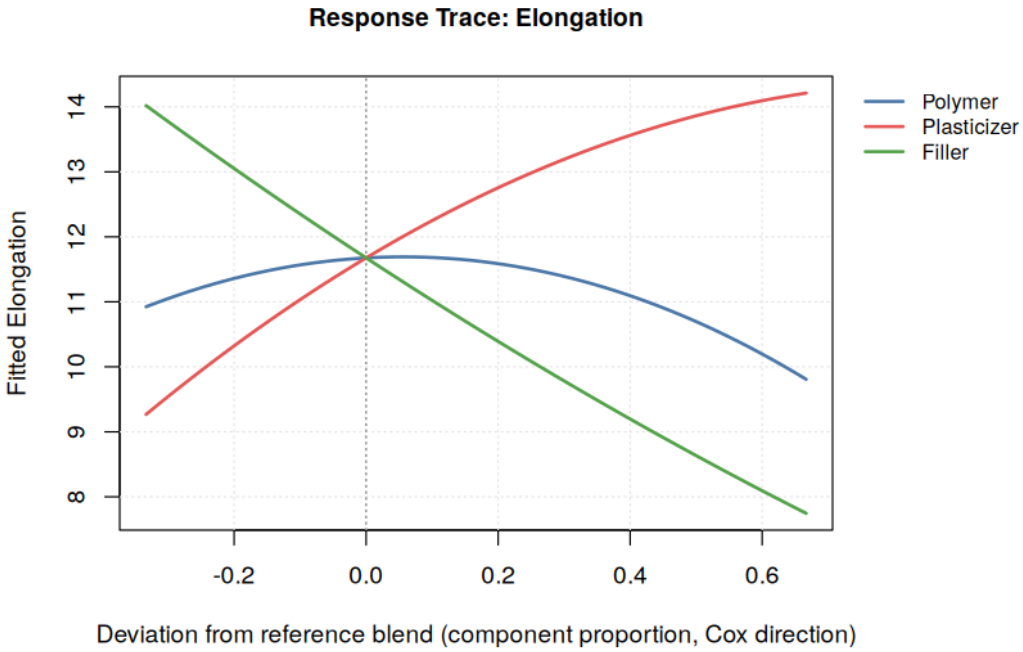


Figure 7.12: Response trace: effect of changing each component along the Cox direction from the reference blend.

How to read the results.

- In the Scheffé quadratic model the linear coefficients are the predicted responses of the pure components: Plasticizer 14.2, Polymer 9.8, Filler 7.7. They have no t-test because testing "pure Polymer gives 0" makes no sense; the *Linear* row of the ANOVA (Sig. 0.000) tests whether the pure components differ.
- Polymer*Plasticizer (coefficient 8.04, Sig. 0.006) is a synergistic blend: the 50:50 mix is better than the average of the two pure components by $8.04/4 = 2.0$ units.
- The trace plot shows Plasticizer increasing elongation, Filler decreasing it, and Polymer having a maximum.
- The optimizer suggests about 20% Polymer, 80% Plasticizer, no Filler, with predicted elongation 14.6 (prediction interval 13.4 to 15.8).
- Residuals are not normal (Sig. 0.020); with 10 runs and 4 error degrees of freedom, treat the p-values as approximate.

7.14 Taguchi signal-to-noise analysis

Example A14

Also known as / search terms: Taguchi analysis, signal-to-noise ratio, robust parameter design: analysis, larger is better

When to use it. Choose settings that give a high response that is also insensitive to noise: each L9 run was measured under three noise conditions (Y1–Y3).

Data. Dataset *taguchi* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag A, B, C, D into **Factors (design columns)**; drag Y1, Y2, Y3 into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Create Design...]**: **Design Type**: select *Taguchi Design – k≥1*. Click **[Continue]**.
4. Click **[Mixture/Taguchi...]**: **Signal-to-noise ratio (Taguchi)**: select *Larger is better*. Click **[Continue]**.
5. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO DESIGNTYPE=TAGUCHI FACTORVARS=A B C D
  RESPONSEVAR=Y1 Y2 Y3 ANALYZE=YES
  /FITMODEL SNTYPE=LARGER.
```

Output.

Summary of Results: Taguchi

	Finding
Signal-to-noise	A has the largest effect on the S/N ratio (delta 1.046), then B, C, D. Higher S/N = more robust (less sensitive to noise).
Best settings	For the highest S/N: A = 3, B = 2, C = 2, D = 2.

Plain-language reading of the tables below. Decisions use the alpha level shown; check the charts before acting.

Response Table for Signal-to-Noise Ratios

	A	B	C	D
1	25.781	25.693	25.978	26.165
2	25.84	26.422	26.398	26.189
3	26.827	26.333	26.072	26.095
Delta	1.046	0.7296	0.4198	0.09405
Rank	1	2	3	4

Larger is better: -10 log10(mean(1/Y^2)). Rank 1 = factor with the largest effect on S/N. Levels: A: 1 = 1, 2 = 2, 3 = 3; B: 1 = 1, 2 = 2, 3 = 3; C: 1 = 1, 2 = 2, 3 = 3; D: 1 = 1, 2 = 2, 3 = 3

Response Table for Means

	A	B	C	D
1	19.489	19.294	19.971	20.441
2	19.602	20.968	20.96	20.402
3	21.971	20.8	20.131	20.219
Delta	2.482	1.673	0.9889	0.2222
Rank	1	2	3	4

Response Table for Standard Deviations

	A	B	C	D
1	0.21406	0.29922	0.24683	0.28245
2	0.42215	0.29492	0.30873	0.33239
3	0.29088	0.33295	0.37153	0.31225
Delta	0.2081	0.03803	0.1247	0.04994
Rank	1	4	2	3

Predicted Result at the Best Settings

	A	B	C	D	Predicted S/N	Predicted mean
Best S/N	3	2	2	2	27.388	23.239

Each factor set to the level with the highest mean S/N; prediction assumes the factor effects add (no interactions). For nominal-is-best, use a factor with a large effect on the mean but little effect on S/N to adjust the mean to target, then confirm with a verification run.

How to read the results. For *larger is better* the signal-to-noise ratio is $-10\log_{10}(\text{mean of } 1/Y^2)$: higher is better, and it rewards both a high mean and a small spread. A has the largest effect on S/N (delta 1.05, rank 1): level 3 gives 26.83 against 25.78 at level 1. The best level of each factor (highest S/N) is A3 B2 C2 D2, predicted S/N 27.4 and mean 23.2. D hardly matters (delta 0.09), so set it on cost. The prediction assumes no interactions; confirm it with runs at A3 B2 C2 D2, which is not one of the nine runs.

7.15 Split-plot analysis (REML)

Example A15

Also known as / search terms: split-plot design: analysis, REML, variance components, whole-plot error, mixed model

When to use it. Analyze an experiment with a hard-to-change factor correctly: Oven was set once per whole plot, so its effect must be tested against the variation between whole plots, not between runs.

Data. Dataset *ovens* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag Oven, Press, Mix into **Factors (design columns)**; drag Y into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Model...]**: **Model terms**: type `Oven Press Mix Oven*Press Oven*Mix Press*Mix`. Click **[Continue]**.
4. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=Oven Press Mix RESPONSEVAR=Y
    ↪ ANALYZE=YES
  /FITMODEL TERMS="Oven Press Mix Oven*Press Oven*Mix Press*Mix".
```

Output.

Variance Components (Split-Plot): Y

	Variance	% of total	StdDev
Whole plot	0.69666	89.7	0.83466
Subplot (error)	0.080186	10.3	0.28317
Total	0.77684	100.0	0.88139

Estimated by REML from 6 whole plots and 24 runs. Whole-plot error df = 4, subplot error df = 13.

Coded Coefficients (Split-Plot): Y

	Effect	Coefficient	SE Coef.	df	t	Sig.	Stratum
Constant		22.137	0.3456	4	64.05	0.000	
Oven	2.6966	1.3483	0.3456	4	3.90	0.018	Whole plot
Press	1.3379	0.66894	0.0578	13	11.57	0.000	Subplot
Mix	0.60477	0.30238	0.0578	13	5.23	0.000	Subplot
Oven*Press	0.12927	0.064633	0.0578	13	1.12	0.284	Subplot
Oven*Mix	-0.13442	-0.067208	0.0578	13	-1.16	0.266	Subplot
Press*Mix	0.15733	0.078667	0.0578	13	1.36	0.197	Subplot

Generalized least squares estimates with REML variance components. Whole-plot terms are tested against whole-plot error, subplot terms against subplot error.

Tests of Fixed Effects (Split-Plot): Y

	Stratum	df	Error df	F	Sig.
Oven	Whole plot	1	4	15.22	0.018
Press	Subplot	1	13	133.93	0.000
Mix	Subplot	1	13	27.37	0.000
Oven*Press	Subplot	1	13	1.25	0.284
Oven*Mix	Subplot	1	13	1.35	0.266
Press*Mix	Subplot	1	13	1.85	0.197

Wald F tests. An ordinary analysis that ignores the whole plots would test the hard-to-change factors against the smaller subplot error and overstate their significance.

How to read the results.

- **Variance components** (REML): the whole-plot variance (0.70) is 90% of the total: oven setups differ much more than runs within a setup (0.08).
- **Tests.** Oven is a whole-plot factor, tested with 4 error degrees of freedom: Sig. 0.018. Press and Mix are subplot factors, tested against the small subplot error with 13 df: both highly significant. The interactions are not significant.
- An ordinary analysis would have used one error term for everything, testing Oven far too optimistically (the subplot standard error 0.058 instead of 0.346). The procedure recognizes the `WholePlot` column automatically; name another column with `WHOLEPLOT` if needed.
- The Lab of the HTML report is not available for split-plot analyses, because it refits ordinary least-squares models.

7.16 Analyze variability (dispersion effects)

Example A16

Also known as / search terms: analyze variability, dispersion effects, robust design: variability, location and dispersion

When to use it. Some factors change the *spread* of the response rather than (or as well as) its average. With replicated settings the procedure models the log standard deviation and finds those factors: the basis for making a process robust.

Data. Dataset *spread* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag A, B into **Factors (design columns)**; drag Y into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Model...]**: tick **Analyze variability (factors that change the spread)**. Click **[Continue]**.
4. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=A B RESPONSEVAR=Y ANALYZE=YES
  /FITMODEL VARIABILITY=YES.
```

Output.

Summary of Results: Y

	Finding
Overall	The model is not statistically significant (Sig. 0.141): the factors studied do not clearly change Y at alpha = 0.05.
Model fit	The model explains 17.4% of the variation in Y (adjusted 8.6%). Predicted R-squared 0.0%: the model predicts new runs poorly; treat predictions with caution.
Significant terms	0 of 3 terms are significant at alpha = 0.05. Not significant: A, B, A*B.
Residuals	Residuals are consistent with a normal distribution (Anderson-Darling Sig. 0.536); 2 run(s) with a large standardized residual (> 2).

Plain-language reading of the tables below. Decisions use the alpha level shown; check the charts before acting.

Analysis of Variability: Y

	Coef (ln StdDev)	SE Coef	Z	Sig.	Ratio of StdDev
Constant	0.41834	0.1437	2.91	0.004	
A	-0.01904	0.1437	-0.13	0.895	0.9626
B	0.36099	0.1437	2.51	0.012	2.058

Model for ln(StdDev) from 4 replicated settings (0 single runs not used). Ratio of StdDev = how many times larger the spread is at the high level than at the low level (coded -1 to +1). Sig. < 0.05: the factor changes the variability.

Tests of Effects on Variability: Y

	df	Chi-square	Sig.
A	1	0.018	0.895
B	1	6.311	0.012
Residual (lack of fit)	1	1.038	0.308

Wald chi-square tests on the log-variance scale (variances of the log variances are known from theory for normal data). A significant residual chi-square means the spread differs in ways the model does not explain.

Standard Deviation by Setting: Y

A	B	N	StdDev	Fitted StdDev
10	1	8	0.86521	1.0794
10	2	8	2.3868	2.2219
20	1	8	1.1161	1.0391
20	2	8	1.7145	2.1389

Observed and fitted standard deviation at each replicated setting.

How to read the results.

- The model for the *mean* finds nothing (Sig. 0.141): neither A nor B moves the average.
- The model for the *spread* does: B has Sig. 0.012 and a ratio of standard deviations of 2.06: at B = 2 the run-to-run standard deviation is about twice as large as at B = 1. A has no effect on the spread (ratio 0.96).
- The observed standard deviations confirm it: 0.87 and 1.12 at B = 1, 2.39 and 1.71 at B = 2.
- Conclusion: run at B = 1. A mean-only analysis would have missed this completely.

7.17 Uncertainty map

Example A17

Also known as / search terms: uncertainty map, prediction interval: half-width, extrapolation, model reliability

When to use it. See where in the factor region the model's predictions are reliable and where they are not, before trusting an optimum near the edge.

Data. Dataset *machining* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag Speed, Feed, Depth into **Factors (design columns)**; drag Rough into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Output]**: tick **Uncertainty map (where predictions are least reliable)**. Click **[Continue]**.
4. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=Speed Feed Depth RESPONSEVAR=
    ↪ Rough ANALYZE=YES
  /OUTPUT UNCERTAINTYMAP=YES.
```

Output.

Prediction Uncertainty: Rough

	Value
S (residual standard deviation)	0.14065
Half-width at the center	0.33844
Half-width at the runs: smallest	0.33844
Half-width at the runs: largest	0.40494
Half-width in the region: largest	0.7117
Limit (largest at the runs)	0.40494
Share of the region above the limit	43.33%

Half-width of the 95% prediction interval for a single new run (+/- this amount around the prediction). Settings where it exceeds the limit are where the model is least reliable: confirm there with extra runs before relying on predictions. The region is the range spanned by the design; other factors at their center / first level.

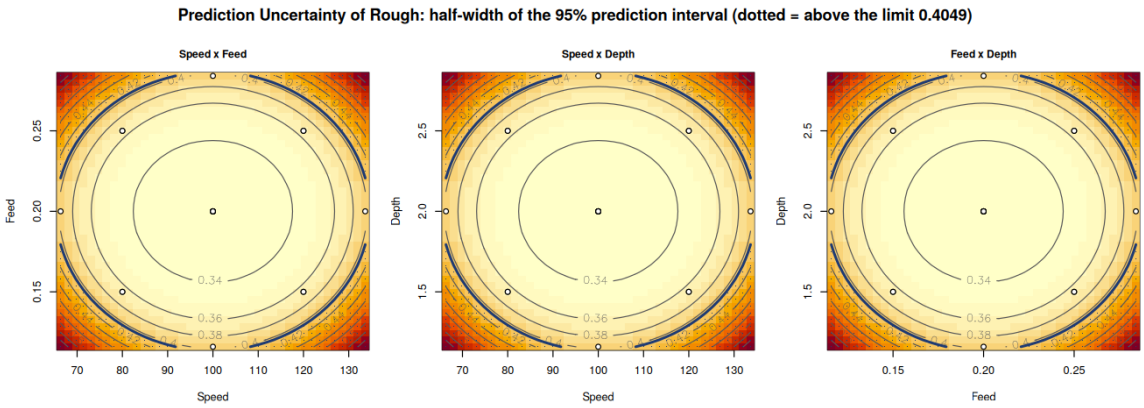


Figure 7.13: Uncertainty map in the SPSS output (the HTML report has the interactive version, Figure 11.5).

How to read the results. The half-width of the 95% prediction interval for one new run is ± 0.34 at the centre and up to ± 0.40 at the runs of the experiment; the limit is set to that largest value. Over the whole region (including the corners beyond the axial points) it reaches ± 0.71, and 43% of the region is less certain than any tested point. Light areas on the map are well supported by data; dark areas, typically the corners of a CCD, are where predictions are least reliable. If the optimizer's best settings fall in a dark area, confirm them with extra runs before relying on them.

7.18 Legacy raw-unit analysis with variable selection

Example A18

Also known as / search terms: legacy model, raw-unit regression, variable selection: AIC, VIF, Lenth's method: legacy table

When to use it. The earlier (2.x) analysis in raw units, kept for compatibility: AIC-based stepwise selection, VIF, residual tests and the effects table.

Data. Dataset *reactor* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag Temp, Time, Press into **Factors (design columns)**; drag Yield into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Analysis]**: tick **Perform Variable Selection**. **Selection Method**: select *Stepwise*. **Stepwise Direction**: select *Both*. tick **Selection Criterion Trace**. Click **[Continue]**.
4. Click **[Model...]**: **Analysis model**: select *Legacy (raw units, all interactions)*. Click **[Continue]**.
5. Click **[Output]**: tick **Effects & Significance Table**. tick **Residual Normality & Autocorrelation Tests**. tick **Multicollinearity (VIF) Table**. Click **[Continue]**.
6. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=Temp Time Press RESPONSEVAR=
    ↪ Yield ANALYZE=YES
  VARSELECT=YES VARSELMETHOD=STEPWISE STEPDIR=BOTH SELECTIONTRACE=YES
  /FITMODEL ANALYSISMODEL=LEGACY
  /OUTPUT EFFECTSTABLE=YES RESIDUALTESTS=YES VIFTABLE=YES.
```

Output.

Variable Selection: Stepwise Steps

Step	Action	Df	AIC
1	start model		48.194
2	- Temp:Time:Press	1	46.227
3	- Temp:Press	1	44.255
4	- Time:Press	1	42.65

AIC-based stepwise selection | Direction: both | Final model: Yield ~ Temp + Time + Press + Temp:Time

Effects & Significance (Lenth's Method)

	Effect	Estimate	|Effect|	Significant (ME)	Significant (SME)
Temp	Temp	9.5465	9.5465	No	No
Time	Time	6.3378	6.3378	No	No
Temp:Time	Temp:Time	5.1662	5.1662	No	No
Press	Press	-4.6442	4.6442	No	No
Time:Press	Time:Press	-0.7075	0.7075	No	No
Temp:Time:Press	Temp:Time:Press	-0.202	0.202	No	No
Temp:Press	Temp:Press	0.1897	0.1897	No	No

Effects in coded units (numeric factors -1 to +1; effect = 2 x coefficient). PSE=6.9664, Margin of Error=26.2223, Simultaneous MoE=62.7552 (Lenth, 1989, Technometrics). Useful when residual degrees of freedom are too low for standard t/F tests.

Multicollinearity: Variance Inflation Factors

Term	VIF
Temp	25
Time	250
Press	250
Temp:Time	270
Temp:Press	270
Time:Press	450
Temp:Time:Press	466

VIF ~= 1 indicates an orthogonal (well-balanced) design. VIF > 5 indicates moderate, VIF > 10 indicates severe multicollinearity among model terms.

Residual Diagnostics: Normality & Autocorrelation

	Value
Shapiro-Wilk W	0.9139
Shapiro-Wilk p-value	0.0755
Durbin-Watson Statistic	0.4858

Shapiro-Wilk tests normality of residuals (p<0.05 suggests non-normality). Durbin-Watson tests autocorrelation (~2 = none). Possible positive autocorrelation

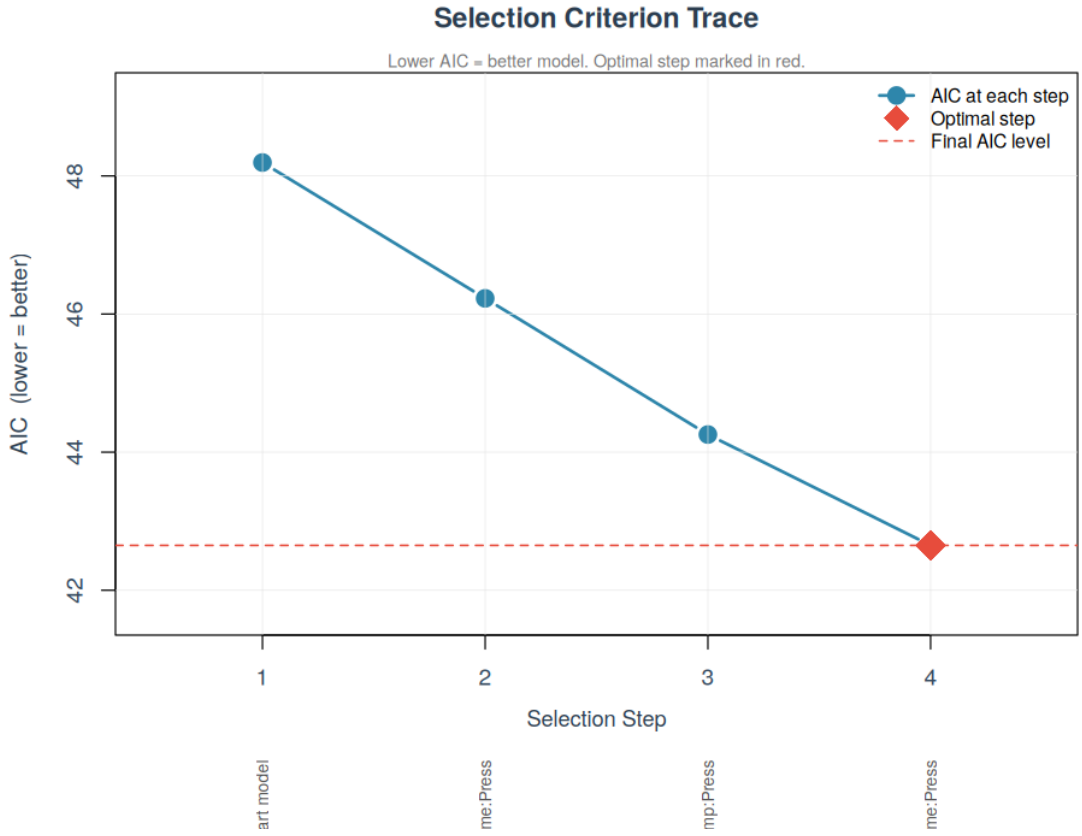


Figure 7.14: AIC at each selection step.

How to read the results.

- Stepwise selection by AIC removed the three-factor and two two-factor interactions; the final model is Yield ~ Temp + Time + Press + Temp:Time, the same terms the coded analysis found significant (Catalyst was not in this model's factor list).
- **VIF** values of 25 to 466 are not a design problem: in raw units the interaction columns (Temp × Time) are strongly correlated with their main effects. In coded units (the default analysis) all VIFs are 1. This is one reason the coded model is recommended.

- The effects table now reports effects in coded units (Temp 9.55, Time 6.34, as in Example 7.1). Lenth's method flags none of them because 4 of the 7 effects are large: the method assumes that most effects are negligible (effect sparsity), which is not the case here. With 12 error degrees of freedom, use the ANOVA instead.

Chapter 8

Optimizing responses

Once the models fit, the optimizer finds the factor settings that best meet your goals.

Desirability. Each response gets a goal: *Maximize*, *Minimize* or *Target*. Its predicted value is turned into a desirability d between 0 (unacceptable) and 1 (fully satisfactory):

- Maximize: $d = 0$ at or below the lower limit, $d = 1$ at or above the target, in between $d = \left(\frac{y-L}{T-L}\right)^w$.
- Minimize: $d = 1$ at or below the target, $d = 0$ at or above the upper limit, in between $d = \left(\frac{U-y}{U-T}\right)^w$.
- Target: $d = 1$ at the target, falling to 0 at the lower and upper limits.

The weight w (0.1–10) shapes the curve: 1 is a straight line, larger values demand being close to the target. The composite desirability is the weighted geometric mean $D = \left(\prod d_i^{r_i}\right)^{1/\sum r_i}$ with importances r_i. Because it is a geometric mean, a single unacceptable response ($d = 0$) makes $D = 0$.

In the dialog. Tick Optimize Response in **[Analysis]** and choose the goal (one response) or type one goal per response in Optimization goals (multi-response). Limits, targets, weights, importance, held factors, the number of solutions, confirmation runs, overlay and robustness are in **[Optimizer...]**. Blank limits default to the observed range of each response.

Profit or cost. Instead of desirability, Profit / cost formula in **[Optimizer...]** optimizes any formula of the predicted responses and the factor settings, with optional requirements (Examples 8.4, 8.5).

Where the optimizer searches. Inside the region covered by the data (axial points included; for mixtures, within the component ranges). When a model contains the centre-point term, only the cube corners and the centre are evaluated, because that model is not defined between them (Example 8.3).

8.1 Single response: minimize surface roughness

Example O01

Also known as / search terms: response optimizer, desirability function, minimize response, optimal settings

When to use it. Find the factor settings that give the smallest surface roughness.

Data. Dataset *machining* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.

2. In the main dialog, drag Speed, Feed, Depth into **Factors (design columns)**; drag Rough into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).

3. Click **[Analysis]**: tick **Optimize Response**; **Optimization Goal**: Minimize. Click **[Continue]**.

4. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=Speed Feed Depth RESPONSEVAR=
    ↪ Rough ANALYZE=YES
  OPTIMIZERESPONSE=YES OPTIMIZATIONGOAL=MINIMIZE.
```

Output.

Summary of Optimization

	Finding
Best settings	Speed = 100, Feed = 0.13272, Depth = 1.159.
Desirability	Composite desirability D = 1.000 (1 = every goal fully met, 0 = at least one response unacceptable). The goals are met well.
Next step	Confirm with a few runs at these settings; the prediction intervals below show the range to expect.

Plain-language reading of the tables below. Decisions use the alpha level shown; check the charts before acting.

Optimization Criteria

	Goal	Lower	Target	Upper	Weight	Importance
Rough	Minimize		2.3638	5.7613	1	1

Individual desirability d rises from 0 at the lower (or upper) bound to 1 at the target; weight shapes the curve; composite D is the importance-weighted geometric mean of all d.

Factor Search Region

	Search Region
Speed	[66.36, 133.64]
Feed	[0.1159, 0.2841]
Depth	[1.159, 2.841]

Optimal Settings

Speed	Feed	Depth	Rough fit	Composite D
100	0.13272	1.159	1.961	1.0000

Predicted Responses at the Optimal Settings

	Fit	SE Fit	95% CI	95% PI	Desirability d
Rough	1.961	0.1707	(1.5807, 2.3413)	(1.4683, 2.4538)	1.0000

CI = confidence interval for the mean response; PI = prediction interval for a single new run. Composite D = 1.0000.

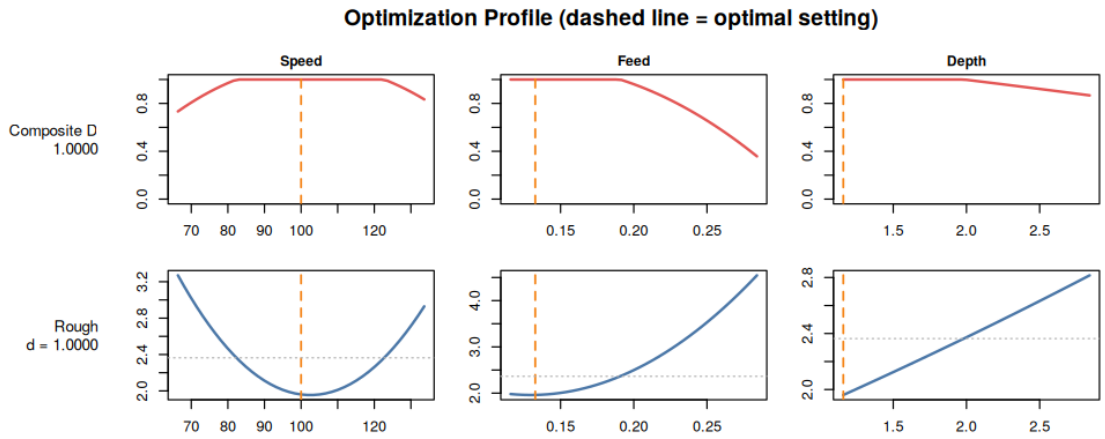


Figure 8.1: Optimization profile: composite desirability and fitted response along each factor; dashed line = optimal setting.

How to read the results. The optimizer searched the region covered by the design, including the axial points (Speed 66.4–133.6). Best settings: Speed 100, Feed 0.133, Depth 1.16, predicted roughness 1.96 μm, below the best observed run (2.36), so the desirability is 1. The 95% prediction interval (1.47, 2.45) is the range for a single run there. Depth is at the lower edge of the searched region (its axial setting 1.16): a still smaller depth might be better, but that lies outside the data, and the prediction uncertainty near the edge is larger (Example 7.17). Confirm with a few runs.

8.2 Several responses: desirability with limits, weights, importance

Example O02

Also known as / search terms: multiple response optimization, desirability function: composite, importance, weights, confirmation runs, robustness, overlaid contour plot, trade-off

When to use it. Two responses with opposite wishes: low roughness and long tool life. The desirability method turns each response into a 0–1 score d and combines them into the composite desirability D.

Data. Dataset *machining* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag Speed, Feed, Depth into **Factors (design columns)**; drag Rough, Life into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Analysis]**: tick **Optimize Response**. **Optimization goals (multi-response)**: type `MINIMIZE MAXIMIZE`. Click **[Continue]**.
4. Click **[Optimizer...]**: **Lower limits**: type `0 40`. **Upper limits**: type `4 0`. **Weights (0.1 - 10)**: type `1 2`. **Importance (0.1 - 10)**: type `1 3`. **Number of solutions to show**: type `3`. **Confirmation runs planned**: type `3`. tick **Robustness check of the optimum**. **Setting error (% of half-range)**: type `5`. tick **Overlaid contour plot**. Click **[Continue]**.
5. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=Speed Feed Depth RESPONSEVAR=
    ↪ Rough Life ANALYZE=YES
  OPTIMIZERESPONSE=YES OPTIMIZATIONGOALS=MINIMIZE MAXIMIZE
  /OPTIMIZE LOWERS=0 40 UPPERS=4 0 WEIGHTS=1 2 IMPORTANCE=1 3 SOLUTIONS=3
    CONFIRMRUNS=3 ROBUST=YES ROBUSTSD=5 OVERLAY=YES.
```

Output.

Optimization Criteria

	Goal	Lower	Target	Upper	Weight	Importance
Rough	Minimize		2.3638	4	1	1
Life	Maximize	40	61.655		2	3

Individual desirability d rises from 0 at the lower (or upper) bound to 1 at the target; weight shapes the curve; composite D is the importance-weighted geometric mean of all d.

Optimal Settings

Speed	Feed	Depth	Rough fit	Life fit	Composite D
90.676	0.1159	1.159	2.0566	62.241	1.0000
86.657	0.1159	1.159	2.1434	63.093	1.0000
86.544	0.13272	1.159	2.2115	62.244	1.0000

Predicted Responses at the Optimal Settings

	Fit	SE Fit	95% CI	95% PI	Desirability d
Rough	2.0566	0.2144	(1.5788, 2.5344)	(1.4852, 2.628)	1.0000
Life	62.241	2.129	(57.498, 66.984)	(56.568, 67.913)	1.0000

CI = confidence interval for the mean response; PI = prediction interval for a single new run. Composite D = 1.0000.

Confirmation Runs (n = 3) at the Optimal Settings

	Predicted	95% interval for the average of 3 runs
Rough	2.0566	(1.5457, 2.5675)
Life	62.241	(57.169, 67.313)

If the average of the confirmation runs falls inside this interval, the model's prediction is confirmed; if not, the model does not describe the process well at these settings.

Robustness of the Optimal Settings

	Mean prediction	SD from setting error	SD incl. run-to-run	90% range of runs	Runs within limits
Rough	2.0635	0.0195	0.138	(1.8347, 2.2921)	100.0%
Life	62.098	0.282	1.43	(59.798, 64.443)	100.0%
Composite D / all responses	0.9994	0.00337			100.0%

2000 simulated runs; each numeric setting varies with SD = 5% of its half-range around the optimum (held factors fixed), plus residual variation (model S). A setting is robust when the SD from setting error is small and most runs stay within the limits.

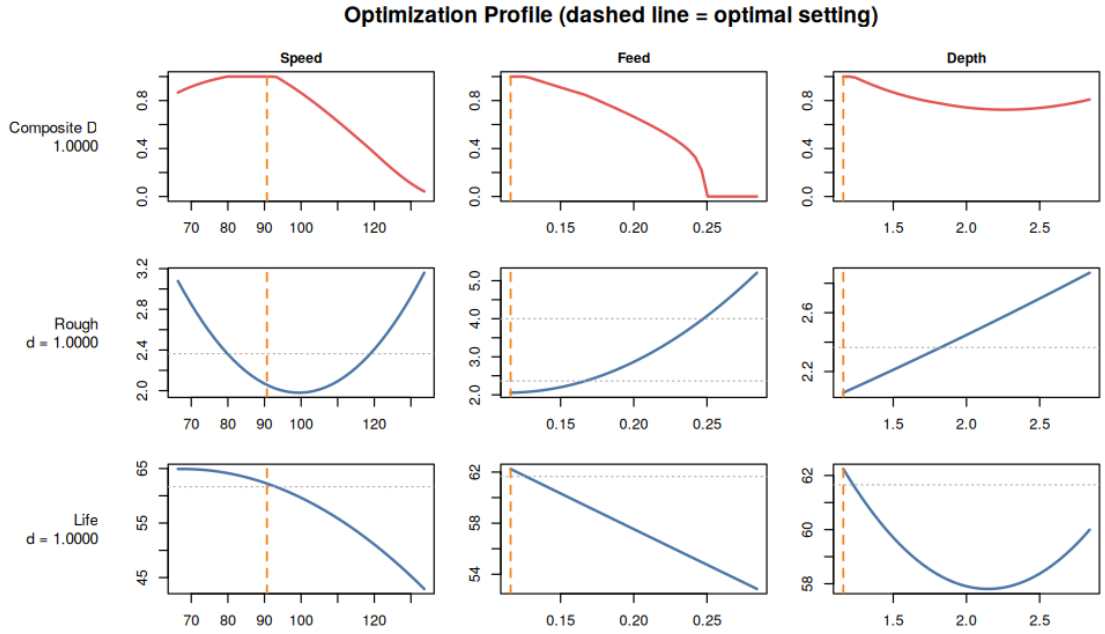


Figure 8.2: Optimization profile for both responses.

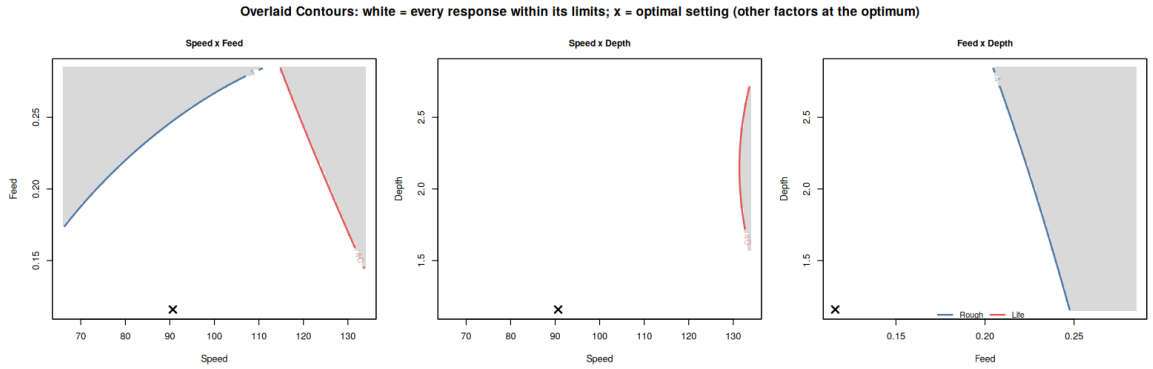


Figure 8.3: Overlaid contours: white = every response within its limits; × = optimum.

How to read the results.

- **Criteria.** Roughness: minimize, acceptable up to 4, fully satisfactory at 2.36 (the best observed). Life: maximize, acceptable from 40, fully satisfactory at 61.7. Life has weight 2 (its d rises slowly until close to the target: a stricter requirement) and importance 3 (it counts three times as much in D).

- **Solutions.** Three distinct settings all reach $D = 1$: there is a region in which both goals are fully met. The best: Speed 90.7, Feed 0.116, Depth 1.16 with roughness 2.06 and life 62.2.

- **Confirmation runs.** If the average of 3 runs at these settings lies within (1.55, 2.57) for roughness and (57.2, 67.3) for life, the model is confirmed.

- **Robustness.** Simulating setting errors of 5% of the half-range: the setting error alone moves roughness by SD 0.02 and life by 0.28, small compared with run-to-run variation (0.14 and 1.43); 100% of simulated runs meet the limits. The optimum is robust.

8.3 Target value and held factor

Example O03

Also known as / search terms: target value, held factor, nominal-the-best, centre-point model restriction

When to use it. Hit a target yield of 64 while keeping cost low, with the catalyst fixed at B.

Data. Dataset *reactor* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag Temp, Time, Press, Catalyst into **Factors (design columns)**; drag Yield, Cost into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Analysis]**: tick **Optimize Response**. **Optimization goals (multi-response)**: type `TARGET MINIMIZE`. **Target values (multi-response)**: type `64 16`. Click **[Continue]**.
4. Click **[Optimizer...]**: **Lower limits**: type `60 0`. **Upper limits**: type `68 22`. **Hold factors at**: type `Catalyst=B`. Click **[Continue]**.
5. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=Temp Time Press Catalyst
    ↪ RESPONSEVAR=Yield Cost ANALYZE=YES
  OPTIMIZERESPONSE=YES OPTIMIZATIONGOALS=TARGET MINIMIZE OPTIMIZATIONTARGETS
    ↪ =64 16
  /OPTIMIZE LOWERS=60 0 UPPERS=68 22 HOLD="Catalyst=B".
```

Output.

Summary of Optimization

	Finding
Best settings	Temp = 175, Time = 20, Press = 2, Catalyst = B.
Desirability	Composite desirability D = 0.429 (1 = every goal fully met, 0 = at least one response unacceptable). The goals conflict; a clear compromise was needed.
Trade-off	Weakest goal: Cost (d = 0.361); best met: Yield (d = 0.510).
Next step	Confirm with a few runs at these settings; the prediction intervals below show the range to expect.

Plain-language reading of the tables below. Decisions use the alpha level shown; check the charts before acting.

Optimization Criteria

	Goal	Lower	Target	Upper	Weight	Importance
Yield	Target	60	64	68	1	1
Cost	Minimize		16	22	1	1

Individual desirability d rises from 0 at the lower (or upper) bound to 1 at the target; weight shapes the curve; composite D is the importance-weighted geometric mean of all d.

Factor Search Region

	Search Region
Temp	[150, 200]
Time	[10, 30]
Press	[1, 3]
Catalyst	held at B

At least one model contains the center-point (curvature) term, which is only defined at the cube corners and the center point; only those settings were evaluated.

Optimal Settings

Temp	Time	Press	Catalyst	Yield fit	Cost fit	Composite D
175	20	2	B	65.96	19.833	0.4291

Predicted Responses at the Optimal Settings

	Fit	SE Fit	95% CI	95% PI	Desirability d
Yield	65.96	0.4983	(64.811, 67.11)	(63.568, 68.352)	0.5099
Cost	19.833	0.2839	(19.178, 20.488)	(18.47, 21.196)	0.3611

CI = confidence interval for the mean response; PI = prediction interval for a single new run. Composite D = 0.4291.

How to read the results.

- **Target goal.** Yield should be 64; d falls to 0 at the limits 60 and 68. Cost should be minimized, $d = 1$ at 16 or below and 0 at 22.
- **Held factor.** Catalyst is held at B (Factor Search Region).
- **Restricted search.** The models contain the centre-point (curvature) term, which is only defined at the corners and the centre, so only those settings were evaluated (see the caption of the Factor Search Region). The best of them is the centre: Yield 66.0 ($d = 0.51$) and Cost 19.8 ($d = 0.36$), $D = 0.43$: a compromise. To search between corners and centre you need a model for the curvature: a response-surface design (Example 6.21), or, if curvature is unimportant for the decision, `CENTERTERM=NO` as in Example 8.5.

8.4 Profit goal with a requirement

Example O04

Also known as / search terms: profit optimization, cost optimization, economic optimization, requirements

When to use it. Optimize what matters to the business directly: here profit = 2 × tool life − 10 × roughness − 0.05 × speed (a speed-dependent energy cost), with the requirement that roughness stays at or below 3.5.

Data. Dataset *machining* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag Speed, Feed, Depth into **Factors (design columns)**; drag Rough, Life into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Optimizer...]**: **Profit / cost formula**: type `2*Life - 10*Rough - 0.05*Speed`. **Formula goal**: select *Maximize (profit)*. **Requirements (optional), e.g. Yield>=60 Cost<=25**: type `Rough<=3.5`. Click **[Continue]**.
4. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=Speed Feed Depth RESPONSEVAR=
    ↪ Rough Life ANALYZE=YES
  /OPTIMIZE PROFIT='2*Life - 10*Rough - 0.05*Speed' PROFITGOAL=MAXIMIZE
    ↪ PROFITLIMITS="Rough<=3.5".
```

Output.

Profit Optimization: Objective

	Setting
Formula	2*Life - 10*Rough - 0.05*Speed
Goal	Maximize
Requirements	Rough <= 3.5
Held factors	none

Responses in the formula are their predicted values in original units; numeric factors are in natural units.

Best Settings by Profit

Speed	Feed	Depth	Rough fit	Life fit	Profit	Meets requirements
82.2679	0.1159	1.159	2.27536	63.8373	100.8075	Yes

Ranked by the formula. The search covers the region spanned by the design (axial points included).

Profit at the Best Settings

	Value
Predicted profit (from mean responses)	100.8075
Median for a single run	100.871
5th percentile for a single run	89.61207
95th percentile for a single run	112.0085

The single-run range simulates each response in the formula from its prediction distribution at these settings (20,000 draws, responses varying independently). If the responses tend to move together, the true range can be wider or narrower.

How to read the results. The formula uses the predicted responses (original units) and the factor settings (natural units). The best settings, Speed 82.3, Feed 0.116, Depth 1.16, give predicted profit 100.8 per unit, meeting the roughness requirement ($2.28 \leq 3.5$). A single run's profit will vary: 90% of runs should fall between 89.6 and 112.0 (simulated from the prediction distributions). Compare with Example 8.2: the profit criterion prefers a lower speed because life is worth twice as much and speed has a cost. Allowed in formulas: response and factor names, numbers, + − * / ^, parentheses, exp, log, log10, sqrt, abs, min, max, ifelse and comparisons; nothing else can be called.

8.5 Cost goal with a categorical factor in the formula

Example O05

Also known as / search terms: cost optimization: with categorical factor, ifelse in formula, minimum yield requirement

When to use it. Minimize cost, including an energy term in temperature and a surcharge of 0.5 for catalyst B, while guaranteeing a yield of at least 62.

Data. Dataset *reactor* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag Temp, Time, Press, Catalyst into **Factors (design columns)**; drag Yield, Cost into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Model...]**: untick **Include center-point (curvature) term**. Click **[Continue]**.
4. Click **[Optimizer...]**: **Profit / cost formula**: type `Cost + 0.01*Temp + ifelse(Catalyst=="B", 0.5, 0)`. **Formula goal**: select *Minimize (cost)*. **Requirements (optional), e.g. Yield>=60 Cost<=25**: type `Yield>=62`. Click **[Continue]**.

5. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=Temp Time Press Catalyst
   ↪ RESPONSEVAR=Yield Cost ANALYZE=YES
  /FITMODEL CENTERTERM=NO
  /OPTIMIZE PROFIT='Cost + 0.01*Temp + ifelse(Catalyst=="B", 0.5, 0)'
   ↪ PROFITGOAL=MINIMIZE
    PROFITLIMITS="Yield>=62".
```

Output.

Profit Optimization: Objective

	Setting
Formula	Cost + 0.01*Temp + ifelse(Catalyst=="B", 0.5, 0)
Goal	Minimize
Requirements	Yield >= 62
Held factors	none

Responses in the formula are their predicted values in original units; numeric factors are in natural units.

Best Settings by Profit

Temp	Time	Press	Catalyst	Yield fit	Cost fit	Profit	Meets requirements
151.808	30	1	B	62.003	16.6147	18.63277	Yes

Ranked by the formula. The search covers the region spanned by the design (axial points included).

Profit at the Best Settings

	Value
Predicted profit (from mean responses)	18.63277
Median for a single run	18.64257
5th percentile for a single run	17.41628
95th percentile for a single run	19.8674

The single-run range simulates each response in the formula from its prediction distribution at these settings (20,000 draws, responses varying independently). If the responses tend to move together, the true range can be wider or narrower.

How to read the results. `CENTERTERM=NO` removes the curvature term so that the search covers the whole region and not only corners and centre. Text factors are used in formulas as quoted text: `ifelse(Catalyst=="B", 0.5, 0)`. The cheapest settings meeting the yield requirement: Temp 151.8, Time 30, Press 1, Catalyst B, with yield exactly at its limit (62.0) and total cost 18.6. Catalyst B is chosen despite its surcharge because it raises yield, allowing a lower (cheaper) temperature. Caution: this dataset has strong curvature, which the model now ignores; the Next Steps table flags the resulting residual problems. Confirm the settings with runs, and consider a response-surface experiment.

Chapter 9

Sequential experimentation and Next Steps

After every analysis (unless `/OUTPUT ADVISOR=NO`) a **Next Steps** table states what the results imply for the next experiment. Each row has a finding, a recommended step and, where possible, the syntax for that step, ready to paste. The rules:

Finding	Recommendation
Curvature at the centre points	augment to a central composite design (syntax when all factors are numeric)
Significant lack of fit	add axial points; or add back terms the design can estimate; or Box–Cox; or narrow the region
Factors with no significant term for any response	refit without them; fix them in the next experiment
Best settings on the edge of the region	a new experiment shifted half a range in that direction
First-order model, no curvature	the path of steepest ascent/descent, step by step
Too few error degrees of freedom	replicate, add centre points, or refit with the active effects
Non-normal residuals, run-order pattern	check flagged runs; Box–Cox; keep randomizing; block by time
Over-fitting (predicted R-squared much below R-squared)	backward elimination
An optimum was found	confirmation runs, robustness check

Syntax that refits a model reproduces it exactly (same coding and terms), so the numbers match the current output. The recommendations follow the evidence; your knowledge of the process still decides.

9.1 Next steps: path of steepest ascent

Example N01

Also known as / search terms: path of steepest ascent, sequential experimentation: steepest ascent, first-order model, Next Steps

When to use it. A first experiment far from the optimum: the response is planar (no curvature), so the fastest way to improve is to move along the gradient. The Next Steps table computes the path.

Data. Dataset *pilot* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.

2. In the main dialog, drag Temp, Time, Speed into **Factors (design columns)**; drag Conversion into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=Temp Time Speed RESPONSEVAR=
    ↪ Conversion ANALYZE=YES.
```

Output.

Summary of Results: Conversion

	Finding
Overall	The model is statistically significant (Sig. 0.001): the factors explain real changes in Conversion.
Model fit	The model explains 99.7% of the variation in Conversion (adjusted 98.9%). Predicted R-squared 93.6%: it should predict new runs well.
Significant terms	2 of 6 terms are significant at alpha = 0.05: Temp, Time (strongest first). Not significant: Speed, Temp*Time, Temp*Speed, Time*Speed.
Largest effects	Averaged over the other factors: Temp from 80 to 100 changes Conversion by 12.02; Time from 30 to 50 changes Conversion by 6.167.
Curvature	No significant curvature (Sig. 0.880): a straight-line (first-order) model is adequate in this region.
Lack of fit	No evidence of lack of fit (Sig. 0.473).
Residuals	Residuals are not normal (Anderson-Darling Sig. 0.043); possible run-order pattern (Durbin-Watson 3.22).

Plain-language reading of the tables below. Decisions use the alpha level shown; check the charts before acting.

Coded Coefficients: Conversion

	Effect	Coefficient	SE Coef.	t	Sig.	VIF
Constant		54.955	0.22421	245.11	0.000	
Temp	12.015	6.0077	0.22421	26.80	0.000	1.00
Time	6.167	3.0835	0.22421	13.75	0.001	1.00
Speed	0.447	0.2235	0.22421	1.00	0.392	1.00
Temp*Time	-0.1105	-0.05525	0.22421	-0.25	0.821	1.00
Temp*Speed	0.2005	0.10025	0.22421	0.45	0.685	1.00
Time*Speed	0.077	0.0385	0.22421	0.17	0.875	1.00
Center Point		-0.070667	0.42933	-0.16	0.880	1.00

Coded units. Effect = 2 x coefficient. Error df = 3; terms with Sig. < 0.05 are significant at alpha = 0.05.

Next Steps

Inactive factors. Speed: no term containing this factor is significant for any response (alpha = 0.05).

→ *Refit without them to check the simpler model; in the next experiment fix them at the most convenient or cheapest level and spend the runs on the active factors. Not significant is not proof of no effect: with few runs a small effect can be missed.*

```
STATS DOE ANALYSIS GENERATEDESIGN=NO VARNAMES=Temp Time Speed RESPONSEVAR=
    ↪ Conversion
  LOWS=80 30 10 HIGHS=100 50 20
  ANALYZE=YES
  /FITMODEL TERMS="Conversion: Temp Time Temp*Time CTPT" SELECTION=NONE
    ↪ CENTERTERM=YES ALPHA=0.05 CONFLEVEL=95.
```

Path of steepest ascent. Conversion follows a first-order (planar) model, so the fastest improvement is along the gradient of the fitted plane.

→ *Run single trials along the path until Conversion stops improving, then run a new factorial around the best point. Path (one coded unit of Temp each step, other factors in proportion to their coefficients): step 1: Temp=100, Time=45.133, Speed=15; step 2: Temp=110, Time=50.265, Speed=15; step 3: Temp=120, Time=55.398, Speed=15; step 4: Temp=130, Time=60.53, Speed=15; step 5: Temp=140, Time=65.663, Speed=15.*

Residuals. Conversion: residuals are not normal (Anderson-Darling Sig. 0.043).

→ *Check the flagged runs for recording errors first; then try a Box-Cox transformation.*

```
STATS DOE ANALYSIS GENERATEDESIGN=NO VARNAMES=Temp Time Speed RESPONSEVAR=
    ↪ Conversion
  LOWS=80 30 10 HIGHS=100 50 20
  ANALYZE=YES
  /FITMODEL TERMS="Conversion: Temp Time Speed Temp*Time Temp*Speed Time*
    ↪ Speed CTPT" SELECTION=NONE CENTERTERM=YES ALPHA=0.05 CONFLEVEL=95
    ↪ BOXCOX=AUTO.
```

Generated from the results above. Each syntax can be pasted into a syntax window; dataset names such as design_ccd are suggestions. Judgement still applies: check that each step makes sense for your process.

How to read the results. Temp (coefficient 6.1) and Time (3.0) increase conversion; Speed does nothing; there is no curvature. Next Steps gives the *path of steepest ascent*: for each coded unit of Temp (10 degrees), Time moves by about half a coded unit, in proportion to their coefficients. Run single trials along the path (step 1: Temp 100, Time 45.1; step 2: Temp 110, Time 50.3; . . . ; Speed stays at its centre, 15) until conversion stops improving, then run a new factorial with centre points around the best point. The table also gives syntax to refit without Speed.

9.2 Next steps after a screening experiment with curvature

Example N02

Also known as / search terms: Next Steps, advisor, sequential experimentation

When to use it. The Next Steps table after a screening experiment with two responses and an optimization: what does the evidence say to do next?

Data. Dataset *reactor* (Appendix A).

Dialog steps.

1. Open **Analyze > Descriptive Statistics > Design of Experiments**.
2. In the main dialog, drag Temp, Time, Press, Catalyst into **Factors (design columns)**; drag Yield, Cost into **Responses (measured results)**; leave **Analyze the responses (model, ANOVA, charts)** ticked (the default).
3. Click **[Analysis]**: tick **Optimize Response**. **Optimization goals (multi-response)**: type `MAXIMIZE MINIMIZE`. Click **[Continue]**.
4. Click **[OK]** (or **[Paste]** to see the syntax first).

Syntax.

```
STATS DOE ANALYSIS GENERATEDESIGN=NO FACTORVARS=Temp Time Press Catalyst
    ↪ RESPONSEVAR=Yield Cost ANALYZE=YES
  OPTIMIZERESPONSE=YES OPTIMIZATIONGOALS=MAXIMIZE MINIMIZE.
```

Output.

Summary of Optimization

	Finding
Best settings	Temp = 200, Time = 30, Press = 1, Catalyst = B.
Desirability	Composite desirability D = 0.707 (1 = every goal fully met, 0 = at least one response unacceptable). The goals are met reasonably; some compromise is needed.
Trade-off	Weakest goal: Cost (d = 0.500); best met: Yield (d = 1.000).
Next step	Confirm with a few runs at these settings; the prediction intervals below show the range to expect.

Plain-language reading of the tables below. Decisions use the alpha level shown; check the charts before acting.

Optimal Settings

Temp	Time	Press	Catalyst	Yield fit	Cost fit	Composite D
200	30	1	B	75.785	19.91	0.7074

Next Steps

Curvature. The center points differ from the corners for Yield (Sig. 0.000): the response bends inside the region, so a two-level model cannot locate the best settings.

→ *Add runs at intermediate levels of the numeric factors (for example a response-surface design) so the curvature can be modelled.*

Residuals. Yield: residuals are not normal (Anderson-Darling Sig. 0.023).

→ *Check the flagged runs for recording errors first; then try a Box-Cox transformation.*

```
STATS DOE ANALYSIS GENERATEDESIGN=NO VARNAMES=Temp Time Press Catalyst
    ↪ RESPONSEVAR=Yield
  LOWS=150 10 1 0 HIGHS=200 30 3 0
  ANALYZE=YES
  /FITMODEL TERMS="Yield: Temp Time Press Catalyst Temp*Time Temp*Press Temp*
    ↪ Catalyst Time*Press Time*Catalyst Press*Catalyst CTPT" SELECTION=NONE
    ↪ CENTERTERM=YES ALPHA=0.05 CONFLEVEL=95 BOXCOX=AUTO.
```

Confirmation. The optimizer found settings with composite desirability D = 0.707.

→ *Run 3 or more confirmation runs at the optimal settings; their average should fall inside the prediction interval.*

```
STATS DOE ANALYSIS GENERATEDESIGN=NO VARNAMES=Temp Time Press Catalyst
    ↪ RESPONSEVAR=Yield Cost
  LOWS=150 10 1 0 HIGHS=200 30 3 0
  ANALYZE=YES OPTIMIZERESPONSE=YES OPTIMIZATIONGOALS=MAXIMIZE MINIMIZE
    ↪ OPTIMIZATIONTARGETS=75.723 15.554
  /FITMODEL TERMS="Yield: Temp Time Press Catalyst Temp*Time Temp*Press Temp*
    ↪ Catalyst Time*Press Time*Catalyst Press*Catalyst CTPT; Cost: Temp Time
    ↪ Press Catalyst Temp*Time Temp*Press Temp*Catalyst Time*Press Time*
    ↪ Catalyst Press*Catalyst CTPT" SELECTION=NONE CENTERTERM=YES ALPHA=0.05
    ↪ CONFLEVEL=95
  /OPTIMIZE LOWERS=51.882 15.554 UPPERS=75.723 24.274 WEIGHTS=1 1 IMPORTANCE
    ↪ =1 1 CONFIRMRUNS=3.
```

Generated from the results above. Each syntax can be pasted into a syntax window; dataset names such as design_ccd are suggestions. Judgement still applies: check that each step makes sense for your process.

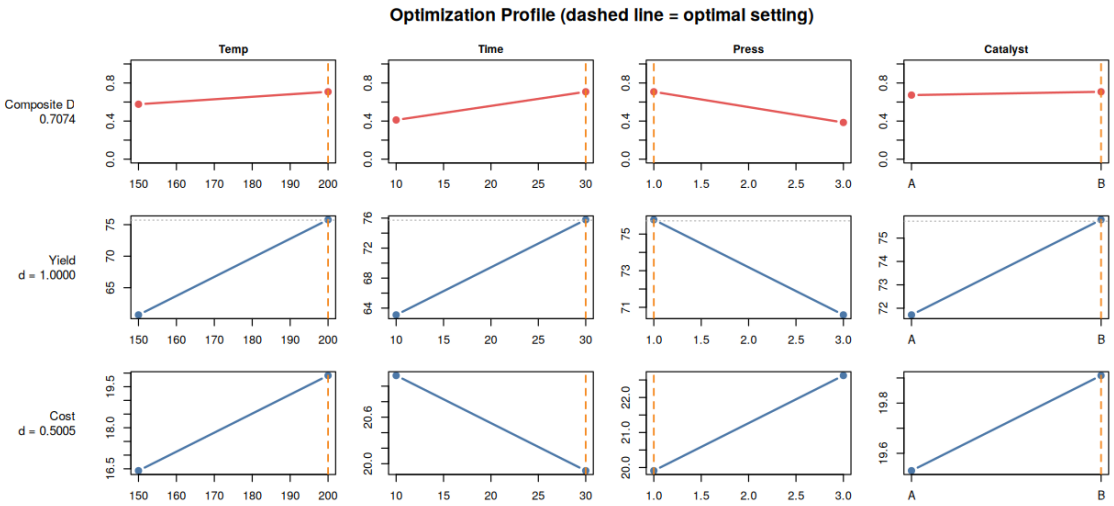


Figure 9.1: Optimization profile.

How to read the results. Three recommendations, each with its reason and syntax: (1) curvature in Yield: the best settings found by a two-level model are not reliable, add intermediate levels; (2) non-normal residuals for Yield: check run 20, then try Box–Cox; (3) confirm the optimum (Temp 200, Time 30, Press 1, Catalyst B: Yield 75.8, Cost 19.9, $D = 0.71$) with 3 or more runs. The syntax reproduces the models exactly and runs the optimizer with the same goals; paste it into a syntax window. The advisor states what the data imply; engineering judgment still decides.

Chapter 10

Complete workflows

Each workflow below chains the worked examples into a complete study. Follow the example numbers for the dialog steps and syntax.

10.1 From screening to optimum (Six Sigma Improve, process development)

1. **Plan**: estimate the run-to-run standard deviation from historical data and the smallest effect worth detecting; check the number of runs with **[Power...]** (Example 6.19).
2. **Screen** 7–15 candidate factors with a Plackett–Burman or definitive screening design (Examples 6.4, 6.5); analyze it with main effects and Lenth's method (Example 7.6).
3. **Characterize** the 3–4 active factors with a full factorial and 3–4 centre points (Example 6.1); analyze effects, interactions and curvature (Example 7.1); simplify the model (Examples 7.2, 7.3).
4. If the Next Steps table reports a **first-order model** with no curvature, follow the path of steepest ascent (Example 9.1) and repeat step 3 near the best point.
5. If it reports **curvature**, augment the factorial to a central composite design (Example 6.21), run the new rows, and fit the quadratic model (Example 7.7).
6. **Optimize** with desirability (Examples 8.1, 8.2) or a profit formula (Example 8.4); check robustness and the uncertainty map (Example 7.17).
7. **Confirm**: run the best settings 3 or more times; their average must lie in the confirmation interval (`CONFIRMRUNS`).

10.2 QbD design space (pharmaceutical development)

1. Identify critical process parameters by risk assessment, then screen them (step 2 above).
2. Run a response-surface design for the critical parameters and all critical quality attributes as responses (Example 6.6 or 6.7).
3. Fit and validate the models (Example 7.7): no lack of fit, high predicted R-squared, residual checks pass.
4. Give each quality attribute its acceptance limits in **[Optimizer...]** and tick Overlaid contour plot: the white region meets every limit (Example 8.2).

5. Inspect the uncertainty map (Example 7.17): the design space should be drawn where predictions are reliable; the robustness simulation estimates how many future runs fall inside the limits.
6. Confirm at the proposed set point and at the edges of the claimed space.

10.3 Formulation development

1. Components without limits: simplex-lattice or simplex-centroid (Examples 6.12, 6.13); with lower/upper limits: extreme vertices (Example 6.14). Add check blends (`AUGMENTMIX=YES`) and replicate a few blends to estimate pure error.
2. Analyze with the Scheffé quadratic or special cubic model (Example 7.13); read the trace plot.
3. Optimize the blend (Example 7.13) and confirm it.

10.4 Robust product or process design

- **Taguchi route**: a Taguchi array for the control factors, each run repeated under the noise conditions (Example 6.11); choose the levels with the highest signal-to-noise ratio, then adjust the mean with a factor that affects the mean but not S/N (Example 7.14).
- **Modelling route**: any replicated design; model the mean as usual and the spread with Analyze variability (Example 7.16). Choose the levels of the dispersion factors that minimize the spread, then put the mean on target with the other factors (Example 8.3).
- In both routes check sensitivity to setting errors with `ROBUST=YES` (Example 8.2).

10.5 An experiment with a hard-to-change factor

Build the design with Hard-to-change factors (Example 6.18), run the whole plots in `WholePlot` order changing the hard factor only between them, and analyze it: the `WholePlot` column triggers the REML split-plot analysis (Example 7.15). Plan at least two replicates; with one, the whole-plot error cannot be estimated.

10.6 Analyzing existing data

For data that were not created by this procedure: make sure every factor setting is a column, and add `RunOrder` if the run order is known and `CenterPt` or `PtType` if centre or axial runs exist. Name the factors in Factors, or leave the list empty and check the *Factors Used* table (Example 7.12). Happenstance data are not a designed experiment: check the VIF column; large VIFs mean the factors moved together and their effects cannot be separated well.

10.7 Refining a model interactively

Run the analysis with the default model, open the HTML report, refine the model in the Lab (remove terms, **[Reduce model]**), set goals and move the sliders, then copy the syntax from tab 3 of the Lab into a syntax window and run it: the final model and optimum become part of the SPSS output (Chapter 11).

Chapter 11

The HTML report and the Model and Optimizer Lab

Every run writes an interactive HTML report (`/OUTPUT EXPORTHTML=YES`, the default) next to the data or to `HTMLPATH`. Its location is printed in the Viewer. It opens in any modern browser, works offline, and sends nothing anywhere. It needs the R packages plotly, htmlwidgets and htmltools.

11.1 Chart options

The bar at the top of the report applies to every chart:

- Theme: Light, Dark or Print.
- Colour palette (all charts): Standard, Colour-blind safe, Vivid, Soft / pastel, Earth, Warm, Grayscale (print); each chart also has its own Colours menu.
- Text size: **[A-]** and **[A+]**.
- Click any title, axis label, legend entry or note to edit it; drag legends and notes to move them.
- Hover a chart for its toolbar: zoom, pan, reset, download PNG or SVG, download the chart's data as CSV, flip X/Y. Downloads keep the chosen colours.

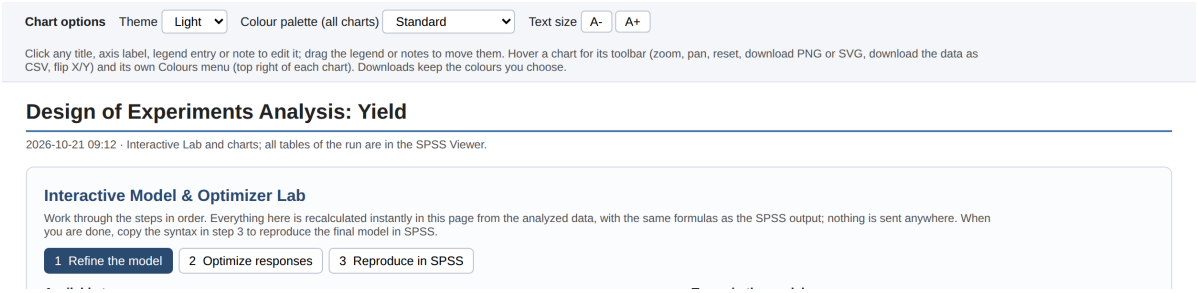


Figure 11.1: Top of an analysis report: chart options.

11.2 The Lab: 1 Refine the model

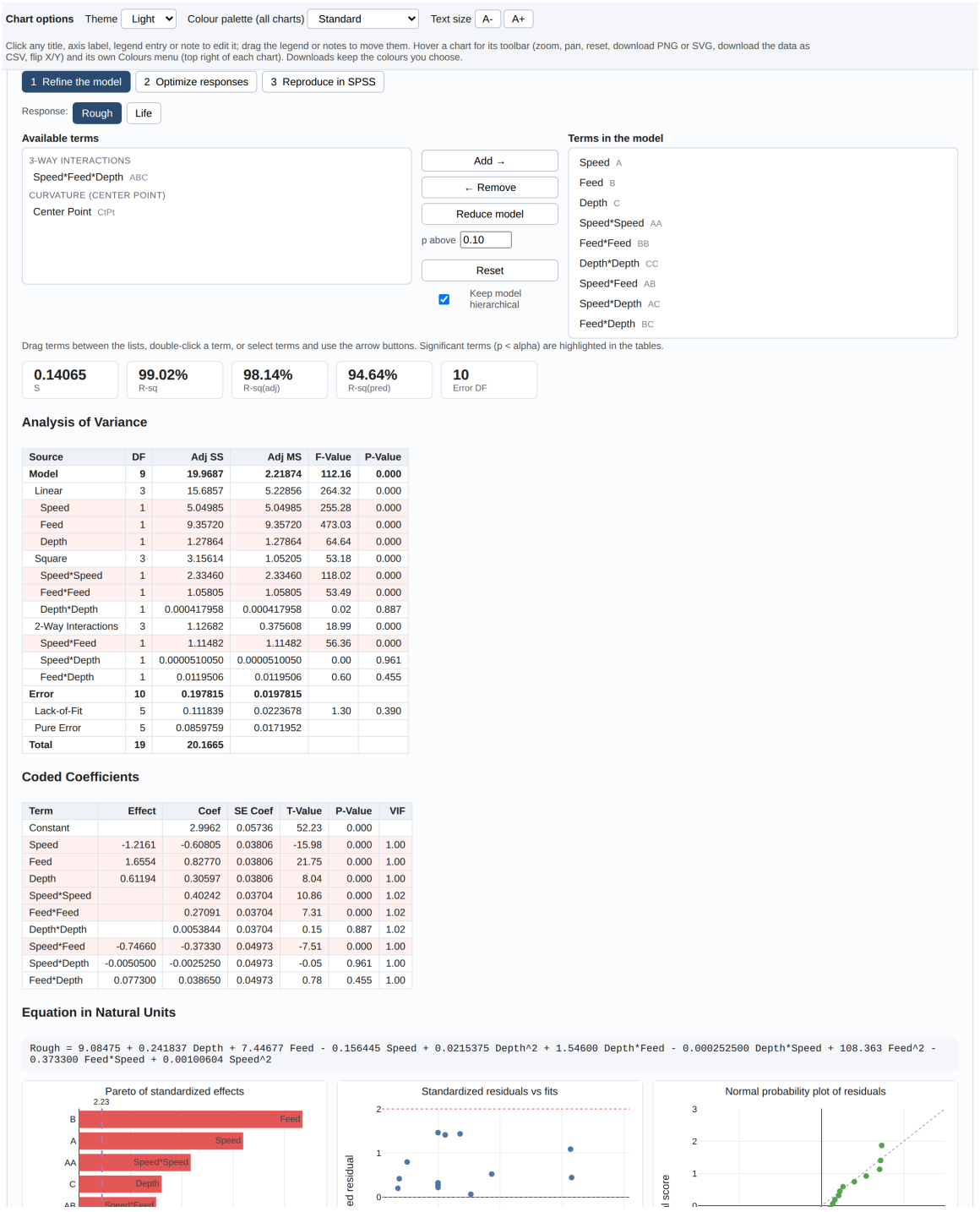


Figure 11.2: Lab, tab 1, for the machining data (two responses).

Response buttons switch between responses; each keeps its own model.

Available terms / Terms in the model drag terms between the lists, double-click a term, or select terms and use **[Add →]** and **[← Remove]**. Letter codes (A, B, AB, AA) are shown next to each term.

[Reduce model] removes the least significant term while its p-value is above p above (0.10 by default): hierarchical backward elimination, as `SELECTION=BACKWARD`.

[Reset] returns to the model fitted by SPSS.

Keep model hierarchical adds the lower-order terms an interaction needs, and removes the interactions of a removed term.

Every change recalculates at once: S, R-sq, R-sq(adj), R-sq(pred), error df, the ANOVA with lack of fit, coded coefficients with effects and VIF, the equation in natural units, the Pareto chart, residuals versus fits and the normal probability plot. Significant terms are highlighted.

11.3 The Lab: 2 Optimize responses

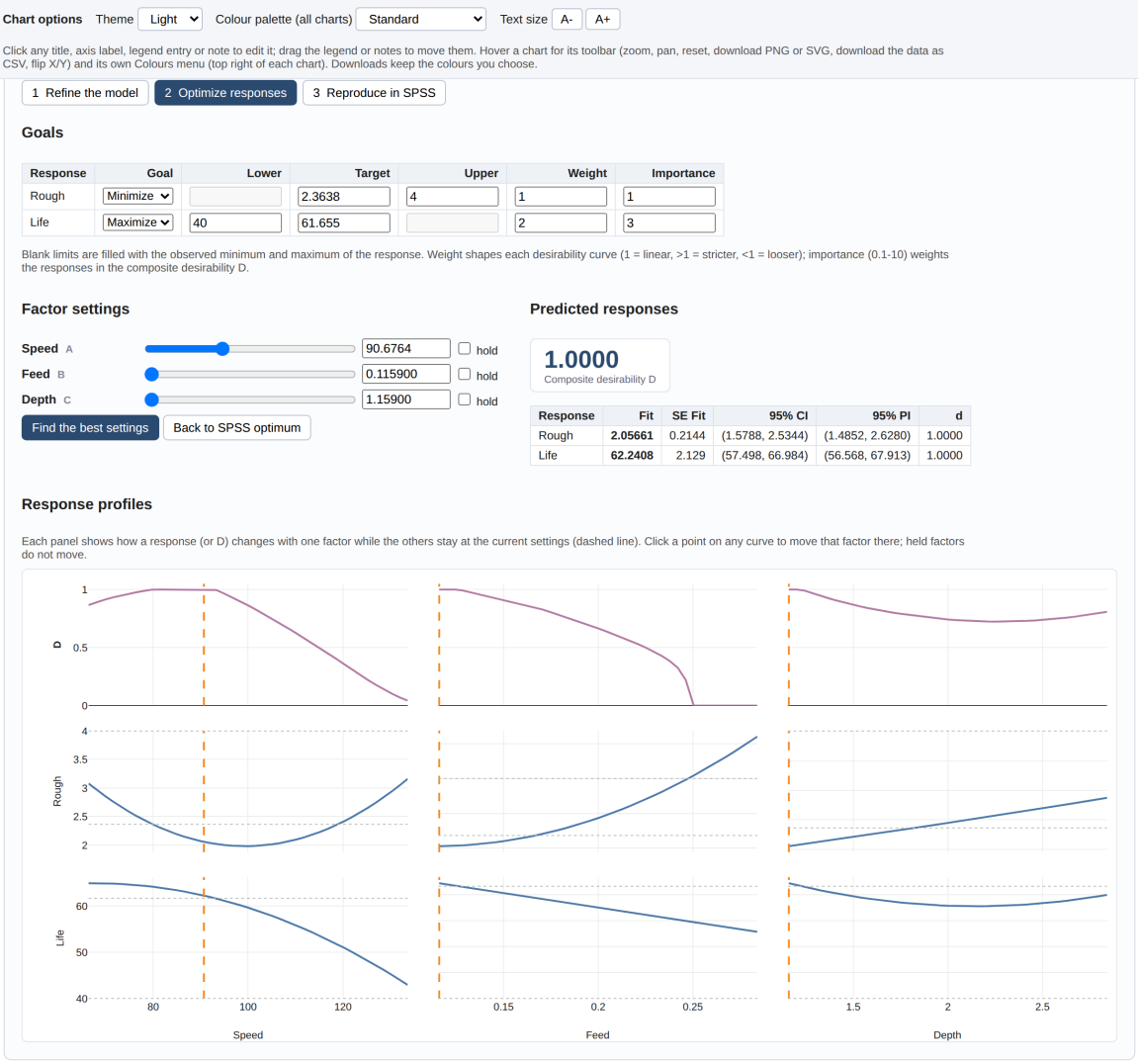


Figure 11.3: Lab, tab 2: goals, sliders, predictions and response profiles.

Goals goal (Maximize, Minimize or Target), lower, target and upper values, weight and importance for each response. Blank limits use the observed range.

Factor settings a slider and a box per factor; tick hold to fix a factor.

[Find the best settings] searches the region (held factors stay fixed).

[Back to SPSS optimum] returns to the settings found by SPSS.

Predicted responses the composite desirability D and, for each response, the fit, its SE, confidence and prediction intervals and d.

Response profiles how each response (and D) changes with one factor while the others stay at the current settings (dashed line). Click a point on a curve to move that factor there.

11.4 The Lab: 3 Reproduce in SPSS

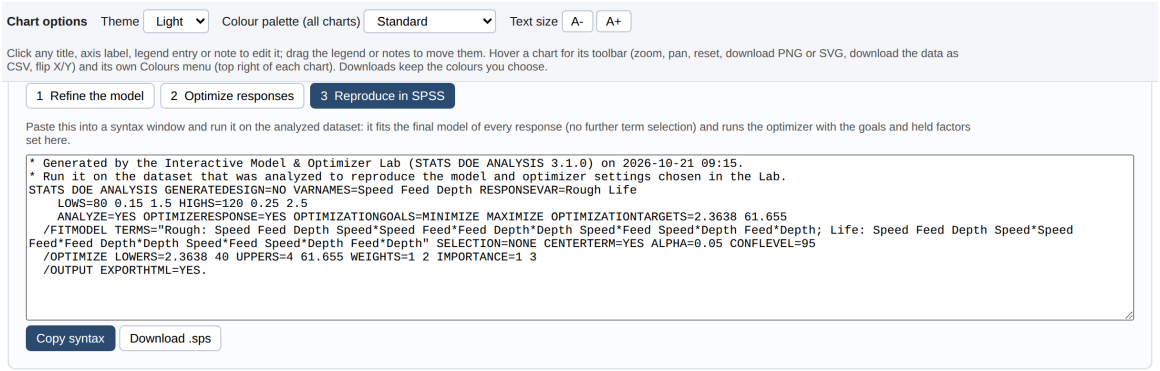


Figure 11.4: Lab, tab 3: the syntax that reproduces the Lab's models and optimum.

The syntax refits each response's final model (`TERMS` per response, `SELECTION=NONE`) and runs the optimizer with the goals set in the Lab. **[Copy syntax]** copies it; **[Download .sps]** saves it. Paste it into a syntax window and run it, so that the final result is part of the SPSS output.

11.5 The self-check

When the page opens, the Lab recomputes every SPSS model (coefficients, sums of squares, p-values and the optimum's desirability) with its own code. If anything differs, a warning badge appears at the top of the Lab; when everything agrees, nothing is shown. Box–Cox models keep the lambda chosen by SPSS. Mixture, Taguchi and split-plot analyses have no Lab. Turn the Lab off with `/OUTPUT LAB=NO`.

11.6 Other sections of the report

The analysis report contains interactive versions of the charts requested in **[Analysis]** (Pareto, main effects, interactions, cube, contour, residuals, surface), the optimizer profiles, the overlaid contour plot and the uncertainty map (Figure 11.5). The design report shows the design points in pairwise scatter plots, coloured by run order (Figure 11.6).

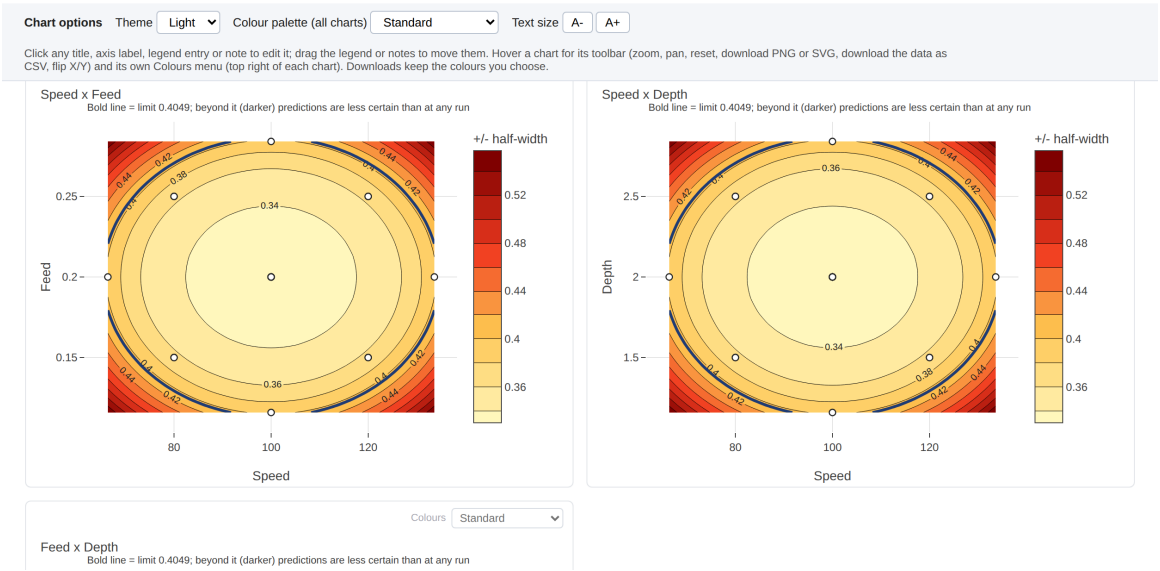


Figure 11.5: Uncertainty map in the HTML report: light = certain, dark = less certain than at any run; the bold line is the limit; circles are the runs.

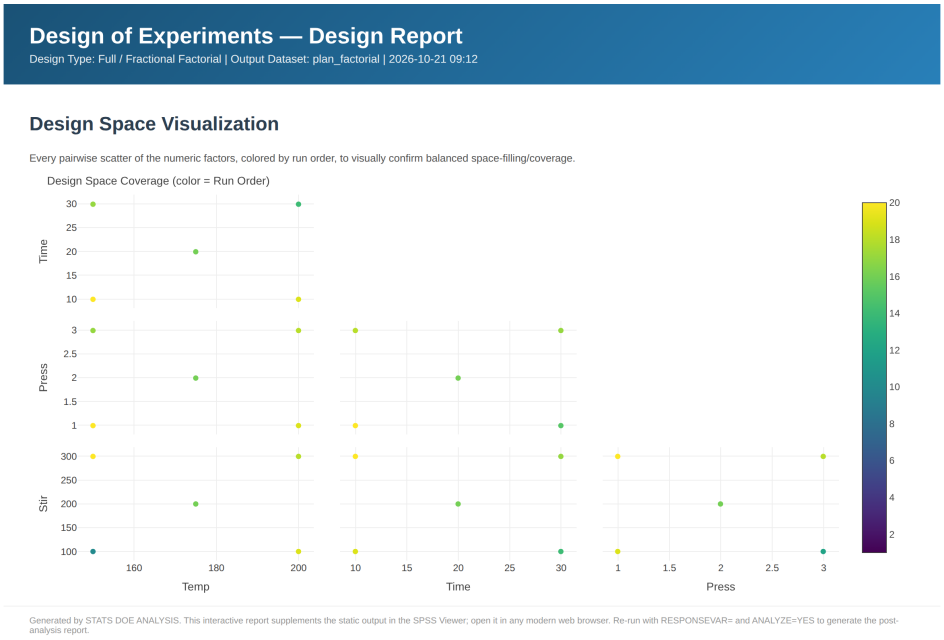


Figure 11.6: Design report of Example 6.1.

Chapter 12

Reading the output: every table

12.1 Design output

Settings and Results Summary Design type, number of runs, factors, output dataset. For optimal designs also the number of candidate points, model, criterion, initial design, centring, repeats and algorithm.

Variable Specifications Per factor: Low, High, Center, Levels, Round (decimals), Factor (categorical) and Mixture flags. Mixture components of a simplex design show no Low and High (they span 0 to the mixture total). The caption names the constraint function, if any.

Design Worksheet The runs (first 20 shown). Columns are described in the introduction to Chapter 6.

Alias / Confounding Structure "Full" for full factorials; named alias chains and the resolution for fractions made with FrF2; "not applicable" plus the *Partial Aliasing* correlation matrix for designs such as Plackett–Burman (rows: main effects; columns: two-factor interactions; values: correlations, 0 = independent, ± 1 = fully confounded).

Power Table / Required Run Size Power for each combination of runs and effect size; the smallest number of runs that reaches the target power.

Augment to Central Composite Design – Summary Original runs kept, axial and centre runs added, total, axial distance, output dataset.

12.2 Analysis output (per response)

Summary of Results The tables below, in words: is the model significant, how much it explains and how well it predicts, which terms matter (strongest first), the largest effects in natural units, curvature, lack of fit, residual problems. Turn off with `INTERPRET=NO`.

Factors Used Only when factors were detected from the data: which columns were used and why.

Factor Coding Letter code, name, type, and the settings coded -1 and $+1$. The caption gives the design kind recognized and the number of runs, centre and axial points.

Box-Cox Transformation Best λ, rounded λ, its 95% confidence interval and the λ used.

Term Selection The steps of backward, forward or stepwise selection, with the p-value of each entered or removed term.

Coded Coefficients Effect ($2\times$ coefficient, two-level terms), coefficient, standard error, t, Sig. and VIF. Coefficients are per coded unit, so their sizes are comparable. Categorical factors with more

than two levels have one row per level but the last ("Factor (1)", ...), deviations from the average. The Center Point coefficient is the difference between the centre runs and the model's prediction at the centre.

Model Summary S, R-squared, adjusted and predicted R-squared, PRESS.

Analysis of Variance Adjusted (Type III) sums of squares: each term is tested given all others. Rows are grouped (Blocks, Linear, Square, 2-Way Interactions, Curvature) with their totals. Error is split into Lack-of-Fit and Pure Error when replicated settings exist.

Regression Equation in Natural Units The model with factors in their original units; one column per level (or combination of levels) of text factors. Use it to calculate predictions by hand; use the coded coefficients to compare effects.

Alias Structure For fractional designs: the aliases of each estimated effect.

Unusual Observations Runs with |standardized residual| > 2 (R) or high leverage (X); row numbers refer to the active dataset.

Residual Checks Anderson–Darling normality test; Durbin–Watson statistic in run order (well below 2 = drift, well above 2 = alternation); largest Cook's distance (above 1 = one run strongly changes the fit).

Prediction For each `PREDICT` setting: fit, SE fit, confidence interval for the mean and prediction interval for one new run; notes on extrapolation.

Pairwise Comparison (Tukey) Fitted means of each level of a categorical factor with letter groupings: levels sharing a letter do not differ significantly (family α).

Canonical Analysis The stationary point of a quadratic model in natural units, the predicted response there, the eigenvalues, the nature (maximum, minimum, saddle) and whether it lies inside the design region.

Analysis of Variability Coefficients of the model for ln(standard deviation), their tests, and the ratio of standard deviations between high and low settings; Tests of Effects on Variability (chi-square); Standard Deviation by Setting (observed and fitted).

Prediction Uncertainty S, the half-width of the 95% prediction interval at the centre, at the runs and over the region, the limit and the share of the region above it.

Split-plot tables Variance Components (whole plot, subplot), Coded Coefficients with the stratum and df of each term, Tests of Fixed Effects.

Mixture tables Mixture Components (ranges), Mixture Regression Coefficients, Model Summary, Analysis of Variance for Mixture.

Taguchi tables Response tables for S/N ratios, means and standard deviations (with delta and rank), S/N ANOVA, Predicted Result at the Best Settings.

12.3 Optimizer output

Summary of Optimization Best settings, composite desirability and its meaning, the trade-off (weakest and best-met goal).

Optimization Criteria Goal, lower, target, upper, weight and importance of each response.

Factor Search Region The range searched for each factor; held factors; the restriction caption for centre-point models.

Optimal Settings The best settings (`SOLUTIONS` rows), fitted responses and D.

Predicted Responses at the Optimal Settings Fit, SE, CI, PI and d of each response.

Confirmation Runs Interval for the average of `CONFIRMRUNS` runs.

Robustness of the Optimal Settings Mean prediction, SD from setting errors, SD including run-to-run variation, the 90% range of single runs and the share within limits.

Profit tables Objective (formula, goal, requirements, held factors), Best Settings by Profit, Profit at the Best Settings (median and 5th–95th percentile of a single run).

12.4 Always at the end

Next Steps Findings, recommendations and syntax (Chapter 9).

Warnings Notes about what was done on your behalf (terms removed, defaults used, files written). Read them: they explain differences from what you asked for. Chapter 14 lists the common ones.

Chapter 13

Syntax reference

13.1 General form

```
STATS DOE ANALYSIS  [main keywords]
  [/OPTIONS   optimal-design search settings]
  [/OUTPUT    charts, tables, HTML report]
  [/POWER     power and sample size]
  [/FITMODEL  analysis model]
  [/OPTIMIZE  optimizer settings]
  [/SAVE DATASET=name]
  [/HELP].
```

Keywords are written `KEYWORD=value`; lists are separated by spaces; YES/NO keywords accept YES or NO. Text with spaces or special characters is quoted: `TERMS`, `PREDICT`, `HOLD`, `LOWLEVELS` and `PROFITLIMITS` in double quotes, `PROFIT` in single quotes (so that text levels inside it can use double quotes). Keywords may appear in any order within their subcommand. The command ends with a period.

The dialog produces exactly this syntax; **[Paste]** shows it. Some keywords are only available in syntax: `EXPORTHTML`, `HTMLPATH`, `EXTERNALPLOTS`, `BACKWARD` (an older form of `SELECTION=BACKWARD`).

> Defaults below are those of the syntax. The dialog starts with Analyze the responses ticked and Create a new design unticked, so a pasted dialog run always contains `ANALYZE` and `GENERATEDESIGN` explicitly. In hand-written syntax for an analysis, write `GENERATEDESIGN=NO ... ANALYZE=YES`.

13.2 Keywords

13.2.1 Main keywords (no subcommand)

Keyword	Values	Default	Meaning
`GENERATEDESIGN`	YES | NO	YES	YES builds a new design (requires /SAVE DATASET); NO works on the active dataset (analysis, augmentation, power).
`VARNAMES`	variables		Names of the design factors, in order. For a new design the order sets the order of every list (LOWS, HIGHS, ...). For an analysis, FACTORVARS does the same job.

Keyword	Values	Default	Meaning
NFACTORS	integer		Number of factors for a new design. Unnamed factors become A, B, C, … (I is skipped) with Low -1 and High $+1$, so only the rows you change need typing.
FACTORSFROMDATA	YES | NO	NO	YES reads the factor table from the active dataset: one row per factor with columns Name, Low, High and optionally Levels and Categorical. Any number of factors.
EVDEGREE	integer	1	Extreme-vertices designs: 0 = vertices and overall centroid; 1 = also the centroids of the edges; 2 = also the centroids of the 2-dimensional faces.
HTC	variables		Hard-to-change factors of a split-plot design (Factorial or Full factorial). Each setup of these factors in each replicate becomes one whole plot.
NAME1..15	variable		Factor table row n (1–15): name of the factor.
LOW1..15	number		Factor table row n: low setting (not needed for a categorical factor).
HIGH1..15	number		Factor table row n: high setting.
LEVELS1..15	integer		Factor table row n (LEVELSn) or list (LEVELS): number of levels.
FACTOR1..15	YES | NO	NO	Factor table row n: YES marks the factor as categorical (coded 1 to Levels unless Low/High are given).
FACTORVARS	variables		Existing factor columns of the active dataset (analysis). Left out, the factors saved with the design or detected from the data are used.
FORMULA	"text"		Optimal designs only: a custom model formula starting with ~, e.g. `"~x1+x2+x1:x2"`. Use instead of MODEL.
FACTORS	YES | NO	NO	One YES/NO per factor: categorical factor flags for the list form.
MIXTURES	YES | NO	NO	One YES/NO per factor: mixture-component flags. Mixture components add to MIXTURESUM in every run.
LOWS	numbers		Low settings, one per factor (a single value applies to all).
HIGHS	numbers		High settings, one per factor (a single value applies to all).
CENTERS	numbers		Centre settings, one per factor; blank uses the midpoint.
ROUNDTOS	integers		Number of decimals each factor's settings are rounded to.

Keyword	Values	Default	Meaning
DESIGNTYPE	OPTIMAL | FACTORIAL | PLACKETTBURMAN | RSM | BOXBEHNKEN | CCD | TAGUCHI | LHS | DSD | FULLFACTORIAL | SIMPLEXLATTICE | SIMPLEXCENTROID | EXTREMEVERTICES	OPTIMAL	Kind of design to build (or, in an analysis, the kind of design the data came from).
LATTICEDEGREE	integer	2	Simplex-lattice designs: proportions step in 1/degree (2 gives 0, 1/2, 1).
AUGMENTMIX	YES | NO	NO	Mixture designs: YES adds axial check blends and the overall centroid.
MODEL	LINEAR | QUAD | CUBIC | CUBICS	LINEAR	Optimal designs: model the design must support – LINEAR, QUAD (quadratic), CUBIC or CUBICS (cubic without the pure cubes).
TRIALS	integer	design rule	Number of runs. Optimal and Latin hypercube designs use it directly; a two-level factorial with fewer runs than 2^k becomes a fraction (needs the R package FrF2).
CONSTANT	YES | NO	YES	Optimal designs: include the constant term (switched off automatically for mixtures).
CONSTRAINTFUNC	file		Optimal designs: file with an R function `dfilter` that returns TRUE for allowed candidate points.
MIXTURESUM	number	1	Total of the mixture components in every run.
RANDOMIZE	YES | NO	NO	YES randomizes the run order (within blocks; for split-plot designs the whole plots and the runs inside them).
BLOCKS	integer		Number of blocks. Factorials are blocked by confounding; other designs block by whole replicates.
REPLICATES	integer		Number of complete copies of the design.
SCREENINGCENTERPTS	YES | NO	NO	YES adds centre-point runs to a two-level design.
SCREENINGCENTERPTSCOUNT	integer		How many centre points to add.
AUGMENTTOCCD	YES | NO	NO	YES adds axial (star) points and centre points to the two-level design in the active dataset, making a face-centred central composite design (saved to /SAVE DATASET).
AUGMENTCENTERPTS	integer		Centre points added by AUGMENTTOCCD.
RESPONSEVAR	variables		Response variable(s) to analyze (Taguchi: the replicate response columns).
ANALYZE	YES | NO	NO	YES fits the model and produces the analysis. (The dialog ticks it by default.)
CREATEPLOTS	YES | NO	NO	YES enables the chart options (main effects, interactions, cube, contour, residual, Pareto, curvature).
OPTIMIZERESPONSE	YES | NO	NO	YES runs the response optimizer (desirability).

Keyword	Values	Default	Meaning
OPTIMIZATIONGOAL	MAXIMIZE | MINIMIZE | TARGET	MAXIMIZE	Goal for a single response, or the default for all responses.
OPTIMIZATIONGOALS	list of MAXIMIZE / MINIMIZE / TARGET		One goal per response: MAXIMIZE, MINIMIZE or TARGET.
OPTIMIZATIONTARGET	number		Target value for a single response with goal TARGET.
OPTIMIZATIONTARGETS	numbers		One target per response: for TARGET the value aimed at; for MAXIMIZE/MINIMIZE the value at which the response is fully satisfactory.
VARSELECT	YES | NO	NO	Legacy model: YES performs AIC-based variable selection.
VARSELMETHOD	STEPWISE | BESTSUBSETS	STEPWISE	Legacy selection method: STEPWISE or BESTSUBSETS.
STEPDIR	BOTH | FORWARD | BACKWARD	BOTH	Legacy stepwise direction.
SELECTIONTRACE	YES | NO	NO	Legacy: chart of the AIC at each stepwise step.
FITPROFILE	YES | NO	NO	Legacy best subsets: predictive fit profile chart.
PARSIMONYPLOT	YES | NO	NO	Legacy best subsets: model parsimony chart.
FACTORMAP	YES | NO	NO	Legacy best subsets: factor inclusion map.

13.2.2 /OPTIONS

Keyword	Values	Default	Meaning
CRITERION	D | A	D	Optimal designs: D (smallest generalized variance of the coefficients) or A (smallest average variance).
CENTER	YES | NO	NO	Optimal designs: centre numeric factors before the design is evaluated.
DESIGNALG	EXACT | APPROX	EXACT	Optimal designs: EXACT design or APPROX (approximate theory) design.
INITIAL	RANDOM | NULLIFICATION	RANDOM	Optimal designs: start from a RANDOM design or by NULLIFICATION (usually better, slower).
REPEATS	integer	5	Optimal designs: number of independent searches; the best design is kept.
NUMCAND	integer	10 x terms	Optimal designs: number of candidate points to choose from.

13.2.3 /OUTPUT

Keyword	Values	Default	Meaning
CONFOUNDING	YES | NO	YES	Design: show the alias / confounding table.
MAINEFFECTS	YES | NO	NO	Main-effects plot.

Keyword	Values	Default	Meaning
INTERACTIONS	YES | NO	NO	Interaction plots.
CUBEPLOT	YES | NO	NO	Cube plot of fitted values.
CONTOURPLOT	YES | NO	NO	Contour plots of the fitted response.
RESIDUALPLOTS	YES | NO	NO	Residual plots (four-in-one).
PARETOPLOT	YES | NO	NO	Pareto chart of standardized effects.
CURVATUREPLOT	YES | NO	NO	Average at the corners against the centre points.
EXTERNALPLOTS	YES | NO	NO	Legacy: also write the charts as files.
EXPORTHTML	YES | NO	YES	Write the interactive HTML report (needed for the Lab, the uncertainty map and the interactive charts).
HTMLPATH	file		File for the HTML report; default: a time-stamped file next to the data.
EFFECTSTABLE	YES | NO	NO	Legacy: effects and Lenth significance table.
RESIDUALTESTS	YES | NO	NO	Legacy: Shapiro–Wilk and Durbin–Watson tests.
VIFTABLE	YES | NO	NO	Legacy: variance inflation factors.
OPTDETAILTABLE	YES | NO	NO	Legacy: optimization detail table.
SCREENINGSUMMARY	YES | NO	NO	Effect-ranking table for screening designs.
LAB	YES | NO	YES	Interactive Model and Optimizer Lab in the HTML report.
ADVISOR	YES | NO	YES	Next Steps table with recommendations and ready-to-run syntax.
UNCERTAINTYMAP	YES | NO	NO	Map of the prediction-interval half-width over pairs of numeric factors.
UNCERTAINTYLIMIT	number	largest at the runs	Limit for the uncertainty map (half-width in response units).

13.2.4 /POWER

Keyword	Values	Default	Meaning
POWERTABLE	YES | NO	NO	Display the power and sample-size tables.
POWEREFFECTSIZES	numbers		Effect sizes in standard-deviation units (effect / σ).
POWERRUNS	integers		Run counts to evaluate.
POWERALPHA	number	0.05	Significance level.
POWERTARGET	number	0.8	Target power for the required-runs table.

13.2.5 /FITMODEL

Keyword	Values	Default	Meaning
ANALYSISMODEL	CODED | LEGACY	CODED	CODED (recommended): coded-factor regression with ANOVA, alias handling and all 3.x tables. LEGACY: the earlier raw-unit model.

Keyword	Values	Default	Meaning
TERMS	"text"		Exact model terms: factor names or letter codes joined with *, plus keywords LINEAR, 2WAY, 3WAY, SQUARES, QUADRATIC, FULL, CTPT. A different model per response: "Y1: A B A*B; Y2: A B C".
TERMSET	TS_DEFAULT | TS_LINEAR | TS_2WAY | TS_3WAY | TS_LINSQ | TS_QUAD | TS_FULL	design default	Standard model: TS_DEFAULT, TS_LINEAR, TS_2WAY, TS_3WAY, TS_LINSQ, TS_QUAD, TS_FULL. TERMS takes precedence.
VARIABILITY	YES | NO	NO	Analyze variability: model ln(standard deviation) of replicated settings.
WHOLEPLOT	variable	WholePlot column	Split-plot analysis: the column numbering the whole plots.
ALPHA	number	0.05	Significance level for highlighting, Pareto reference line and summaries.
BACKWARD	YES | NO	NO	Older form of SELECTION=BACKWARD.
ALPHAREMOVE	number	0.10	Term selection: a term leaves when its p-value is above this.
CENTERTERM	YES | NO	YES	Two-level designs with centre points: include the curvature (centre point) term.
CONFLEVEL	number	95	Confidence level (%) for intervals.
LOWLEVELS	"text"		Which level of each two-level text factor is coded -1, e.g. "Catalyst=A".
EQUATION	YES | NO	YES	Regression equation in natural units.
INTERPRET	YES | NO	YES	Plain-language Summary of Results.
PREDICT	"text"		Predict at settings: "A=1 B=2; A=3 B=4". Gives fit, SE, confidence and prediction intervals.
ALIASTABLE	YES | NO	YES	Alias structure for fractional designs.
DIAGTABLE	YES | NO	YES	Unusual observations and residual checks.
FACTORINFO	YES | NO	YES	Factor coding table.
EFFECTSPLOT	YES | NO	NO	Normal plot of standardized effects.
SELECTION	NONE | BACKWARD | FORWARD | STEPWISE	NONE	Term selection: NONE, BACKWARD, FORWARD or STEPWISE (by p-values, hierarchy kept).
ALPHAENTER	number	0.15	Term selection: a term enters when its p-value is below this.
BOXCOX	NONE | AUTO | LOG | SQRT | INVERSE | CUSTOM	NONE	Box–Cox transformation: NONE, AUTO, LOG, SQRT, INVERSE, CUSTOM (with LAMBDA).
LAMBDA	number		Lambda for BOXCOX=CUSTOM.
CATEGORICAL	variables		Numeric factors to analyze as categories.
GROUPING	YES | NO	NO	Tukey pairwise comparisons with letter groupings for categorical factors.
CANONICAL	YES | NO	NO	Stationary point and canonical analysis of a second-order model.
SURFACEPLOT	YES | NO	NO	3-D surface plots.
MIXMODEL	LINEAR | QUADRATIC | SPECIALCUBIC | FULLCUBIC	QUADRATIC	Mixture model: LINEAR, QUADRATIC, SPECIALCUBIC, FULLCUBIC.
MIXCOMPS	AUTO | PSEUDO | PROPORTIONS	AUTO	Mixture component scale: AUTO, PSEUDO (L-pseudo-components), PROPORTIONS.

Keyword	Values	Default	Meaning
TRACEPLOT	YES | NO	NO	Mixture response trace plot.
SNTYPE	NONE | LARGER | SMALLER | NOMINAL | NOMINAL2	NONE	Taguchi signal-to-noise ratio: LARGER, SMALLER, NOMINAL ($10\log_{10}(\bar{y}^2/s^2)$), NOMINAL2 ($-10\log_{10}s^2$), NONE.

13.2.6 /OPTIMIZE

Keyword	Values	Default	Meaning
LOWERS	numbers		Lower acceptable limit per response (MAXIMIZE and TARGET use it).
UPPERS	numbers		Upper acceptable limit per response (MINIMIZE and TARGET use it).
WEIGHTS	numbers	1	Shape of each desirability curve (0.1–10; 1 = linear).
IMPORTANCE	numbers	1	Relative importance of each response in the composite desirability (0.1–10).
HOLD	"text"		Fix factors during optimization: `"Catalyst=B Time=20"`.
PROFIT	'formula'		Profit or cost formula of responses and factors, in single quotes.
PROFITGOAL	MAXIMIZE | MINIMIZE	MAXIMIZE	MAXIMIZE (profit) or MINIMIZE (cost).
PROFITLIMITS	"text"		Requirements for the profit search, e.g. `"Yield>=60 Cost<=25"`.
SOLUTIONS	integer	1	Number of distinct best settings to list.
OVERLAY	YES | NO	NO	Overlaid contour plot of the region meeting every limit.
ROBUST	YES | NO	NO	Robustness simulation around the optimum.
ROBUSTSD	number	5	Setting error for ROBUST, in % of each factor's half-range.
CONFIRMRUNS	integer		Number of planned confirmation runs: gives the interval for their average.

13.2.7 /SAVE

Keyword	Values	Default	Meaning
DATASET	dataset name		Name of the new dataset that receives the design.

13.2.8 /HELP

Keyword	Values	Default	Meaning

Chapter 14

Messages and troubleshooting

Messages appear in the *Warnings* table. Most say what was done on your behalf; a few stop the run.

Message (abridged)	Meaning and what to do
Setting up	
Output Dataset is required when Generate Design is selected	Type a new dataset name in **[Create Design...]** (`/SAVE DATASET=`).
Create a new design is unticked, so this run only reads the active dataset – no output was requested	Tick Analyze the responses and add a response, or tick Create a new design.
Give the new factors either in the Factor table or with VARNAMES, not both	Use one way of entering factors.
Factor table row *n*: give both Low and High, or leave both empty	Complete the row.
Read the factor table from the active dataset: the active dataset needs columns named Name, Low and High	Rename the columns of the factor sheet.
A fractional factorial (TRIALS=*n*) needs the R package FrF2	Install FrF2, or use Plackett–Burman / definitive screening.
Taguchi design rounded up to the nearest valid orthogonal array	Information: the array size is shown.
HTC applies to Factorial and Full factorial designs	Use one of those design types for a split-plot.
Bounds implied by the others were used for …	A mixture limit could not be reached; the implied limit is used.
INITIAL=NULLIFICATION cannot be combined with DESIGNALG=APPROX	Use INITIAL=RANDOM with approximate designs.
Analysis	
Response variable(s) not found	Check the spelling; variable names are case sensitive in some places.
n row(s) with a missing response or factor value were excluded	Information: runs not yet done are skipped.
TERMS: *X* is not in the model although interactions or squares containing it are	The typed model lacks a lower-order term; add it unless that is intended.
TERMS: '*X*' is not a factor name or letter code and was ignored	Check the spelling or the letter code in the Factor Coding table.
these terms cannot be estimated separately (aliased …) and were removed	The design cannot separate them; see the alias table.

Message (abridged)	Meaning and what to do
the model uses every degree of freedom (no error term)	No p-values; effects are judged with Lenth's method. Remove terms, replicate or add centre points.
the model reproduces every run exactly (S = 0)	Tests are undefined; check that the response is not a copy of a factor.
Box-Cox needs every response value > 0	Add a constant or use another transformation.
term selection needs at least one error degree of freedom	Start from a smaller model or use forward/stepwise selection.
X treated as categorical, as recorded with the design	Information: the design said so; give `CATEGORICAL` to override.
the block variable is confounded with the design and was dropped	The blocks coincide with a factor setting; the block effect cannot be separated.
SPLIT FILE by ... is in effect	The procedure analyzes all cases together; split the data with SELECT IF instead.
X does not group the runs into whole plots	The WholePlot column has one run per whole plot or only one whole plot: a standard analysis was done.
The factors sum to the same total in every run, so they were analyzed as mixture components	Specify `TERMS` (or `ANALYSISMODEL=LEGACY`) to treat them as ordinary factors.
Optimizer and report	
Overlaid contour plot / Robustness check skipped: a center-point (curvature) model ...	That model predicts only at corners and centre; remove the CTPT term (`CENTERTERM=NO`) or use a response-surface design.
Optimizer: HOLD factor '*X*' is not a design factor; ignored	Check the name in `HOLD`.
Optimizer: weight / importance must be between 0.1 and 10; 1 used	Correct the value.
Profit optimization: '*X*' is not allowed in the formula	Only names, numbers, arithmetic and the listed functions are allowed.
PROFITLIMITS: '*X*' is not a response	Use a response name in each requirement.
Interactive HTML report skipped: requires the 'plotly' and 'htmlwidgets' R packages	Install them (Chapter 2).
The interactive Lab is not available for split-plot analyses	By design.

14.1 Common questions

The dialog's Options button is greyed out. It applies to new designs: tick Create a new design. Inside, the optimal-design settings are enabled only for Optimal Design.

My factors were not found. Name them in Factors, or check the *Factors Used* table.

No p-values. The model is saturated (no error df): see Example 7.5.

Why is my optimum only at a corner or the centre? The model has a centre-point term (Example 8.3).

The axial points are outside my range. That is the rotatable CCD (Example 6.6); use augmentation (face-centred) or Box–Behnken.

Results differ slightly between runs of an optimal or Latin hypercube design. These designs

come from random searches, so each run gives a different, equally valid design. Keep the dataset of the design you actually run.

Chapter 15

Methods and formulas

15.1 Coding and the model

A numeric factor with settings u between Low L and High H is coded $x = (u - m)/h$ with $m = (L + H)/2$ and $h = (H - L)/2$. A two-level text factor is coded -1 for its first (or `LOWLEVELS`) level and $+1$ for the other; a text factor with $k > 2$ levels by $k - 1$ effect-coded columns (the last level is -1 in all of them). The model $y = X\beta + \varepsilon$ is fitted by least squares. For two-level terms the *effect* is $2\hat{\beta}$: the change from -1 to $+1$. The centre-point term is an indicator of centre runs; its coefficient is the difference between the centre mean and the corner-based prediction at the centre. Squares of three-level factors are used as x^2.

15.2 Tests and model statistics

- **Adjusted sums of squares** (Type III): the increase in the residual sum of squares when a term (or group of terms) is dropped from the full model. $F = \mathrm{MS}_{\mathrm{term}}/\mathrm{MS}_{\mathrm{error}}$.
- **Lack of fit**: $\mathrm{SS}_{\mathrm{error}} = \mathrm{SS}_{\mathrm{LOF}} + \mathrm{SS}_{\mathrm{PE}}$, pure error from the variation within groups of identical settings; $F = \mathrm{MS}_{\mathrm{LOF}}/\mathrm{MS}_{\mathrm{PE}}$.
- $S = \sqrt{\mathrm{MS}_{\mathrm{error}}}$; $R^2 = 1 - \mathrm{SS}_{\mathrm{error}}/\mathrm{SS}_{\mathrm{total}}$; $R^2_{\mathrm{adj}} = 1 - \mathrm{MS}_{\mathrm{error}}/\mathrm{MS}_{\mathrm{total}}$.
- $\mathrm{PRESS} = \sum_i \left(e_i/(1 - h_{ii})\right)^2$ with leverages h_{ii}; $R^2_{\mathrm{pred}} = 1 - \mathrm{PRESS}/\mathrm{SS}_{\mathrm{total}}$ (shown as 0 when negative).
- $\mathbf{VIF}_j = 1/(1 - R_j^2)$ from regressing column j on the other model columns.
- **Lenth's method** (no error df): $s_0 = 1.5\,\mathrm{median}|c_j|$, $\mathrm{PSE} = 1.5\,\mathrm{median}\{|c_j| : |c_j| < 2.5 s_0\}$, margin of error $\mathrm{ME} = t_{0.975,\,m/3}\,\mathrm{PSE}$, simultaneous margin $\mathrm{SME} = t_{\gamma,\,m/3}\,\mathrm{PSE}$ with $\gamma = (1 + 0.95^{1/m})/2$, for m effects. Standardized effects in the Pareto chart are $|c_j|/\mathrm{PSE}$.
- **Residual checks**: Anderson–Darling statistic of standardized residuals against the normal distribution (estimated mean and SD; Stephens' p-value approximation); Durbin–Watson $\sum(e_t - e_{t-1})^2 / \sum e_t^2$ in `RunOrder` (data order when absent); Cook's distance $D_i = \frac{r_i^2}{p}\frac{h_{ii}}{1-h_{ii}}$.
- **Intervals**: at settings x_0, $\hat{y}_0 \pm t\,\mathrm{SE}_{\mathrm{fit}}$ (confidence) and $\hat{y}_0 \pm t\sqrt{S^2 + \mathrm{SE}_{\mathrm{fit}}^2}$ (prediction), t with the error df. For n confirmation runs: $\hat{y}_0 \pm t\sqrt{S^2/n + \mathrm{SE}_{\mathrm{fit}}^2}$.

15.3 Term selection

Backward: repeatedly remove the term with the largest p-value above α_{remove}, never a term contained in a remaining interaction. Forward: add the term with the smallest p-value below α_{enter}, adding its lower-order terms. Stepwise: forward steps, each followed by a backward check. The legacy analysis uses AIC instead (R's `step`).

15.4 Box–Cox

For each λ on a grid, the model is fitted to $y^{(\lambda)} = (y^\lambda - 1)/(\lambda \dot{y}^{\lambda-1})$ ($\dot{y} \ln y$ at $\lambda = 0$), $\dot{y}$ the geometric mean, and the residual sum of squares is recorded. The best λ minimizes it; the 95% interval contains the λ whose profile log-likelihood is within $\chi^2_{1,0.95}/2$ of the maximum. AUTO uses the best λ rounded to 0.5.

15.5 Tukey comparisons

Fitted (least-squares) means of each level, averaged over the other factors; differences tested with the studentized range distribution at the family α; letters group levels whose differences are not significant.

15.6 Canonical analysis

Writing the quadratic model as $\hat{y} = b_0 + x^\top b + x^\top B x$, the stationary point is $x_s = -\frac{1}{2} B^{-1} b$. The eigenvalues of B decide its nature: all negative = maximum, all positive = minimum, mixed = saddle; an eigenvalue near 0 indicates a ridge.

15.7 Mixtures

Scheffé models without constant: linear $\sum \beta_i x_i$; quadratic adds $\beta_{ij} x_i x_j$; special cubic adds $\beta_{ijk} x_i x_j x_k$; full cubic adds $\delta_{ij} x_i x_j (x_i - x_j)$. With lower bounds a_i, L-pseudo-components $z_i = (x_i - a_i)/(1 - \sum a_i)$ are used. Extreme vertices follow McLean and Anderson (1966): every combination of bounds for all but one component, with the remaining component completing the total, is a vertex if it lies within its own bounds; edge (and face) centroids average the vertices sharing $k-2$ (and $k-3$) active bounds.

15.8 Taguchi signal-to-noise ratios

Larger is better: $-10 \log_{10} \left(\frac{1}{n} \sum 1/y_i^2 \right)$. Smaller is better: $-10 \log_{10} \left(\frac{1}{n} \sum y_i^2 \right)$. Nominal is best: $10 \log_{10} (\bar{y}^2 / s^2)$; NOMINAL2: $-10 \log_{10} s^2$. Response tables give the mean S/N at each level; delta = max − min; the predicted S/N at the best levels is $\bar{\eta} + \sum_j (\bar{\eta}_{j,\text{best}} - \bar{\eta})$.

15.9 Split-plot REML

Model $y = X\beta + Zu + \varepsilon$ with whole-plot effects $u \sim N(0, \sigma_w^2)$ and $\varepsilon \sim N(0, \sigma^2)$. The variance ratio is estimated by maximizing the restricted likelihood (one parameter, profiled exactly); β by

generalized least squares. Whole-plot terms are tested with the between-whole-plot df, subplot terms with the within-whole-plot df (Wald F).

15.10 Analysis of variability

For each setting replicated $n_i \geq 2$ times with variance s_i^2 and $\nu_i = n_i - 1$ df: $z_i = \ln s_i^2 - [\psi(\nu_i/2) - \ln(\nu_i/2)]$ is unbiased for $\ln \sigma_i^2$ with variance $\psi'(\nu_i/2)$ (ψ = digamma). The model $z = X\gamma$ is fitted by weighted least squares with these known variances; Wald chi-square tests; the reported coefficients are on the ln(SD) scale, $c = \gamma/2$, and the ratio of SDs between the high and low settings (coded $+1$ and -1) is e^{2c}.

15.11 Optimization

Desirabilities as in Chapter 8; $D = \left(\prod d_i^{r_i}\right)^{1/\sum r_i}$. The search starts from a grid over the region and refines the best points with bounded quasi-Newton optimization (L-BFGS-B); distinct solutions are separated by a minimum distance. Profit formulas are parsed and checked against a whitelist (names, numbers, arithmetic, the listed functions) and maximized or minimized with requirements handled by a penalty that increases until they are met; the single-run range simulates each response from its prediction distribution (t with error df) in 20,000 draws. Robustness: 2000 simulated runs with normal setting errors (SD = `ROBUSTSD`% of the half-range) plus residual variation.

15.12 Uncertainty map

Half-width of the prediction interval, $t\sqrt{S^2 + \mathrm{SE}_{\mathrm{fit}}(x)^2}$, on a grid over each pair of numeric factors (others at the centre). Default limit: the largest half-width at the runs of the experiment.

15.13 Power

Planning approximation for a two-level effect estimated from N runs: noncentral t with $N - 2$ df and noncentrality $d\sqrt{N}/2$, two-sided at α.

15.14 Steepest ascent

For a first-order model with coded coefficients b_j, the path moves one coded unit in the factor with the largest $|b_j|$ per step and $b_j/|b_{\max}|$ coded units in the others (signs reversed for descent), converted to natural units.

Appendix A

Example datasets

The datasets below are included, with `DATA LIST` commands, in `DOE_3.1.0_manual_examples.sps`. Run the top part of that file once; each example then starts with `DATASET ACTIVATE`.

A.0.1 *reactor*

Chemical reactor screening study: a 2^4 factorial in Temp (150–200), Time (10–30 min), Press (1–3 bar) and the text factor Catalyst (A/B), with 4 centre points (Catalyst A and B twice each). Responses Yield (%), Cost and Visc (viscosity, log-normal errors). Used for factorial analysis, model selection, Box–Cox and optimization.

Temp	Time	Press	Catalyst	StdOrder	RunOrder	Yield	Cost	Visc
150	10	1	A	1	17	56.613	18.784	10.9466
200	10	1	A	2	5	60.036	21.26	23.9905
150	30	1	A	3	1	57.315	17.173	17.708
200	30	1	A	4	10	71.978	19.348	34.8147
150	10	3	A	5	4	51.882	18.761	10.7486
200	10	3	A	6	2	56.482	24.274	27.1672
150	30	3	A	7	20	52.966	17.59	15.3928
200	30	3	A	8	18	66.185	22.141	39.4083
150	10	1	B	9	8	58.284	17.223	11.2425
200	10	1	B	10	7	62.838	21.198	27.6014
150	30	1	B	11	16	60.936	16.887	16.2246
200	30	1	B	12	9	75.723	19.78	35.5461
150	10	3	B	13	19	54.358	18.63	11.8907
200	10	3	B	14	6	59.302	24.045	25.7939
150	30	3	B	15	14	54.606	15.554	17.0107
200	30	3	B	16	15	70.788	23.049	31.7803
175	20	2	A	17	12	63.171	20.313	19.8308
175	20	2	B	18	3	64.551	20.019	20.0362
175	20	2	A	19	13	62.536	20.152	19.6129
175	20	2	B	20	11	67.616	19.609	19.5437

A.0.2 *machining*

Machining study run as a central composite design (rotatable, $\alpha = 1.682$) in Speed (80–120 m/min), Feed (0.15–0.25 mm/rev) and Depth (1.5–2.5 mm) with 6 centre points. PtType: 1 = cube, -1 = axial, 0 = centre. Responses Rough (surface roughness, μm; smaller is better) and Life (tool life, min; larger is better).

Speed	Feed	Depth	PtType	StdOrder	Rough	Life
80	0.15	1.5	1	1	2.7162	61.655
120	0.15	1.5	1	2	2.3638	45.857
80	0.25	1.5	1	3	5.2247	56.894
120	0.25	1.5	1	4	3.0095	36.939
80	0.15	2.5	1	5	3.4678	59.816
120	0.15	2.5	1	6	2.7357	44.098
80	0.25	2.5	1	7	5.7613	55.637
120	0.25	2.5	1	8	3.9056	35.658
133.64	0.2	2	-1	9	3.2364	31.588
100	0.2841	2	-1	10	5.194	45.144
100	0.2	2.841	-1	11	3.5318	52.965
66.36	0.2	2	-1	12	5.1089	56.422
100	0.1159	2	-1	13	2.4072	54.332
100	0.2	1.159	-1	14	2.567	52.905
100	0.2	2	0	15	3.0315	49.666
100	0.2	2	0	16	3.184	49.926
100	0.2	2	0	17	3.0242	48.356
100	0.2	2	0	18	2.8552	50.35
100	0.2	2	0	19	3.0378	51.103
100	0.2	2	0	20	2.8316	51.638

A.0.3 *purify*

Purification step run as a Box–Behnken design in Dose, TempC and Ratio with 3 centre points; response Purity (%).

Dose	TempC	Ratio	StdOrder	Purity
5	40	1	1	73.483
15	40	1	2	82.533
5	60	1	3	62.654
15	60	1	4	69.552
5	50	0.5	5	71.629
15	50	0.5	6	74.095
5	50	1.5	7	69.708
15	50	1.5	8	82.029
10	40	0.5	9	81.493
10	60	0.5	10	69.034
10	40	1.5	11	84.885
10	60	1.5	12	72.884
10	50	1	13	79.969
10	50	1	14	79.485
10	50	1	15	80.308

A.0.4 *materials*

General factorial: Material (Steel, Alu, Ti) × TestTemp (20, 80 °C), two replicates; response Strength.

Material	TestTemp	StdOrder	Strength
Steel	20	1	106.955
Alu	20	2	90.962
Ti	20	3	106.56
Steel	80	4	112.116
Alu	80	5	92.077
Ti	80	6	111.837
Steel	20	7	107.1
Alu	20	8	92.303
Ti	20	9	103.841
Steel	80	10	114.605
Alu	80	11	90.898
Ti	80	12	110.176

A.0.5 *screen5*

Half-fraction 2^{5-1} (resolution V, 16 runs) in coded factors A–E; response Response.

A	B	C	D	E	StdOrder	Response
-1	-1	-1	-1	1	1	47.847
1	-1	-1	-1	-1	2	59.553
-1	1	-1	-1	-1	3	44.23
1	1	-1	-1	1	4	48.705
-1	-1	1	-1	-1	5	44.83
1	-1	1	-1	1	6	48.764
-1	1	1	-1	1	7	45.848
1	1	1	-1	-1	8	60.478
-1	-1	-1	1	-1	9	50.335
1	-1	-1	1	1	10	53.543
-1	1	-1	1	1	11	39.68
1	1	-1	1	-1	12	53.433
-1	-1	1	1	1	13	40.798
1	-1	1	1	-1	14	52.061
-1	1	1	1	-1	15	51.183
1	1	1	1	1	16	53.708

A.0.6 *pb11*

Plackett–Burman design: 11 factors F1–F11 in 12 runs; response Response.

F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	StdOrder	Response
1	1	-1	1	1	1	-1	-1	-1	1	-1	1	10.331
-1	1	1	-1	1	1	1	-1	-1	-1	1	2	10.454
1	-1	1	1	-1	1	1	1	-1	-1	-1	3	12.158
-1	1	-1	1	1	-1	1	1	1	-1	-1	4	7.896
-1	-1	1	-1	1	1	-1	1	1	1	-1	5	8.597
-1	-1	-1	1	-1	1	1	-1	1	1	1	6	6.942
1	-1	-1	-1	1	-1	1	1	-1	1	1	7	14.437
1	1	-1	-1	-1	1	-1	1	1	-1	1	8	12.16
1	1	1	-1	-1	-1	1	-1	1	1	-1	9	13.671
-1	1	1	1	-1	-1	-1	1	-1	1	1	10	5.48
1	-1	1	1	1	-1	-1	-1	1	-1	1	11	9.394
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	12	8.049

A.0.7 *blend*

Polymer film formulation: simplex-centroid design in Polymer, Plasticizer and Filler (proportions summing to 1): the three pure components (each run twice), the three 50:50 blends and the centroid, 10 runs; response Elongation (%).

Polymer	Plasticizer	Filler	Elongation
1	0	0	9.599
0	1	0	13.892
0	0	1	7.479
0.5	0.5	0	14.153
0.5	0	0.5	9.402
0	0.5	0.5	11.058
0.333333333333333	0.333333333333333	0.333333333333333	11.374
1	0	0	9.982
0	1	0	14.498
0	0	1	7.978

A.0.8 *suppliers*

General factorial: Supplier (SupA, SupB, SupC) × Machine (M1, M2, M3) × Temp (10, 20), two replicates; response Thickness.

Supplier	Machine	Temp	StdOrder	Thickness
SupA	M1	10	1	34.196
SupB	M1	10	2	32.026
SupC	M1	10	3	32.104
SupA	M2	10	4	36.22
SupB	M2	10	5	32.22
SupC	M2	10	6	33.269
SupA	M3	10	7	31.372
SupB	M3	10	8	32.916
SupC	M3	10	9	29.383
SupA	M1	20	10	36.477
SupB	M1	20	11	32.962
SupC	M1	20	12	33.709
SupA	M2	20	13	37.955
SupB	M2	20	14	35.398
SupC	M2	20	15	34.817
SupA	M3	20	16	34.031
SupB	M3	20	17	33.63
SupC	M3	20	18	31.988
SupA	M1	10	19	34.661
SupB	M1	10	20	30.282
SupC	M1	10	21	32.017
SupA	M2	10	22	35.397
SupB	M2	10	23	31.881
SupC	M2	10	24	33.186
SupA	M3	10	25	31.855
SupB	M3	10	26	32.103
SupC	M3	10	27	30.294
SupA	M1	20	28	35.488
SupB	M1	20	29	33.383
SupC	M1	20	30	33.223
SupA	M2	20	31	37.478
SupB	M2	20	32	34.942
SupC	M2	20	33	34.771
SupA	M3	20	34	33.794
SupB	M3	20	35	34.018
SupC	M3	20	36	32.057

A.0.9 *shifts*

2^3 factorial in A, B, C with centre points, run in two blocks (Block: Day1/Day2); response Strength.

A	B	C	Block	StdOrder	Strength
-1	-1	-1	Day1	1	18.386
1	-1	-1	Day1	2	22.56
-1	1	-1	Day1	3	14.975
1	1	-1	Day1	4	22.787
-1	-1	1	Day1	5	19.224
1	-1	1	Day1	6	23.518
-1	1	1	Day1	7	15.174
1	1	1	Day1	8	23.061
0	0	0	Day1	9	21.538
0	0	0	Day1	10	21.922
-1	-1	-1	Day2	11	19.823
1	-1	-1	Day2	12	23.949
-1	1	-1	Day2	13	16.234
1	1	-1	Day2	14	24.005
-1	-1	1	Day2	15	19.277
1	-1	1	Day2	16	23.664
-1	1	1	Day2	17	16.208
1	1	1	Day2	18	23.435
0	0	0	Day2	19	23.227
0	0	0	Day2	20	22.006

A.0.10 *pilot*

First pilot study far from the optimum: 2^3 factorial in Temp (80–100), Time (30–50), Speed (10–20) with 3 centre points; response Conversion.

Temp	Time	Speed	Conversion	StdOrder
80	30	10	45.519	1
100	30	10	57.854	2
80	50	10	52.129	3
100	50	10	63.424	4
80	30	20	46.098	5
100	30	20	58.015	6
80	50	20	52.043	7
100	50	20	64.558	8
90	40	15	54.391	9
90	40	15	55.634	10
90	40	15	54.628	11

A.0.11 *taguchi*

Taguchi L9 array in four 3-level factors A–D with three replicate responses Y1–Y3 (larger is better).

A	B	C	D	Y1	Y2	Y3
1	1	1	1	18.2	18	18.2
1	2	2	2	21.01	20.59	20.67
1	3	3	3	19.24	19.83	19.66
2	1	2	3	18.82	18.73	19.49
2	2	3	1	20.59	19.87	19.78
2	3	1	2	19.53	19.43	20.18
3	1	3	2	21.16	20.51	20.54
3	2	1	3	22.31	21.89	22
3	3	2	1	22.98	23.44	22.91

A.0.12 *firstorder*

2^3 factorial in A (10–20), B (1–3) and C (100–200) with 3 centre points, response Y: the starting point for augmentation to a central composite design.

A	B	C	StdOrder	Y
10	1	100	1	48.69
20	1	100	2	54.91
10	3	100	3	44.61
20	3	100	4	50.56
10	1	200	5	48.62
20	1	200	6	55.27
10	3	200	7	44.37
20	3	200	8	50.65
15	2	150	9	54.19
15	2	150	10	53.93
15	2	150	11	54.62

A.0.13 *factorsheet*

A factor table typed into the Data Editor: one row per factor (Name, Low, High), 14 factors. Read by FACTORSFROMDATA=YES.

Name	Low	High
Temp	150	200
Time	10	30
Press	1	3
Speed	80	120
Feed	0.1	0.3
Depth	1	3
Coolant	0	1
Tool	0	1
Angle	5	15
Humid	30	60
Load	1	5
Ratio	0.2	0.8
Mix	1	4
Cure	10	20

A.0.14 *ovens*

Split-plot experiment: Oven temperature is hard to change (whole plots, numbered in WholePlot), Press and Mix are changed inside each oven setup; 3 replicates (Rep), 24 runs in run order (Run); response Y.

Run	Rep	Oven	Press	Mix	WholePlot	Y
1	1	150	1	1	1	21.0267
2	1	150	1	2	1	21.0704
3	1	150	3	1	1	21.8364
4	1	150	3	2	1	22.5795
5	1	200	1	1	2	22.2219
6	1	200	1	2	2	22.8761
7	1	200	3	1	2	23.9798
8	1	200	3	2	2	24.2809
9	2	150	1	1	3	19.4521
10	2	150	1	2	3	20.3932
11	2	150	3	1	3	20.5257
12	2	150	3	2	3	21.4089
13	2	200	1	1	4	23.6602
14	2	200	1	2	4	24.2136
15	2	200	3	1	4	24.5284
16	2	200	3	2	4	25.5917
17	3	150	1	1	5	19.1255
18	3	150	1	2	5	20.0402
19	3	150	3	1	5	20.5500
20	3	150	3	2	5	21.4593
21	3	200	1	1	6	21.9814
22	3	200	1	2	6	21.5589
23	3	200	3	1	6	23.1308
24	3	200	3	2	6	23.8034

A.0.15 *spread*

Replicated 2^2 factorial in A and B with 8 runs per setting; factor B changes the spread of Y. Response Y.

A	B	Y
10	1	49.3131
10	1	51.0764
10	1	49.7107
10	1	51.0674
10	1	51.1135
10	1	48.9070
10	1	49.8021
10	1	49.8954
10	2	49.3057
10	2	47.2339
10	2	47.3280
10	2	51.4208
10	2	46.9746
10	2	48.1923
10	2	52.6524
10	2	52.3816
20	1	50.5012
20	1	49.1460
20	1	48.2738
20	1	50.6256
20	1	50.4433
20	1	51.5422
20	1	51.5217
20	1	49.9893
20	2	49.8241
20	2	50.4233
20	2	50.7513
20	2	52.0566
20	2	54.9661
20	2	51.9167
20	2	51.5575
20	2	49.5890

Appendix B

Glossary

Adjusted R-squared R-squared penalized for the number of terms.

Alias Two effects estimated by the same contrast; see Resolution.

Axial (star) point A run on a factor axis outside (rotatable CCD) or on (face-centred) the cube.

Block Group of runs under similar conditions; its effect is removed from the analysis.

Box–Behnken design Three-level response-surface design without corner runs.

Box–Cox transformation Power transformation y^{λ} chosen from the data.

Canonical analysis Locating and classifying the stationary point of a quadratic model.

Central composite design (CCD) Factorial + axial + centre runs, for quadratic models.

Centre point Run at the middle of every numeric factor's range.

Coded units Factor settings rescaled to -1 (Low) to $+1$ (High).

Composite desirability (D) Weighted geometric mean of individual desirabilities.

Confirmation runs Runs at the chosen settings to verify the model's prediction.

Curvature Non-linear dependence of the response on a factor.

D-optimal design Design maximizing $\det(X^{\top}X)$: most precise coefficients.

Definitive screening design Three-level screening design with $2k + 1$ runs.

Design space Region of factor settings where all quality requirements are met with assurance.

Dispersion effect A factor's effect on the variability of the response.

Effect Change in the response from the low to the high setting.

Extreme vertices Corners of a constrained mixture region.

Face-centred design CCD with axial distance 1.

Factor A controlled input variable.

Fractional factorial A planned subset of a full factorial (2^{k-p}).

Hard-to-change factor A factor that is changed only between groups of runs (split-plot).

Interaction The effect of one factor depends on another factor's setting.

L-pseudo-components Rescaled mixture proportions for constrained regions.

Lack of fit Systematic deviation of the data from the model's form.

Lenth's method Judging effects without error df, using a robust pseudo standard error.

Level A value a factor takes.

Main effect Average effect of one factor.

Mixture design Design for proportions that add to a constant.

Orthogonal Factor columns uncorrelated, so effects are estimated independently (VIF = 1).

Path of steepest ascent Direction of fastest increase of a first-order model.

Plackett–Burman design Two-level screening design with a multiple of 4 runs.

Predicted R-squared Leave-one-out predictive ability of the model.

Prediction interval Range for a single future run at given settings.

Pure error Variation between replicates of identical settings.

Randomization Running runs in random order.

Replicate Independent repetition of a complete run.

Resolution Degree of aliasing in a fractional design (III, IV, V, . . .).

REML Restricted maximum likelihood, for variance components.

Response surface methodology (RSM) Sequential experimentation with quadratic models to find optimal settings.

Robust design Settings that make the response insensitive to noise.

Rotatable design Prediction precision depends only on the distance from the centre.

Saddle point Stationary point that is a maximum in some directions and a minimum in others.

Saturated model Model with as many terms as runs: no error df.

Screening Finding the few important factors among many.

Signal-to-noise ratio (S/N) Taguchi's measure combining mean and variability.

Split-plot design Design with restricted randomization of hard-to-change factors.

Stationary point Point where the fitted surface is flat (maximum, minimum, saddle).

VIF Variance inflation factor: how much correlation among terms inflates a coefficient's variance.

Whole plot Group of runs sharing one setting of the hard-to-change factors.

Appendix C

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